

Digital Design & Computer Arch.

Lecture 17: Branch Prediction

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ETH Zürich
Spring 2026
23 April 2026

We Are Almost Done with All This

- **Microarchitecture Fundamentals**

- Single-cycle Microarchitectures
- Multi-cycle Microarchitectures

- **Pipelining & Precise Exceptions**

- Pipelining
- Pipelined Processor Design
 - Control & Data Dependence Handling
 - Precise Exceptions: State Maintenance & Recovery

Problem
Algorithm
Program/Language
System Software
SW/HW Interface
Micro-architecture
Logic
Devices
Electrons

- **Last+this week: Out-of-Order & Superscalar Execution**

- Out-of-Order Execution
- Dataflow & Superscalar Execution
- Branch Prediction

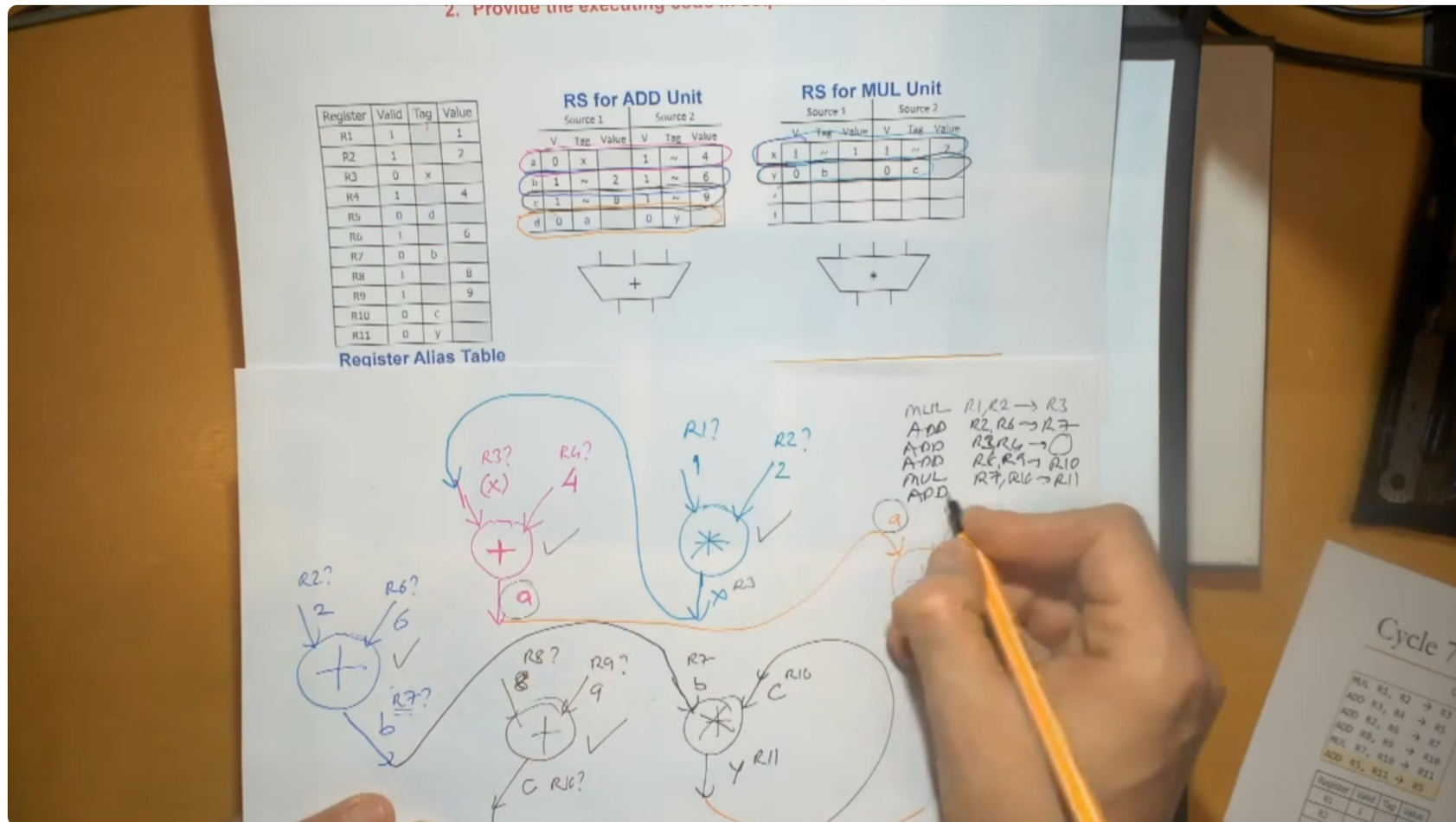
Readings

■ This week

- ❑ Smith and Sohi, “The Microarchitecture of Superscalar Processors,” Proceedings of the IEEE, 1995
- ❑ H&H Chapters 7.8 and 7.9
- ❑ McFarling, “Combining Branch Predictors,” DEC WRL Technical Report, 1993.
- ❑ Optional: Kessler, “The Alpha 21264 Microprocessor,” IEEE Micro 1999.

Optional Lecture 16a: OoO Execution

- Easier to follow online & covers Data Flow Graph construction



Digital Design and Computer Arch. - L16a: Out-of-Order Execution (Spring 2026)



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<https://www.youtube.com/watch?v=cSlzoF4VASs>

Optional Lecture 16b: LD/ST Handling

- How to perform Load & Store Scheduling in OoO execution

Store-Load Forwarding Complexity

- Content Addressable Search (based on Load Address)
- Range Search (based on Address and Size of both the Load and earlier Stores)
- Age-Based Search (for last written values)
- Load data can come from a combination of multiple places
 - ❑ One or more stores in the Store Buffer (SQ)
 - ❑ Memory/cache

69



Digital Design and Comp. Arch. - L16b: Load-Store Handling in Out-of-Order Execution (Spring 2026)



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<https://www.youtube.com/watch?v=AuwdX-5NO9A>

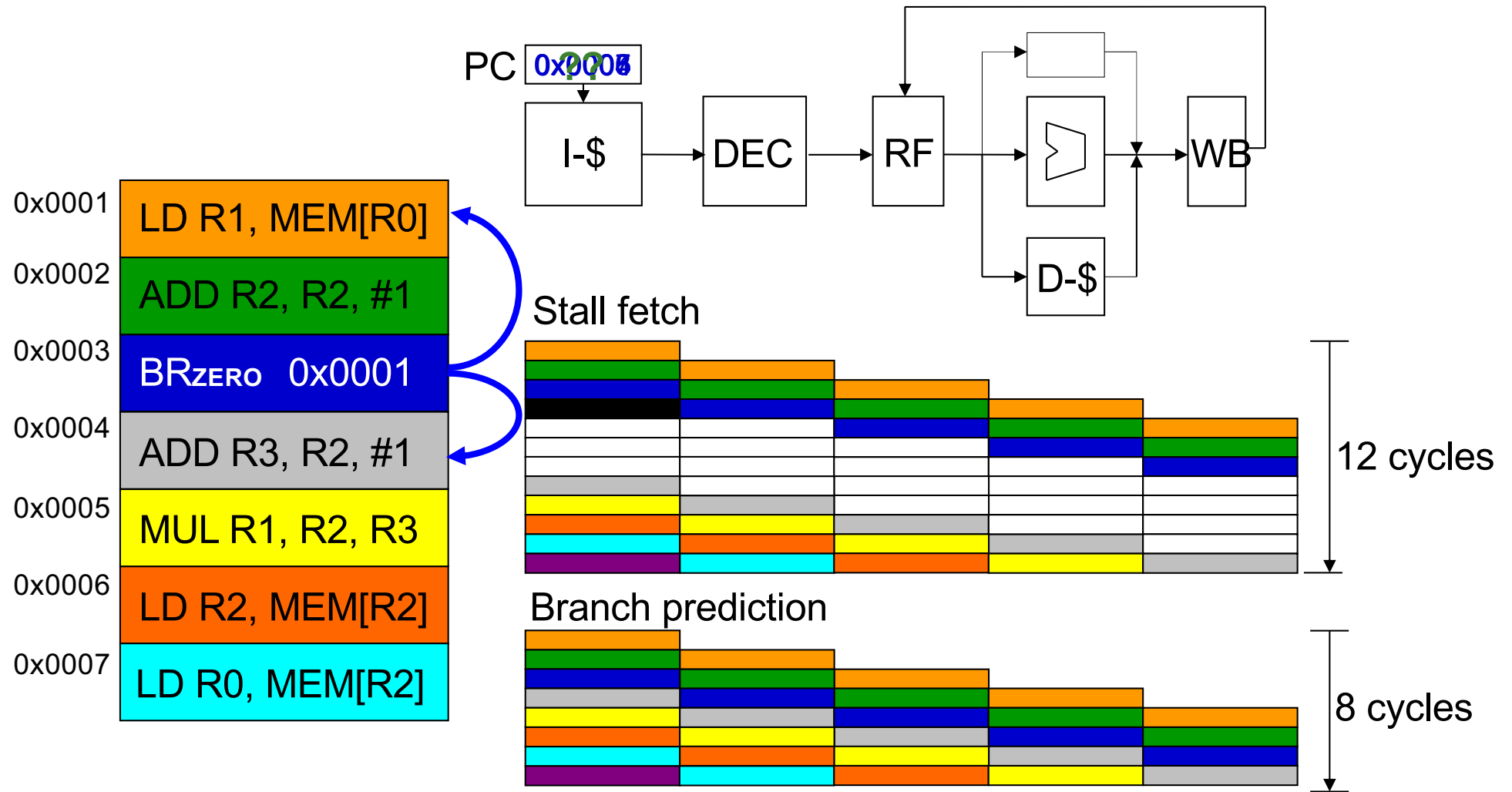
Recall: How to Handle Control Dependences

- Critical to keep the pipeline full with correct sequence of dynamic instructions.
- Potential solutions if the instruction is a control-flow instruction:
 - **Stall** the pipeline until we know the next fetch address
 - Guess the next fetch address (**branch prediction**)
 - Employ delayed branching (**branch delay slot**)
 - Do something else (**fine-grained multithreading**)
 - Eliminate control-flow instructions (**predicated execution**)
 - Fetch from both possible paths (if you know the addresses of both possible paths) (**multipath execution**)

Recall: Importance of The Branch Problem

- Assume $N = 20$ (20 pipe stages), $W = 5$ (5 wide fetch)
- Assume: 1 out of 5 instructions is a branch
- Assume: Each 5 instruction-block ends with a branch
- How long does it take to fetch 500 instructions?
 - 100% accuracy
 - 100 cycles (all instructions fetched on the correct path)
 - No wasted work; $IPC = 500/100$
 - 99% accuracy
 - 100 (correct path) + $20 * 1$ (wrong path) = 120 cycles
 - 20% extra instructions fetched; $IPC = 500/120$
 - 90% accuracy
 - 100 (correct path) + $20 * 10$ (wrong path) = 300 cycles
 - 200% extra instructions fetched; $IPC = 500/300$
 - 60% accuracy
 - 100 (correct path) + $20 * 40$ (wrong path) = 900 cycles
 - 800% extra instructions fetched; $IPC = 500/900$

Branch Prediction: Guess the Next Instruction to Fetch



Recall: Always Guess $\text{NextPC} = \text{PC} + 4$

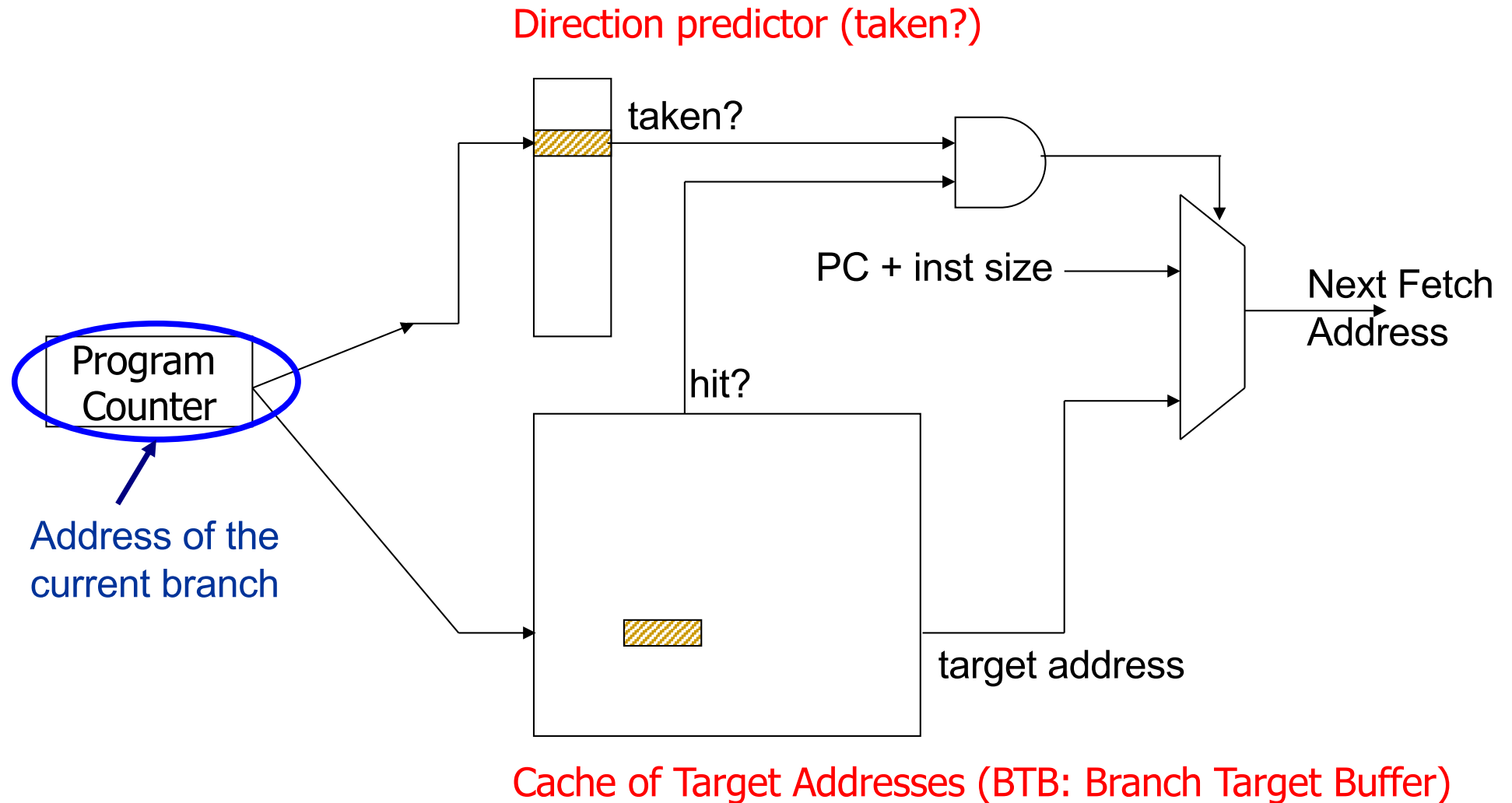
- Always predict the next sequential instruction is the next instruction to be executed
- This is a form of **next fetch address prediction** (and branch prediction)
- **How can you make this more effective?**
- Idea: **Maximize the chances that the next sequential instruction is the next instruction to be executed**
 - Software: **Lay out the control flow graph such that the “likely next instruction” is on the not-taken path of a branch**
 - Profile guided code positioning → Pettis & Hansen, PLDI 1990.
 - Hardware: **???** (how can you do this in hardware...)
 - Cache traces of executed instructions → Trace cache

(Enhanced) Branch Prediction

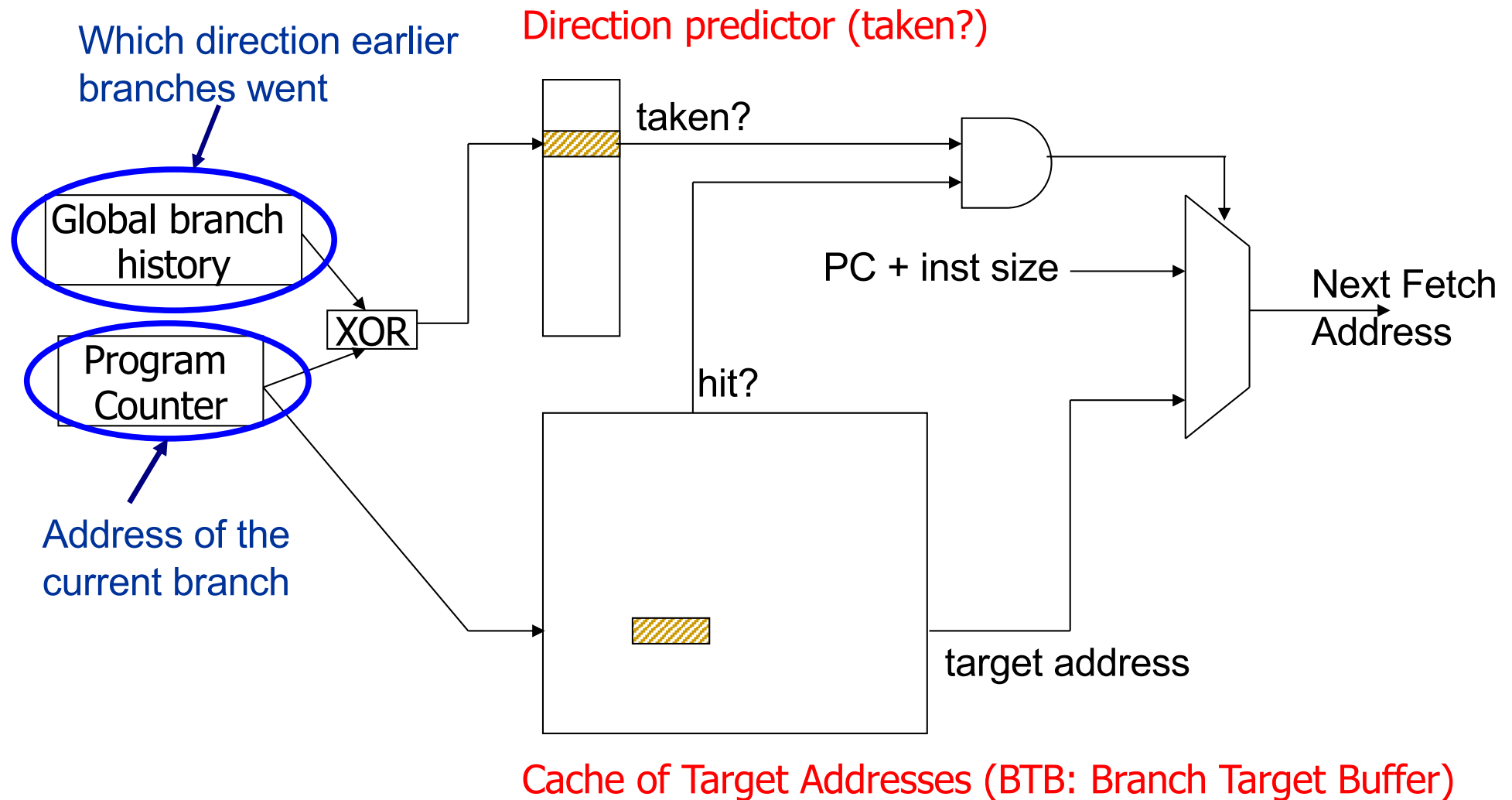
Branch Prediction (A Bit More Enhanced)

- Idea: **Predict the next fetch address** (to be used in the **next cycle**) → no wasted cycle(s) on correct prediction
- Requires three things to be predicted at fetch stage:
 - ❑ Whether the fetched instruction is a branch
 - ❑ (Conditional) branch direction
 - ❑ Branch target address (if taken)
- Observation: Target address remains the same for a conditional direct branch across dynamic instances
 - ❑ Idea: Store the target address from previous instance and access it with the PC
 - ❑ Called **Branch Target Buffer (BTB)** or Branch Target Address Cache

Fetch Stage with BTB and Direction Prediction



More Sophisticated Branch Direction Prediction



Three Things to Be Predicted

- Requires three things to be predicted at fetch stage:
 1. Whether the fetched instruction is a branch
 2. (Conditional) branch direction
 3. Branch target address (if taken)
- Third (3.) can be accomplished using a BTB
 - Remember target address computed last time branch was executed
- First (1.) can be accomplished using a BTB
 - If BTB provides a target address for the program counter, then it must be a branch
 - Or, we can store “branch metadata” bits in instruction cache/memory → partially decoded instruction
- Second (2.): How do we predict the direction?

Simple Branch Direction Prediction Schemes

- Compile time (static)
 - ❑ Always not taken
 - ❑ Always taken
 - ❑ BTFN (Backward taken, forward not taken)
 - ❑ Profile based (likely direction)

- Run time (dynamic)
 - ❑ Last time prediction (single-bit)

More Sophisticated Direction Prediction

■ Compile time (static)

- ❑ Always not taken
- ❑ Always taken
- ❑ BTFN (Backward taken, forward not taken)
- ❑ Profile based (likely direction)
- ❑ Program analysis based (likely direction)

■ Run time (dynamic)

- ❑ Last time prediction (single-bit)
- ❑ Two-bit counter based prediction
- ❑ Two-level prediction (global vs. local)
- ❑ Hybrid
- ❑ Advanced algorithms (e.g., using perceptrons, geometric history)

Static Branch Prediction (I)

■ Always not-taken

- ❑ Simple to implement: no need for BTB, no direction prediction
- ❑ Low accuracy: ~30-40% (for conditional branches)
- ❑ Remember: Compiler can layout code such that the likely path is the “not-taken” path → more effective prediction

■ Always taken

- ❑ No direction prediction
- ❑ Better accuracy: ~60-70% (for conditional branches)
 - Backward branches (i.e., loop branches) are usually taken
 - Backward branch: target address lower than branch PC

■ Backward taken, forward not taken (BTFN)

- ❑ Predict backward (loop) branches as taken, others not-taken

Static Branch Prediction (II)

■ Profile-based

- ❑ Idea: Compiler determines likely direction for each branch using a profile run. Encodes that direction as a hint bit in the branch instruction format.

+ Per branch prediction (more accurate than schemes in previous slide) → accurate if profile is representative!

-- Requires hint bits in the branch instruction format

-- Accuracy depends on dynamic branch behavior:

TTTTTTTTTTTTNNNNNNNNNNNN → 50% accuracy

TNTNTNTNTNTNTNTNTNTNTN → 50% accuracy

-- Accuracy depends on representativeness of profile input set

TTTTTTTTTTTTTTTTTTTTNN → 90% accuracy (or 10%?)

Static Branch Prediction (III)

- **Program-based (or, program analysis based)**
 - ❑ Idea: Use heuristics based on program analysis to determine statically-predicted direction
 - ❑ Example opcode heuristic: Predict BLEZ as NT (negative integers used as error values in many programs)
 - ❑ Example loop heuristic: Predict a branch guarding a loop execution as taken (i.e., execute the loop)
 - ❑ Pointer and FP comparisons: Predict not equal
- + Does not require profiling
- Heuristics might be not representative or good
- Requires compiler analysis and ISA support (ditto for other static methods)
- Ball and Larus, "Branch prediction for free," PLDI 1993.
 - ❑ 20% misprediction rate

Static Branch Prediction (IV)

■ Programmer-based

- Idea: Programmer provides the statically-predicted direction
- Via *pragmas* in the programming language that qualify a branch as likely-taken versus likely-not-taken

- + Does not require profiling or program analysis
- + Programmer may know some branches and their program better than other analysis techniques
- Requires programming language, compiler, ISA support
- Burdens the programmer

Pragmas

- Idea: Keywords that enable a programmer to convey hints to lower levels of the transformation hierarchy
- `if (likely(x)) { ... }`
- `if (unlikely(error)) { ... }`
- Many other hints and optimizations can be enabled with pragmas
 - E.g., whether a loop can be parallelized
 - **#pragma omp parallel**
 - **Description**
 - The omp parallel directive explicitly instructs the compiler to parallelize the chosen segment of code

Static Branch Prediction

- All previous techniques can be combined
 - ❑ Profile based
 - ❑ Program based
 - ❑ Programmer based

- How would you do that?

- What is the common disadvantage of all three techniques?
 - ❑ Cannot adapt to dynamic changes in branch behavior
 - This can be mitigated by a dynamic compiler, but not at a fine granularity (and a dynamic compiler has its overheads...)
 - What is a Dynamic Compiler?
 - ❑ A compiler that generates code at runtime
 - ❑ Java JIT (just in time) compiler, Microsoft CLR (common lang. runtime)

Recall: Rosetta 2 Binary Translator

Rosetta 2 [\[edit\]](#)

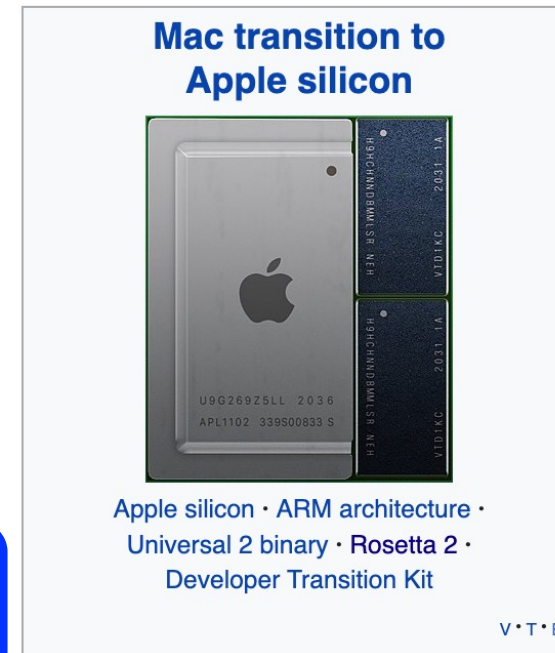
In 2020, Apple announced Rosetta 2 would be bundled with [macOS Big Sur](#), to aid in the [Mac transition to Apple silicon](#). The software permits many applications compiled exclusively for execution on x86-64-based processors to be translated for execution on Apple silicon.^{[2][8]}

In addition to the [just-in-time](#) (JIT) translation support, Rosetta 2 offers [ahead-of-time compilation](#) (AOT), with the x86-64 code fully translated, just once, when an application without a universal binary is installed on an Apple silicon Mac.^[9]

Rosetta 2's performance has been praised greatly.^{[10][11]} In some benchmarks, x86-64-only programs performed better under Rosetta 2 on a Mac with an Apple M1 SOC than natively on a Mac with an Intel x86-64 processor. One of the key reasons why Rosetta 2 provides such high level of translation efficiency is the support of x86-64 [memory ordering](#) in Apple M1 SOC.^[12]

Although Rosetta 2 works for most software, some software doesn't work at all^[13] or is reported to be "sluggish".^[14] A lot of software can be made compatible with the new Macs by the vendor recompiling the software, often a simple task; while for some software (such as software that includes [assembly language](#) code, or that generates [machine code](#)), the changes to make them work aren't simple and cannot be automated.

Similar to the first version, Rosetta 2 does not normally require user intervention. When a user attempts to launch an x86-64-only application for the first time, macOS prompts them to install Rosetta 2 if it is not already available. Subsequent launches of x86-64 programs will execute via translation automatically. An option also exists to force a universal binary to run as x86-64 code through Rosetta 2, even on an ARM-based machine.^[15]

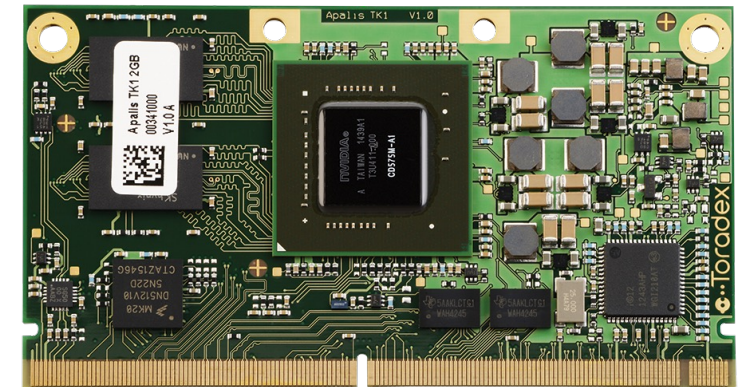
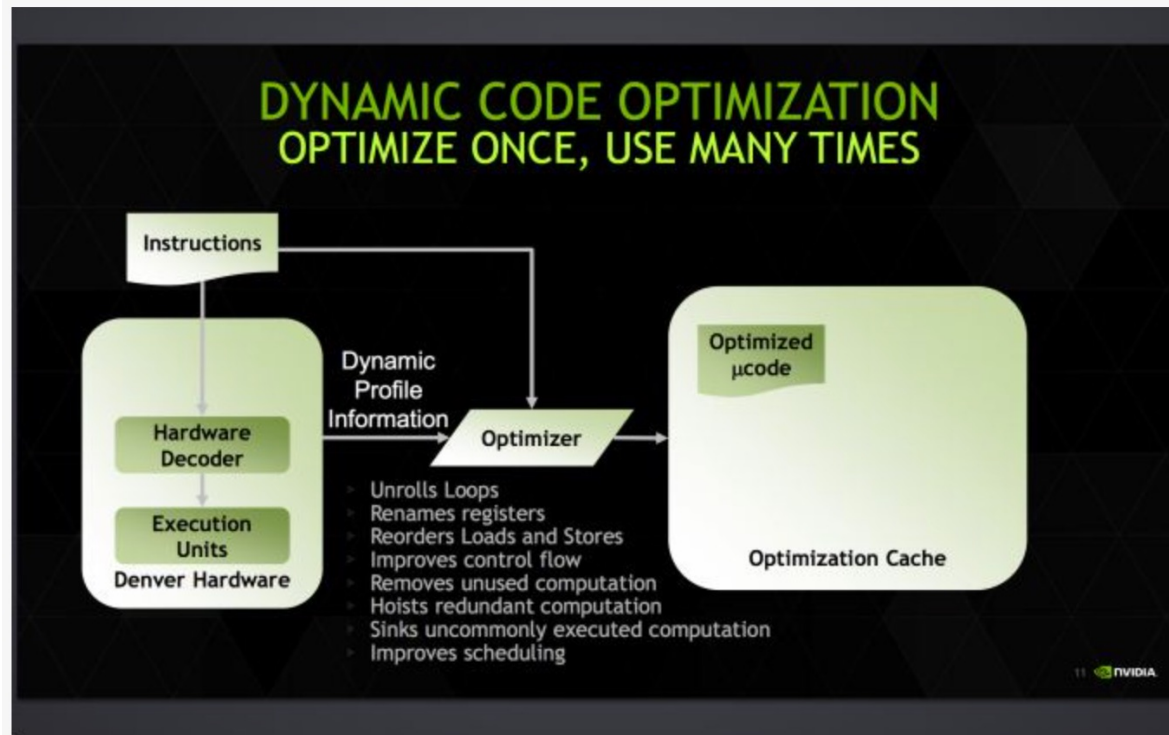


Recall: NVIDIA Denver Dynamic Code Optimizer

The Secret of Denver: Binary Translation & Code Optimization

As we alluded to earlier, NVIDIA's decision to forgo a traditional out-of-order design for Denver means that much of Denver's potential is contained in its software rather than its hardware. The underlying chip itself, though by no means simple, is at its core a very large in-order processor. So it falls to the software stack to make Denver sing.

Accomplishing this task is NVIDIA's dynamic code optimizer (DCO). The purpose of the DCO is to accomplish two tasks: to translate ARM code to Denver's native format, and to optimize this code to make it run better on Denver. With no out-of-order hardware on Denver, it is the DCO's task to find instruction level parallelism within a thread to fill Denver's many execution units, and to reorder instructions around potential stalls, something that is no simple task.



More Sophisticated Direction Prediction

■ Compile time (static)

- ❑ Always not taken
- ❑ Always taken
- ❑ BTFN (Backward taken, forward not taken)
- ❑ Profile based (likely direction)
- ❑ Program analysis based (likely direction)

■ Run time (dynamic)

- ❑ Last time prediction (single-bit)
- ❑ Two-bit counter based prediction
- ❑ Two-level prediction (global vs. local)
- ❑ Hybrid
- ❑ Advanced algorithms (e.g., using perceptrons, geometric history)

Dynamic Branch Prediction

- Idea: Predict branches based on dynamic information (collected at run-time)
- Advantages
 - + Prediction based on history of the execution of branches
 - + It can adapt to dynamic changes in branch behavior
 - + No need for static profiling: input set representativeness problem goes away
- Disadvantages
 - More complex (requires additional hardware)

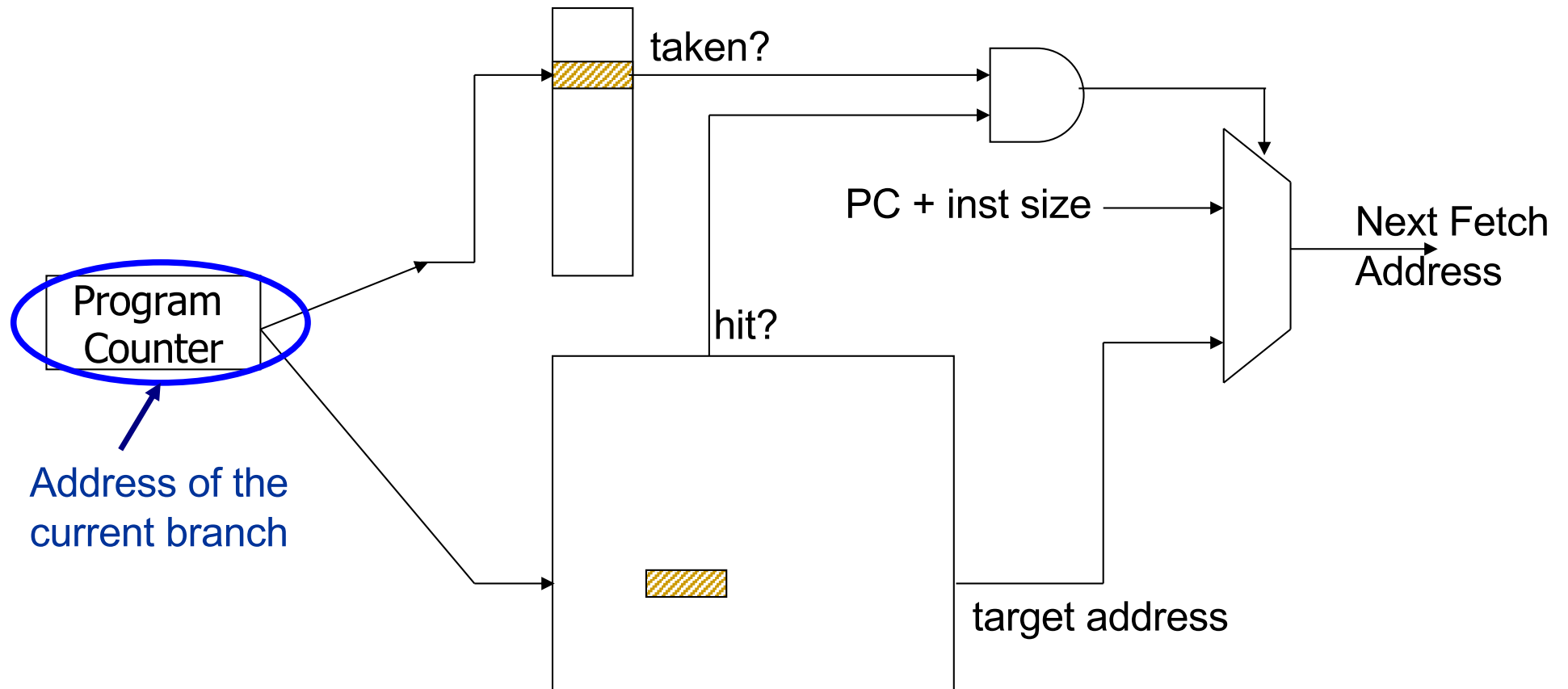
Last Time Predictor

- Last time predictor
 - Idea: Guess branch will take the same direction as its last instance
 - Single bit per branch (stored in BTB)
 - Indicates which direction branch went last time it executed

TTTTTTTTTTNNNNNNNNNN → 90% accuracy
- Always mispredicts the last iteration and the first iteration of a loop branch
 - Accuracy for a loop with N iterations = $(N-2)/N$
- + Loop branches for loops with large N (number of iterations)
~100% accuracy
- Loop branches for loops with small N (number of iterations)
TNTNTNTNTNTNTNTNTN → 0% accuracy

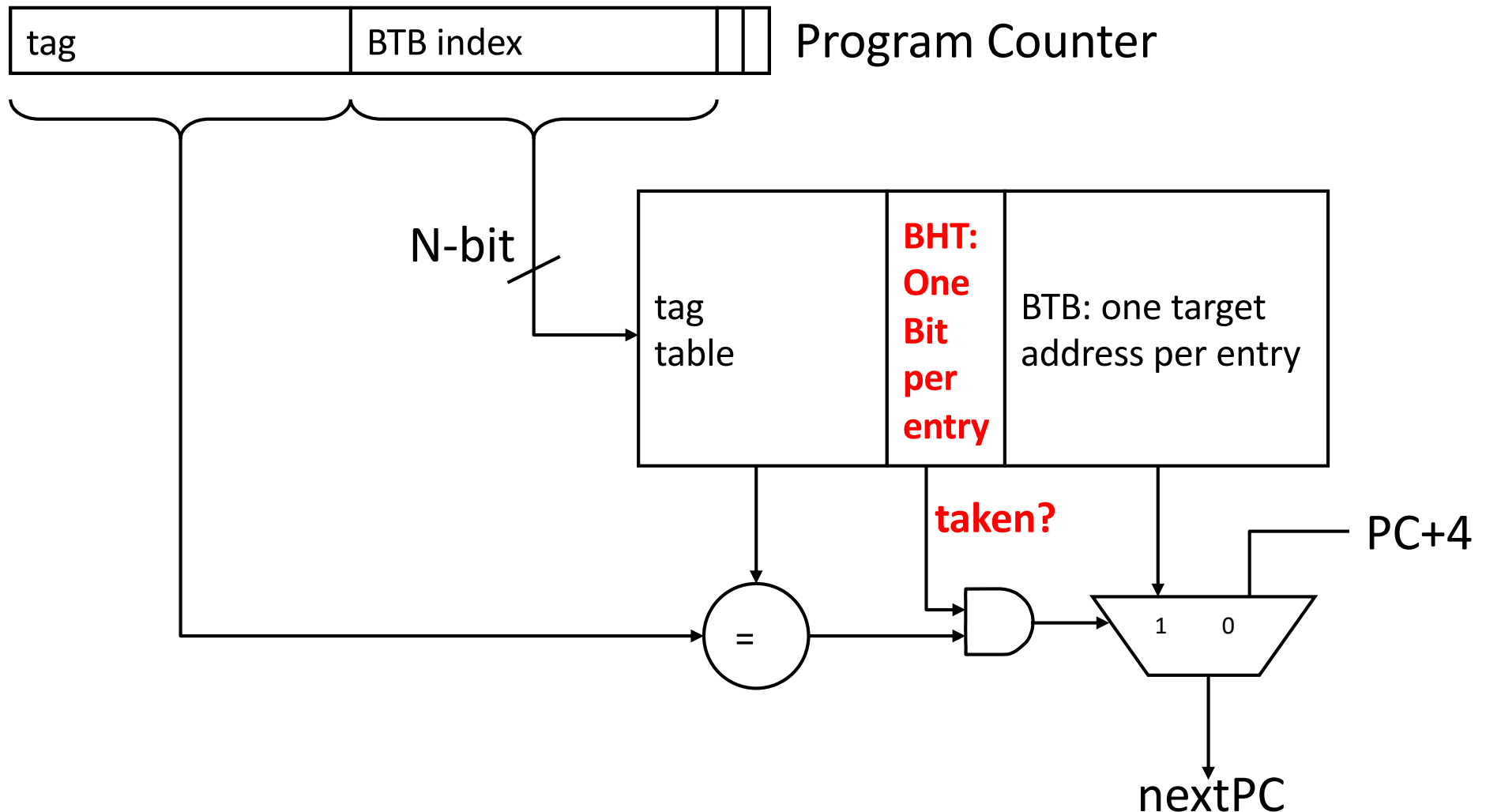
Implementing the Last-Time Predictor (I)

Direction predictor (**taken last time?**)



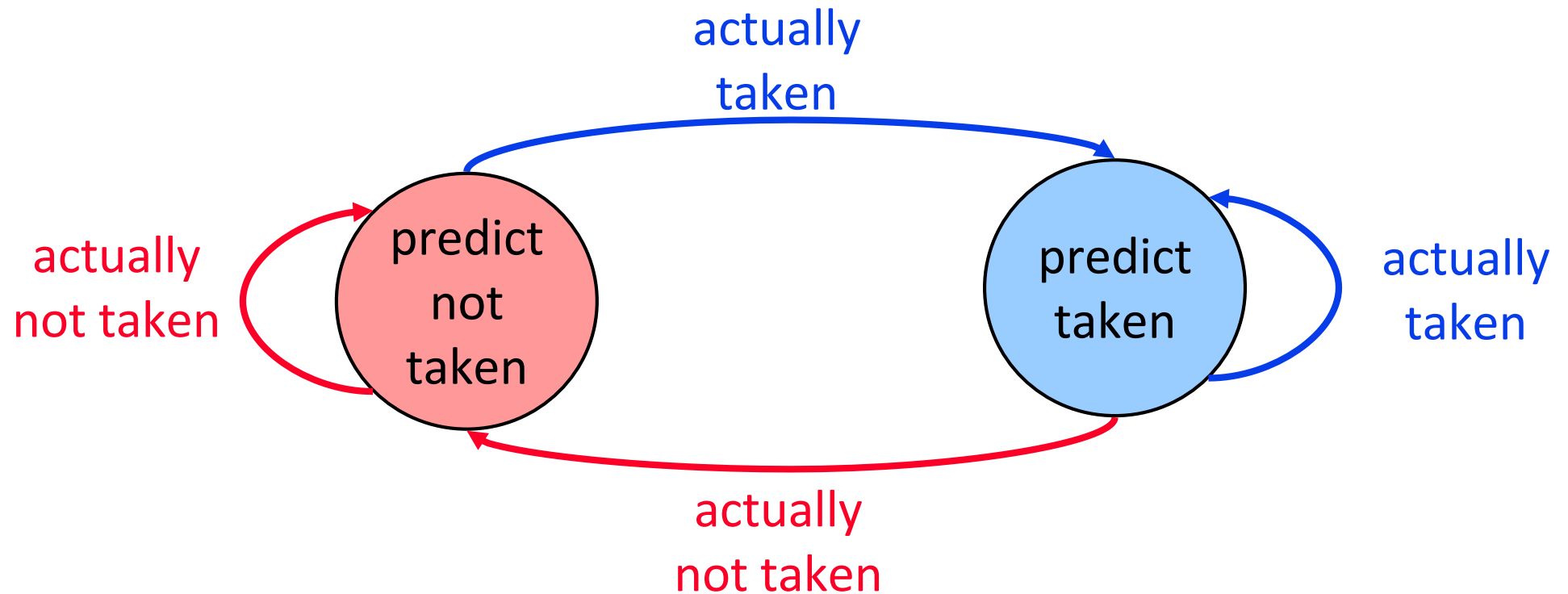
Cache of Target Addresses (BTB: Branch Target Buffer)

Implementing the Last-Time Predictor (II)



The 1-bit BHT (Branch History Table) entry is updated with the correct outcome after each execution of a branch

State Machine for Last-Time Prediction



Improving the Last Time Predictor

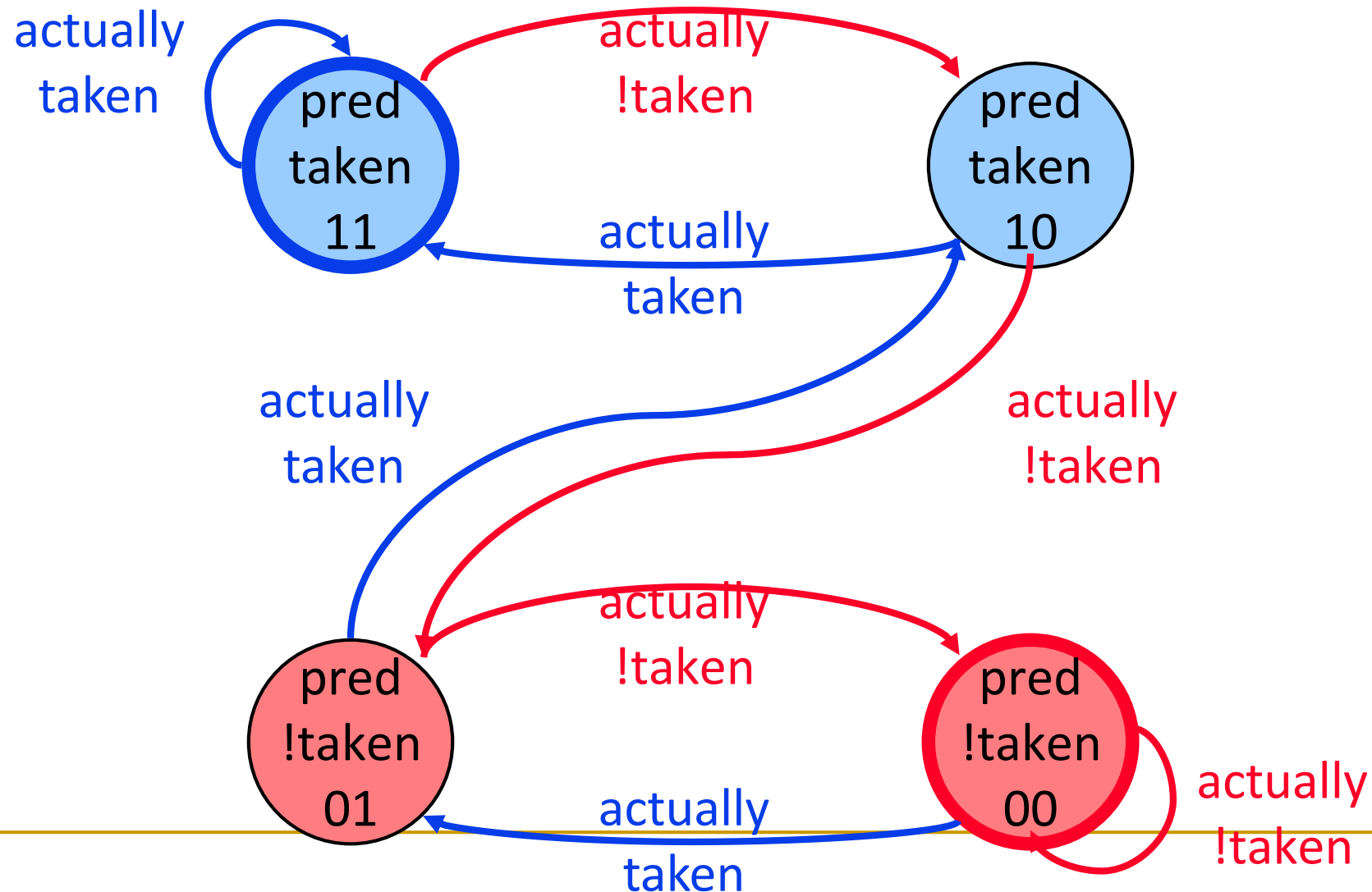
- Problem: A last-time predictor changes its prediction from $T \rightarrow NT$ or $NT \rightarrow T$ too quickly
 - even though the branch may be mostly taken or mostly not taken
- Solution Idea: Add hysteresis to the predictor so that prediction does not change on a single different outcome
 - Use two bits to track the history of predictions for a branch instead of a single bit
 - Can have 2 states each for T or NT instead of 1 state for each
- Smith, "A Study of Branch Prediction Strategies," ISCA 1981.

Two-Bit Counter Based Prediction

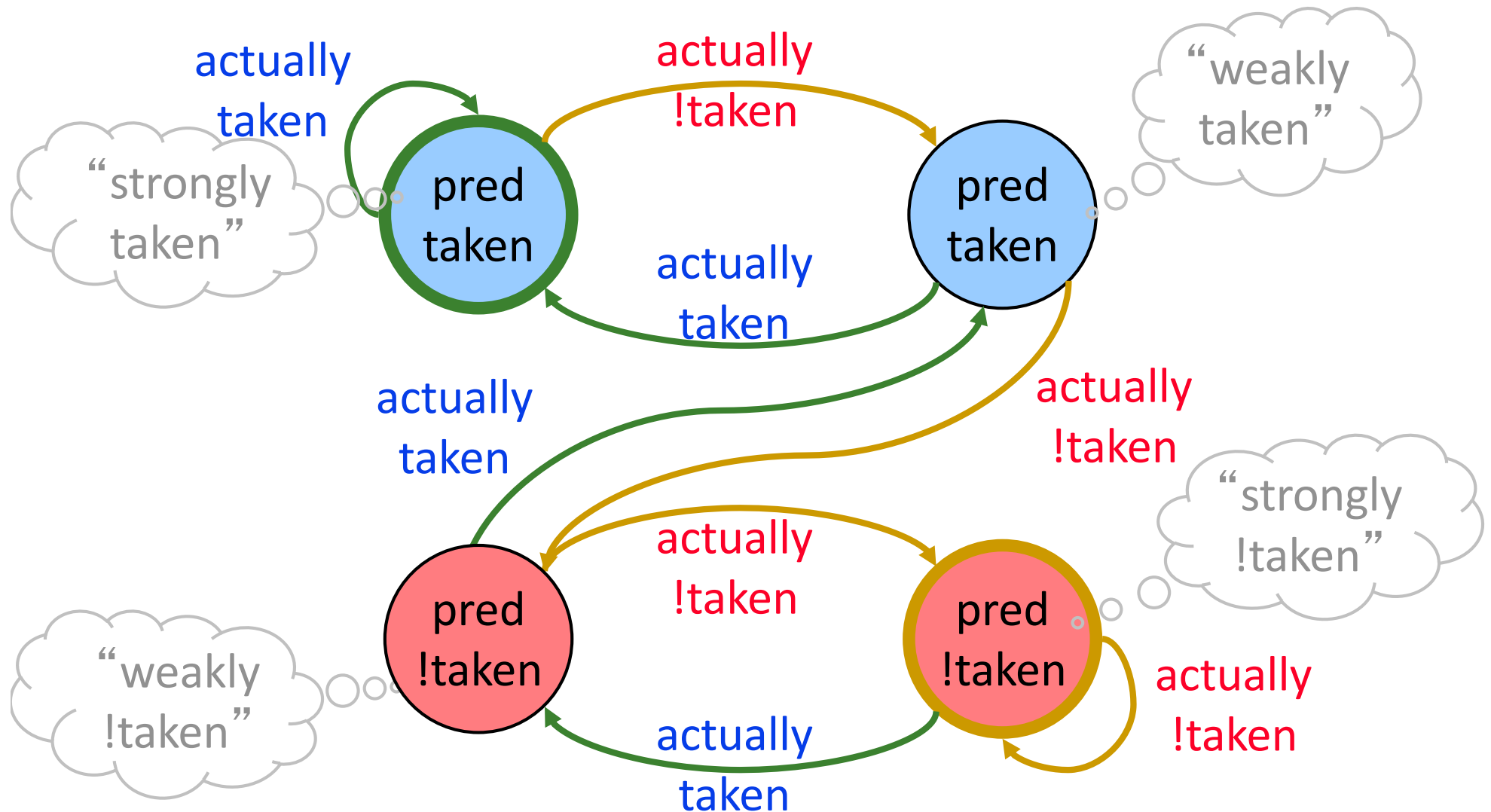
- Each branch associated with a two-bit counter (2BC)
- One more bit provides hysteresis
- A strong prediction does not change with one single different outcome
- Also called **bimodal prediction**

State Machine for 2-bit Saturating Counter

- Counter using *saturating arithmetic*
 - ▣ Arithmetic with maximum and minimum values



Hysteresis Using a 2-bit Counter



Change "strong" prediction only after 2 consecutive mistakes

Two-Bit Counter Based Prediction

- Each branch associated with a two-bit counter (stored in BTB)
- One more bit provides hysteresis
- A strong prediction does not change with one single different outcome

- Accuracy for a loop with N iterations = $(N-1)/N$

TTTTTTTTTTTTTTTTTTN → 95% accuracy

TNTNTNTNTNTNTNTNTN → 50% accuracy

(assuming counter initialized to weakly taken)

+ Better prediction accuracy

-- More hardware cost (but counter can be part of a BTB entry)

Is This Good Enough?

- ~85-90% accuracy for **many** programs with 2-bit counter based prediction (also called **bimodal prediction**)
- Is this good enough?
- How big is the branch problem?

Let's Do the Exercise Again

- Assume $N = 20$ (20 pipe stages), $W = 5$ (5 wide fetch)
- Assume: 1 out of 5 instructions is a branch
- Assume: Each 5 instruction-block ends with a branch
- How long does it take to fetch 500 instructions?
 - 100% accuracy
 - 100 cycles (all instructions fetched on the correct path)
 - No wasted work; $IPC = 500/100$
 - 90% accuracy
 - 100 (correct path) + $20 * 10$ (wrong path) = 300 cycles
 - 200% extra instructions fetched; $IPC = 500/300$
 - 85% accuracy
 - 100 (correct path) + $20 * 15$ (wrong path) = 400 cycles
 - 300% extra instructions fetched; $IPC = 500/400$
 - 80% accuracy
 - 100 (correct path) + $20 * 20$ (wrong path) = 500 cycles
 - 400% extra instructions fetched; $IPC = 500/500$

Can We Do Better: Two-Level Prediction

- Last-time and 2BC predictors exploit “last-time” predictability

- Realization 1: A branch’s outcome can be correlated with other branches’ outcomes
 - Global branch correlation

- Realization 2: A branch’s outcome can be correlated with past outcomes of the same branch (other than the outcome of the branch “last-time” it was executed)
 - Local branch correlation

Global Branch Correlation (I)

- Recently executed branch outcomes in the execution path are correlated with the outcome of the next branch

```
if (cond1)
...
if (cond1 AND cond2)
```

- If first branch not taken, second also not taken

if (x<1) ...	branch Y: if (cond1) a = 2;
if (x>1) ...	...
	branch X: if (a == 0)

- If first branch taken, second definitely not taken

Global Branch Correlation (II)

branch Y: if (cond1)

...

branch Z: if (cond2)

...

branch X: if (cond1 AND cond2)

- If Y **and** Z both taken, then X also taken
- If Y **or** Z not taken, then X also not taken

Global Branch Correlation (III)

- Eqntott, SPEC'92 workload
 - Generates truth table from Boolean expression

```
if (aa==2)                ;; B1
    aa=0;
if (bb==2)                ;; B2
    bb=0;
if (aa!=bb) {             ;; B3
    ....
}
```

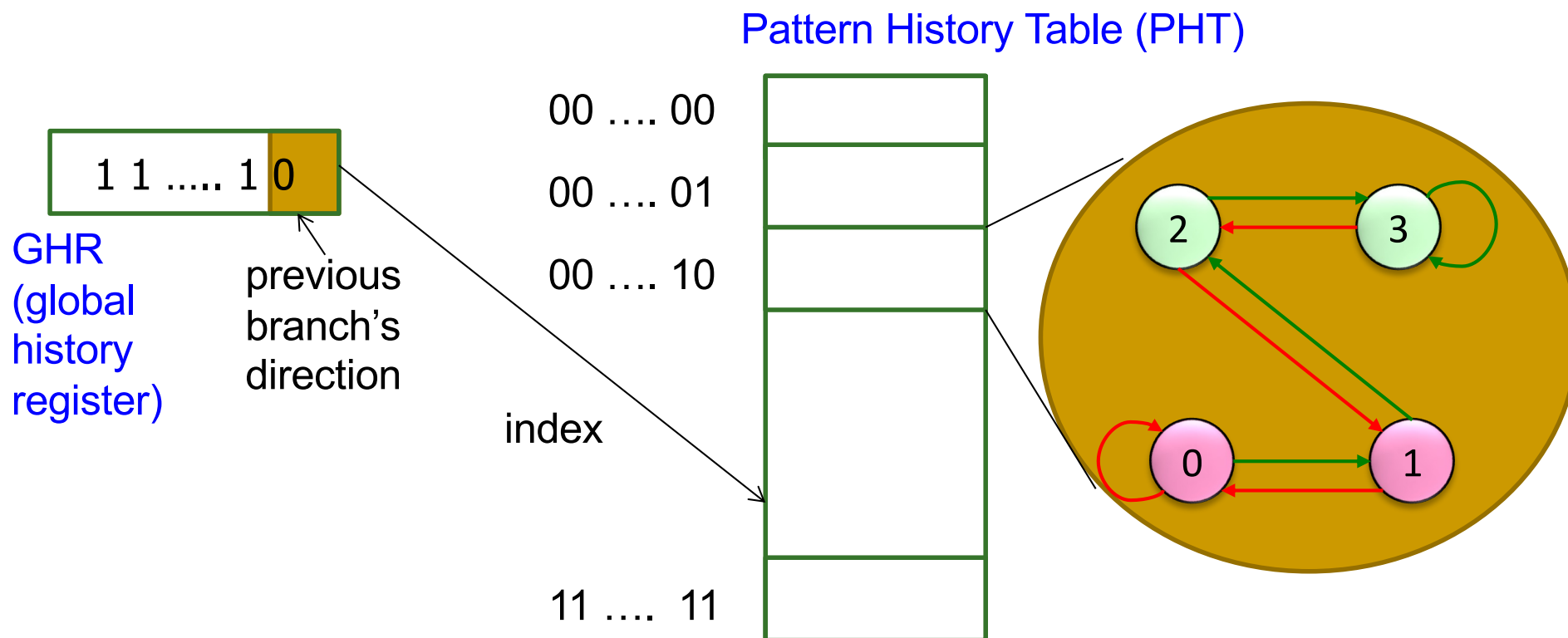
If **B1** is taken (i.e., aa=0@**B3**) and **B2** is taken (i.e. bb=0@**B3**)
then **B3** is not taken

Capturing Global Branch Correlation

- Idea: Associate branch outcomes with “global T/NT history” of all branches
- Make a prediction based on the outcome of the branch the last time the same global branch history was encountered
- Implementation:
 - Keep track of the “global T/NT history” of all branches in a register → Global History Register (GHR)
 - Use GHR to index into a table that recorded the outcome that was seen for each GHR value in the recent past → Pattern History Table (table of 2-bit counters)
- Global history/branch predictor
- Uses two levels of history (GHR + history at that GHR)

Two Level Global Branch Prediction

- First level: **Global branch history register** (N bits)
 - The directions of last N branches (global history)
- Second level: **Table of saturating counters for each history entry**
 - The direction the branch took the last time the same history was seen



How Does the Global Predictor Work?

```
for (i=0; i<100; i++)  
    for (j=0; j<3; j++)
```

After the initial startup time, the conditional branches have the following behavior, assuming GR is shifted to the left:

test	value	GR	result
j<3	j=1	1101	taken
j<3	j=2	1011	taken
j<3	j=3	0111	not taken
i<100		1110	usually taken

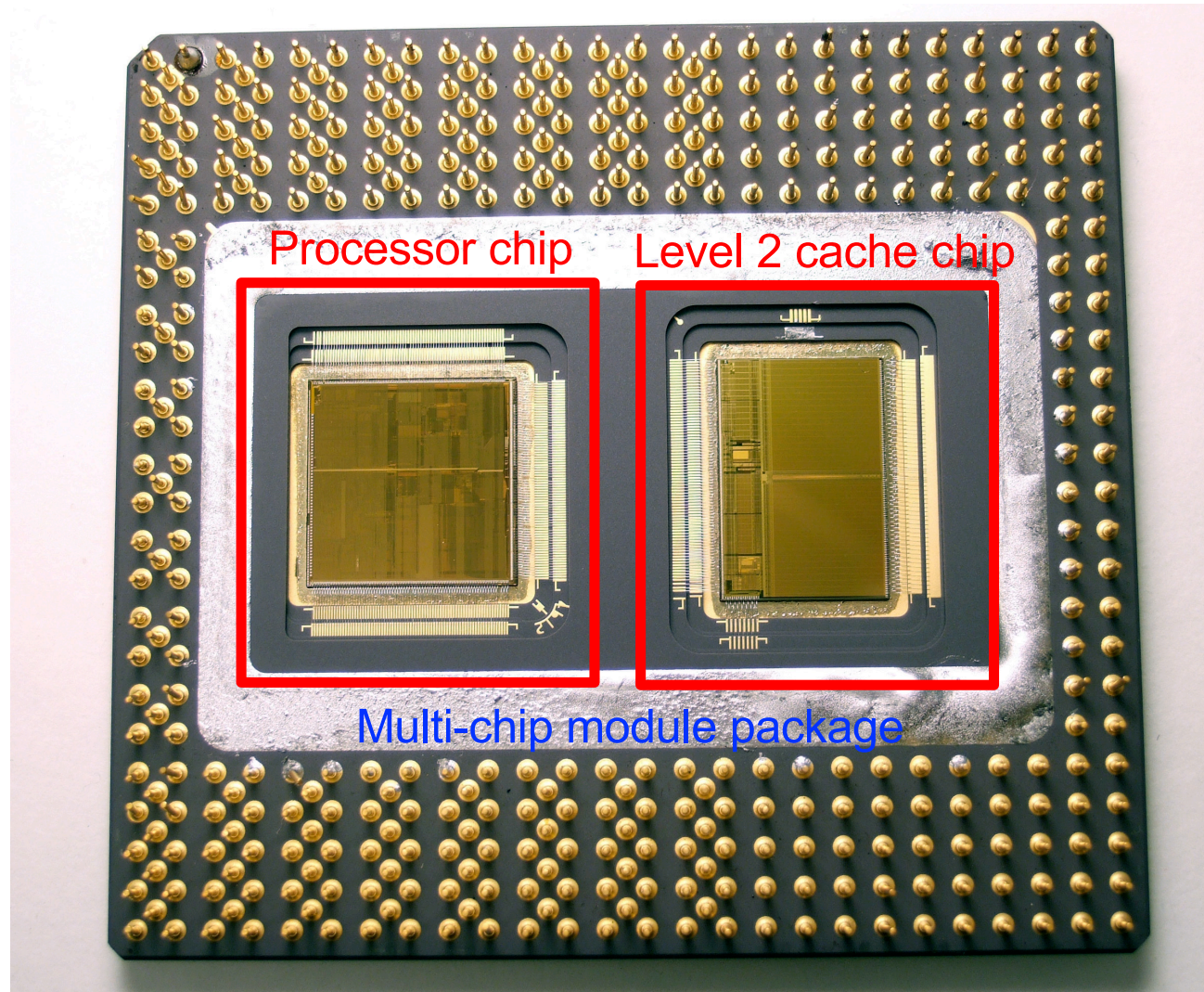
This branch tests i
Last 3 branches test j
History: TTTN
Predict taken for i
Next history: TTNT
(shift in last outcome)

- McFarling, “Combining Branch Predictors,” DEC WRL TR 1993.

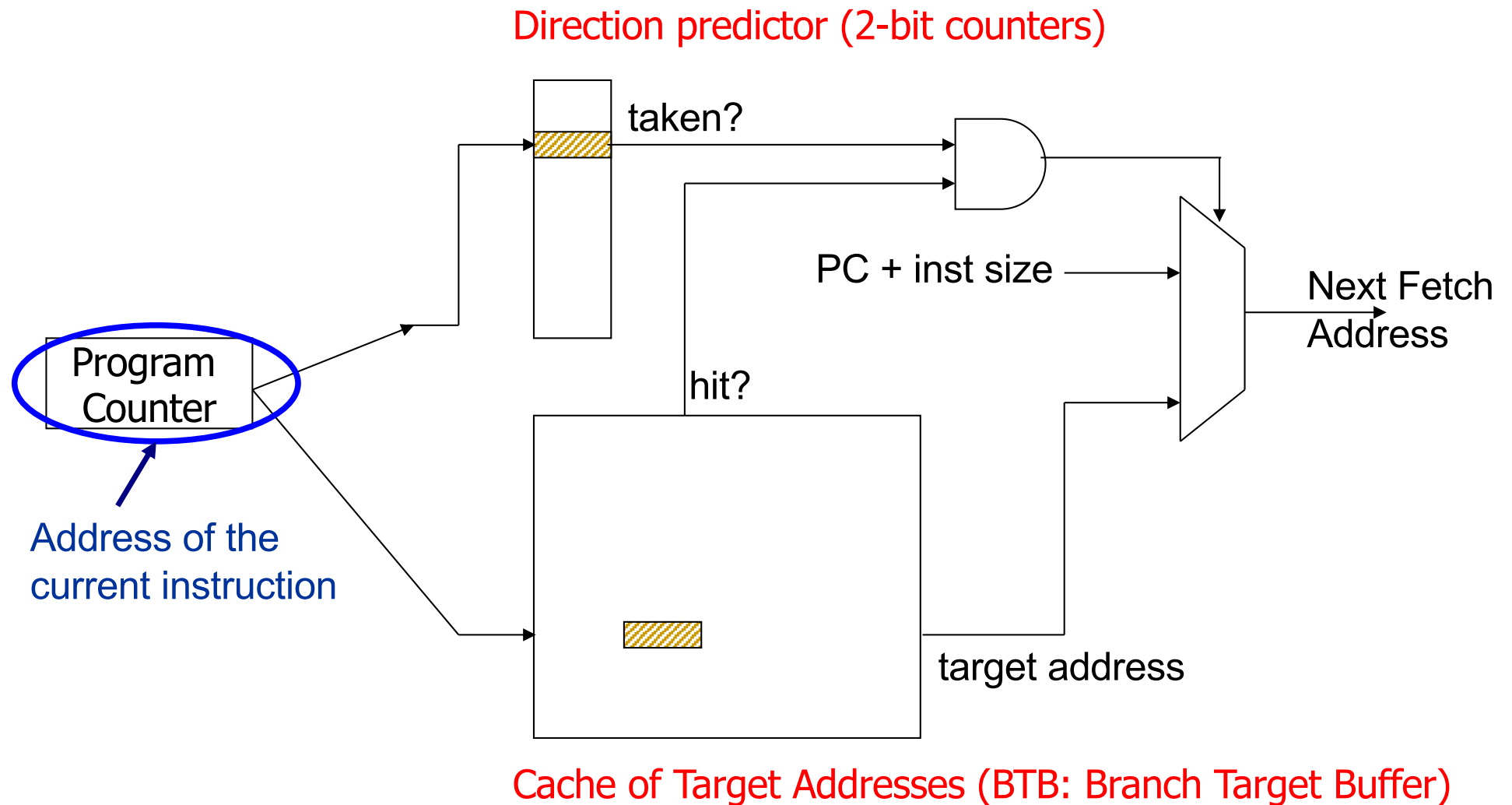
Intel Pentium Pro Branch Predictor

- Two-level global branch predictor
- 4-bit global history register
- Multiple pattern history tables (of 2-bit counters)
 - ❑ Which pattern history table to use is determined by lower order bits of the branch address
- First widely commercially successful out-of-order execution machine
 - ❑ Out-of-order + superscalar + 2-level branch prediction + precise exceptions using reorder buffer

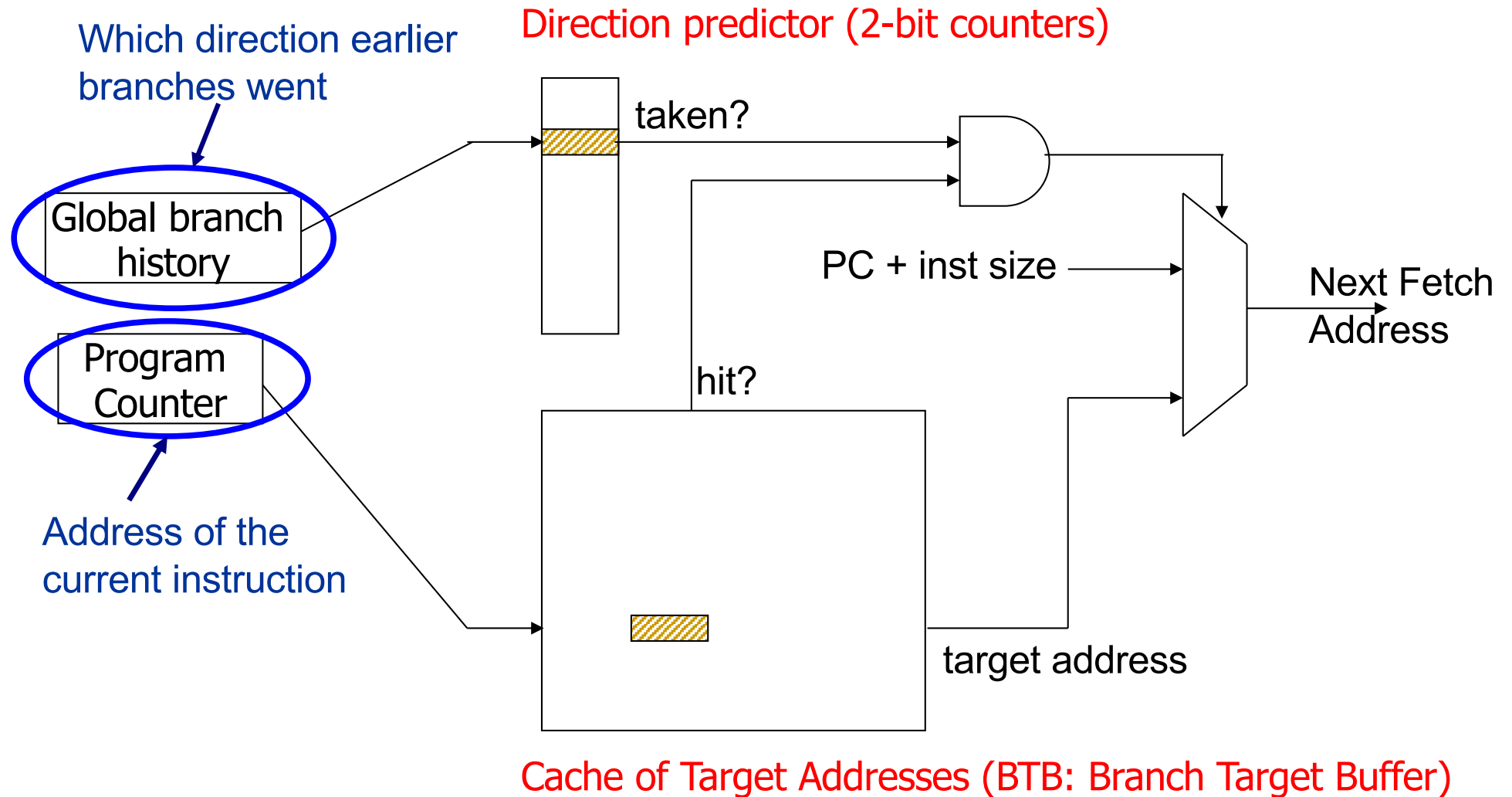
Intel Pentium Pro (1995)



Review: One-Level Branch Predictor

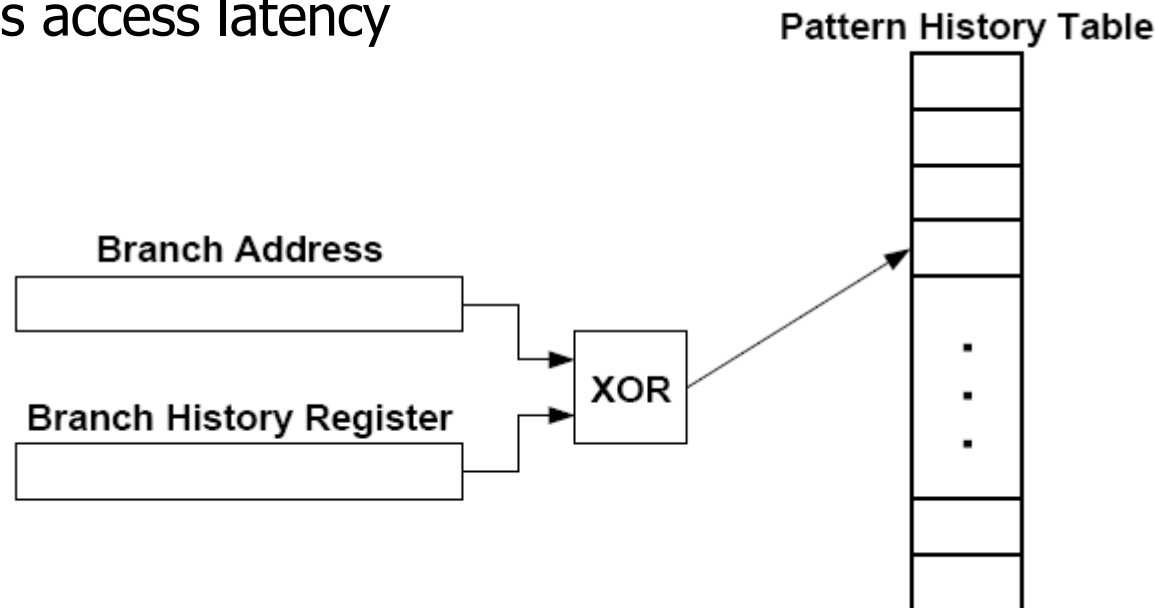


Two-Level Global History Branch Predictor



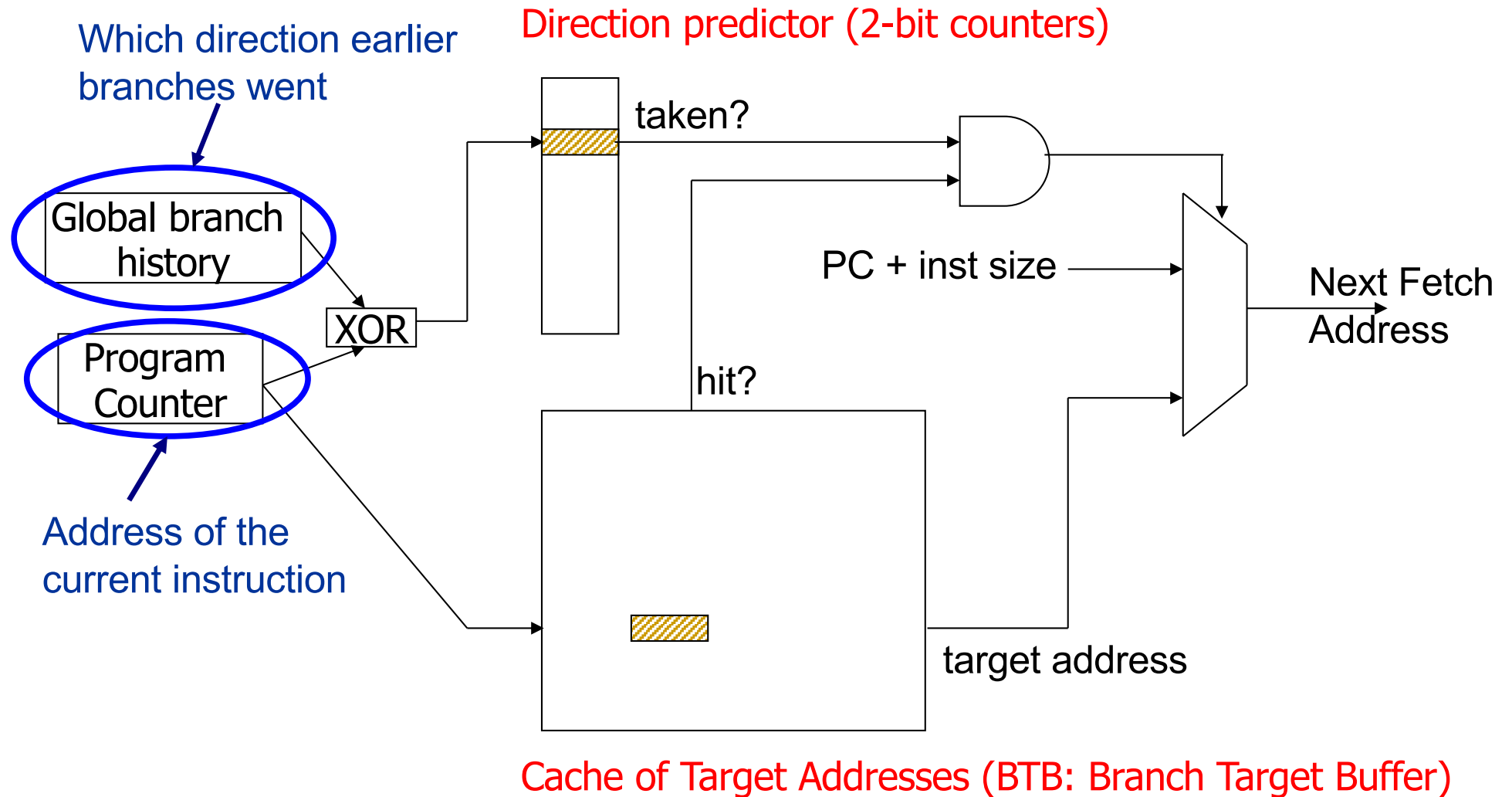
Improving Global Predictor Accuracy

- Idea: Add more context information to the global predictor to take into account which branch is being predicted
 - Gshare predictor: GHR hashed with the Branch PC
 - + More context information used for prediction
 - + Better utilization of the two-bit counter array
 - Increases access latency



- McFarling, “Combining Branch Predictors,” DEC WRL Tech Report, 1993.

Two-Level Gshare Branch Predictor



Can We Do Better: Two-Level Prediction

- Last-time and 2BC predictors exploit only “last-time” predictability for a given branch
- Realization 1: A branch’s outcome can be correlated with other branches’ outcomes
 - Global branch correlation
- Realization 2: A branch’s outcome can be correlated with past outcomes of the same branch (in addition to the outcome of the branch “last-time” it was executed)
 - Local branch correlation

Local Branch Correlation

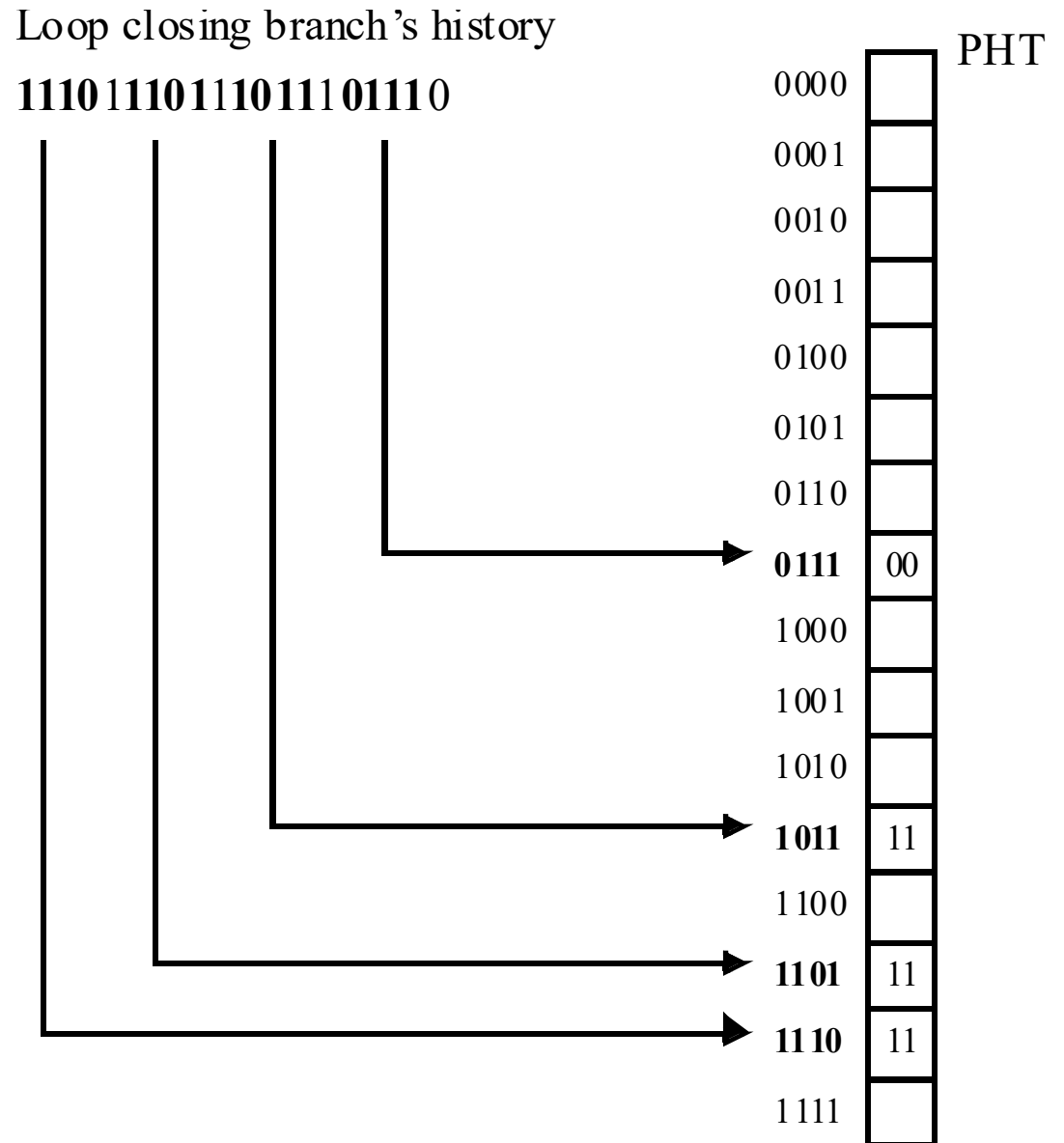
for (i=1; i<=4; i++) { }

If the loop test is done at the end of the body, the corresponding branch will execute the pattern $(1110)^n$, where 1 and 0 represent taken and not taken respectively, and n is the number of times the loop is executed. Clearly, if we knew the direction this branch had gone on the previous three executions, then we could always be able to predict the next branch direction.

- McFarling, “Combining Branch Predictors,” DEC WRL TR 1993.

More Motivation for Local History

- To predict a loop branch “perfectly”, we want to identify the last iteration of the loop
- By having a separate PHT entry for each local history, we can distinguish different iterations of a loop
- Works for “short” loops

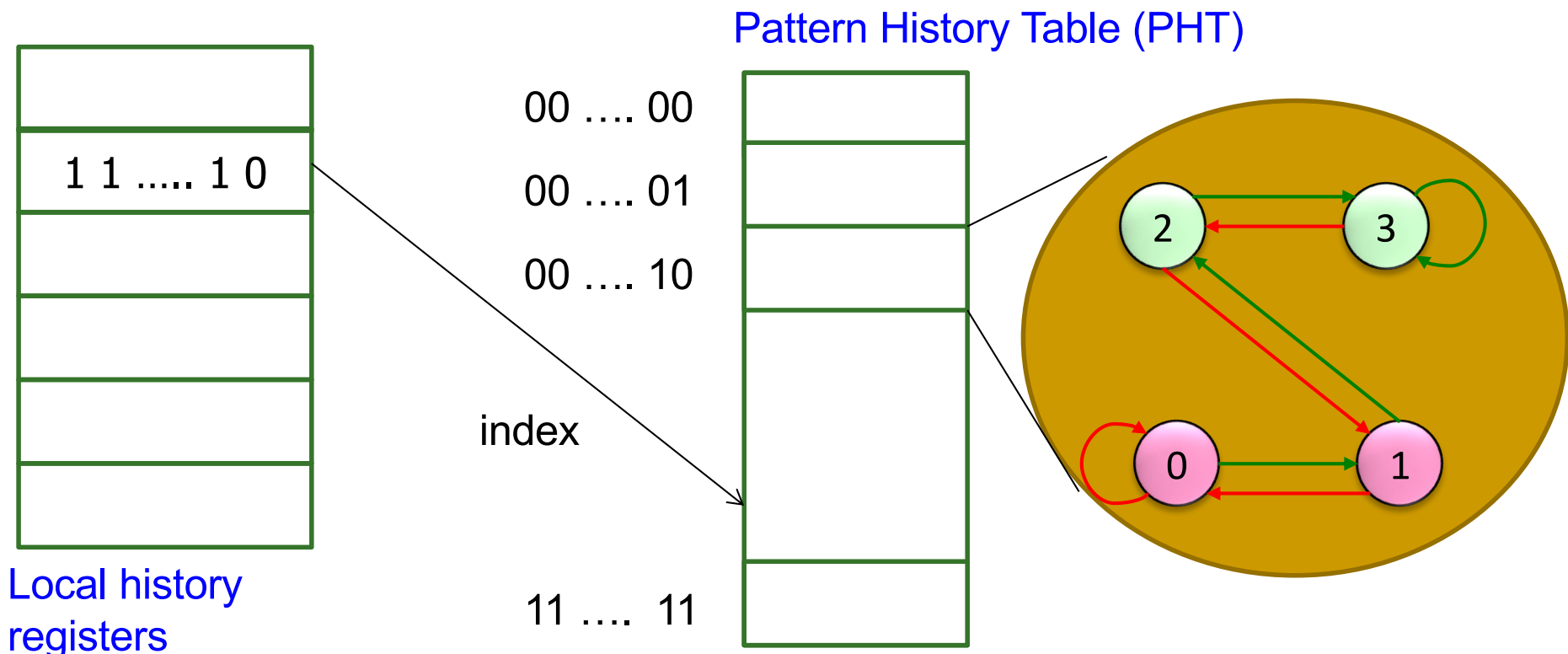


Capturing Local Branch Correlation

- Idea: Have a per-branch history register
 - Associate the predicted outcome of a branch with “T/NT history” of the same branch
- Make a prediction based on the outcome of the branch the last time the same local branch history was encountered
- Called the local history/branch predictor
- Uses two levels of history
 - per-branch history register + history at that history register value

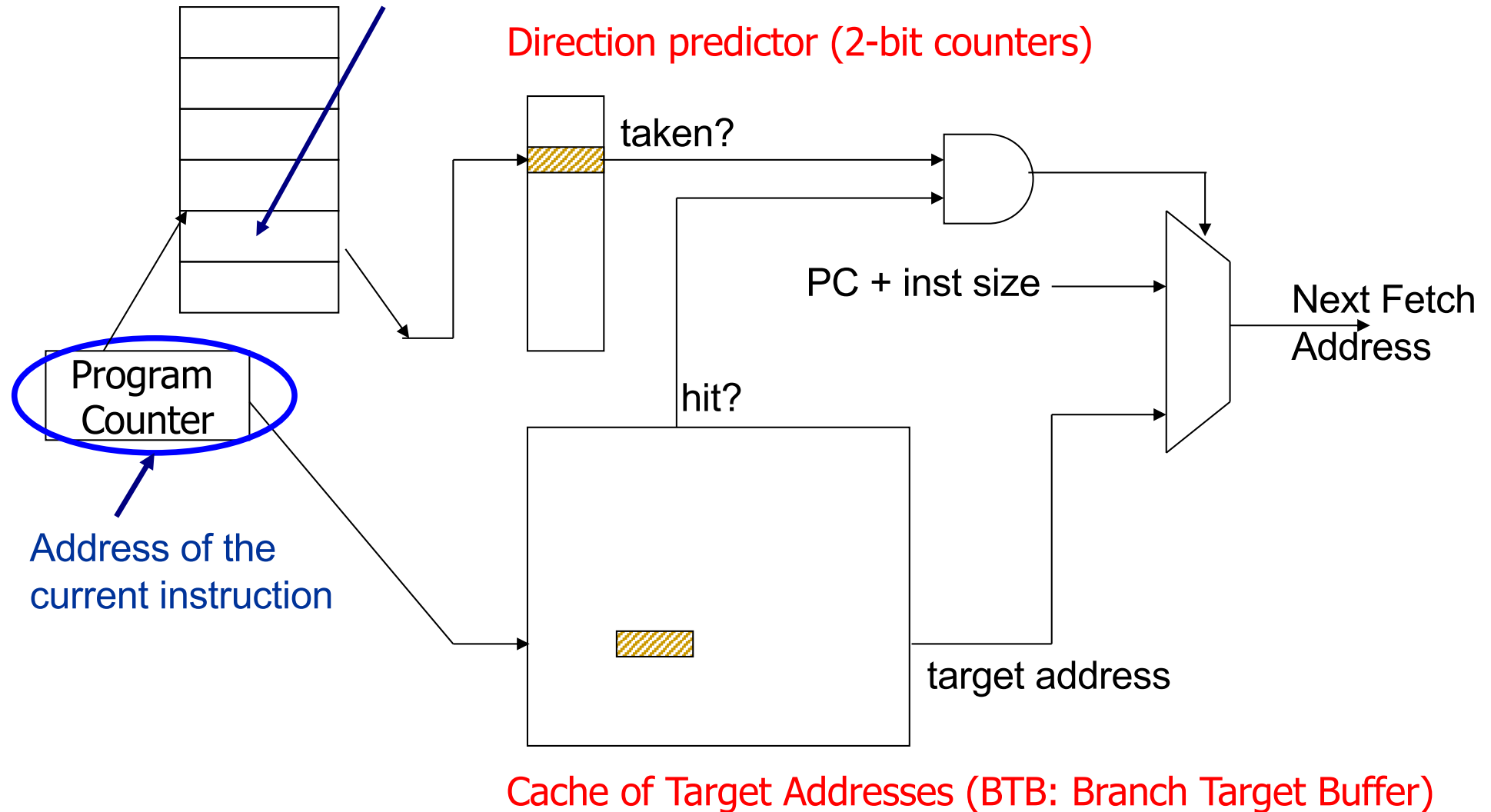
Two Level Local Branch Prediction

- First level: A set of local history registers (N bits each)
 - Select the history register based on the PC of the branch
- Second level: Table of saturating counters for each history entry
 - The direction the branch took the last time the same history was seen



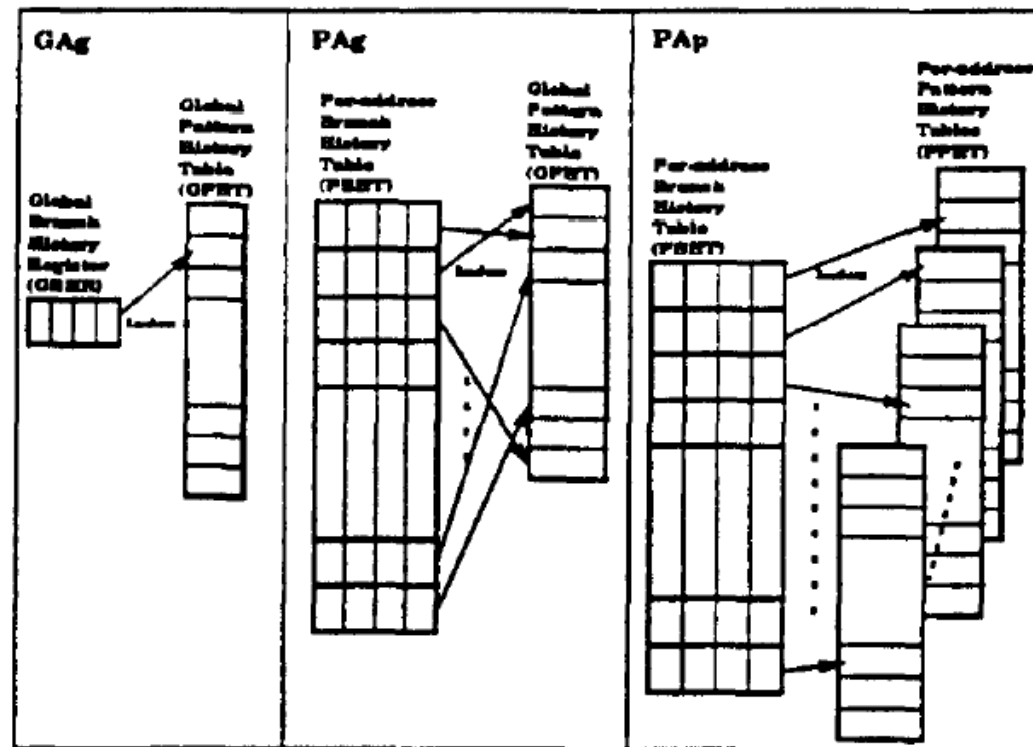
Two-Level Local History Branch Predictor

Which directions earlier instances of *this branch* went



Aside: Two-Level Predictor Taxonomy

- BHR can be global (G), per set of branches (S), or per branch (P)
- PHT counters can be adaptive (A) or static (S)
- PHT can be global (g), per set of branches (s), or per branch (p)



- Yeh and Patt, “Two-Level Adaptive Training Branch Prediction,” MICRO 1991.

Aside: A Retrospective

Retrospective on “Two-Level Adaptive Training Branch Prediction”

TSE-YU YEH

Apple

YALE N. PATT

University of Texas at Austin

..... Looking back at the world of computer architecture in 1991, when we wrote our MICRO paper,¹ we believe our motivation for addressing branch prediction is best illustrated by our earlier 1991 ISCA paper,² which showed that instructions per cycle can be greater than two. Conventional wisdom at the time had argued otherwise. We demonstrated in that paper that there was enough instruction-level parallelism in nonscientific workloads to support our contention. We further showed that the HPS microarchitecture (a superscalar out-of-order execution engine) could improve the execution rate greatly by exploiting instruction-level parallelism, provided the penalty caused by incorrect branch predictions was minimized.

In fact, the lack of an aggressive, highly accurate branch predictor was a major stumbling block in enabling our high-performance microarchitecture to achieve its potential. The HPS microarchitecture required the execution core to be kept fully supplied with micro-ops, organized as a dataflow graph that allowed lots

of concurrent execution. This meant a powerful and effective branch predictor that could fetch, decode, and execute instructions speculatively, and only very infrequently have to throw away the speculative work because of a branch misprediction. The computer architecture community had pretty well given up on branch prediction providing any major benefits. Most accepted as fact that Jim Smith's 2-bit saturating counter was as good as it was going to get.³ His branch predictor was published in ISCA in 1981 and was still the best-performing branch predictor 10 years later. It was the branch predictor in Pentium, Intel's “brainiac” microprocessor, the top of the x86 line at the time, released in 1991.

Tse-Yu spent the summer of 1990 at Motorola, working for Michael Shebanow, one of the original inventors of the HPS microarchitecture. Michael was interested in branch prediction, and in fact almost did his PhD dissertation at Berkeley on branch prediction. Tse-Yu and Michael had many conversations about

microarchitecture tradeoffs that summer, and often the discussions would gravitate to branch prediction. When Tse-Yu returned to Ann Arbor for the fall semester, he was excited about the possibility of a branch predictor that could support the HPS microarchitecture.

After many after-midnight meetings over several months in Yale's office, combined with a whole lot of simulation, the two-level adaptive branch predictor was born. We knew the branch predictor would have to be dynamic; that was a no-brainer. The branch predictor would have to use historical information based on the input data that the program was processing, and it would have to accommodate phase changes. But that was not enough. What historical information would yield good branch prediction? Somehow we stumbled on the answer: it was not the direct record of a branch's behavior that was needed, but rather whether the branch was taken at previous instances of time having the same history. That is, if the

branch's history was 011000101 (0 means not taken, 1 means taken), there is little one can infer. But, if the last time the history was 011000101 the branch was taken, and the time before that when the history was 011000101 the branch was also taken, it was a good bet that the branch would be taken at this time.

With that insight, we set out to design the two-level predictor. Several issues had to be resolved.

First, how should the direct record of branches be stored? Three choices presented themselves: a global history register to record the behavior of each dynamic branch; a separate history register for each static branch; and a half-way measure, in which we partitioned branches into equivalence classes and allocated one history register for each class. We dubbed the three schemes G, P, and S, respectively. The obvious tradeoffs were interference versus correlation and the cost of the history registers.

Second, how should the information be stored that records what happens as a result of a previous instance of the current history? We decided to store results as entries in a table, indexed by the bits of history in the history registers. The same three choices applied here—a global pattern table, a per-static-branch pattern table, and the in-between scheme. We dubbed the three choices g, p, and s. Thus, we had nine choices to investigate: GAg, GAs, GAP, SAg, SAs, SAP, PAg, PAs, and PAP.

Third, what decision structure should we use given the entries in the pattern tables? We examined several state machines, finally deciding on the 2-bit counter scheme of Jim Smith's 1981 ISCA paper.

Finally, how many bits of history should be kept in the history registers? Using n bits of history meant a pattern table of size 2^n . Adding one bit of history meant doubling the size of each pattern table.

We introduced the Two-Level Adaptive Branch Predictor at MICRO in 1991,¹ demonstrating that a lot more performance could be obtained with a branch predictor that was significantly more

accurate than the previous best case. Our predictor showed that maintaining two levels of branch history would provide much greater accuracy than was possible with the simpler branch prediction schemes. Other researchers soon agreed that the benefit from branch prediction could be substantial, and that maintaining two levels of branch history was the correct mechanism. The result was a wave of papers on dynamic branch prediction, successively improving on our basic two-level predictor design. We followed our first paper with a more comprehensive paper describing alternative implementations of the two-level scheme in ISCA 1992,⁴ and Kimming So and colleagues at IBM published their correlation predictor in ASPLOS later that same year.⁵ Soon after, Scott McFarling introduced gshare,⁶ a variation on the GAs two-level predictor and hybrid branch prediction.

At our industrial affiliates meeting the next year, Tse-Yu presented our branch predictor. Intel engineers in the audience were excited about its possibilities. They had decided to increase the pipeline of their next processor Pentium Pro from 5 stages to 13 to cut down the cycle time to be more competitive with all the emerging RISC designs. More pipeline stages meant a greater branch-misprediction penalty, so Intel badly needed a better branch predictor. Intel's interest sparked substantial interaction between us, resulting in Tse-Yu spending the summer of 1992 at Intel in Hillsboro, Oregon, where the Pentium Pro was being designed. Intel decided to opt for the PAs version of the two-level branch predictor and did all the hard engineering to make it implementable in the Pentium Pro. With the announcement of that product, the computer architecture landscape had changed forever in favor of aggressive branch prediction.

joined Sibyte, which was acquired by Broadcom; later he moved to PA Semi and eventually to Apple. He has been responsible for many microprocessor designs over the years. He is currently an engineering director at Apple, where he is responsible for CPU verification. Yale Patt retired from Michigan in 1999 to become the Ernest Cockrell, Jr. Centennial Chair in Engineering at the University of Texas at Austin. He continues to thrive on directing PhD students in microarchitecture research and teaching both freshmen and graduate students.

References

1. T.-Y. Yeh and Y.N. Patt, “Two-Level Adaptive Training Branch Prediction,” *Proc. 24th Ann. Int'l Symp. Microarchitecture*, 1991, pp. 51–61.
2. M. Butler et al., “Single Instruction Stream Parallelism Is Greater Than Two,” *Proc. 18th Ann. Int'l Symp. Computer Architecture*, 1991, pp. 276–286.
3. J.E. Smith, “A Study of Branch Prediction Strategies,” *Proc. 8th Ann. Symp. Computer Architecture*, 1981, pp. 135–148.
4. T.-Y. Yeh and Y.N. Patt, “Alternative Implementations of Two-Level Adaptive Branch Prediction,” *Proc. 19th Ann. Int'l Symp. Computer Architecture*, 1992, pp. 124–134.
5. S.-T. Pan, K. So, and J.T. Rahmeh, “Improving the Accuracy of Dynamic Branch Prediction Using Branch Correlation,” *Proc. 5th Int'l Conf. Architectural Support for Programming Languages and Operating Systems*, 1992, pp. 76–84.
6. S. McFarling, “Combining Branch Predictors,” tech. note TN-36, Western Research Library, 1993.

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JULY/AUGUST 2016

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Can We Do Better?

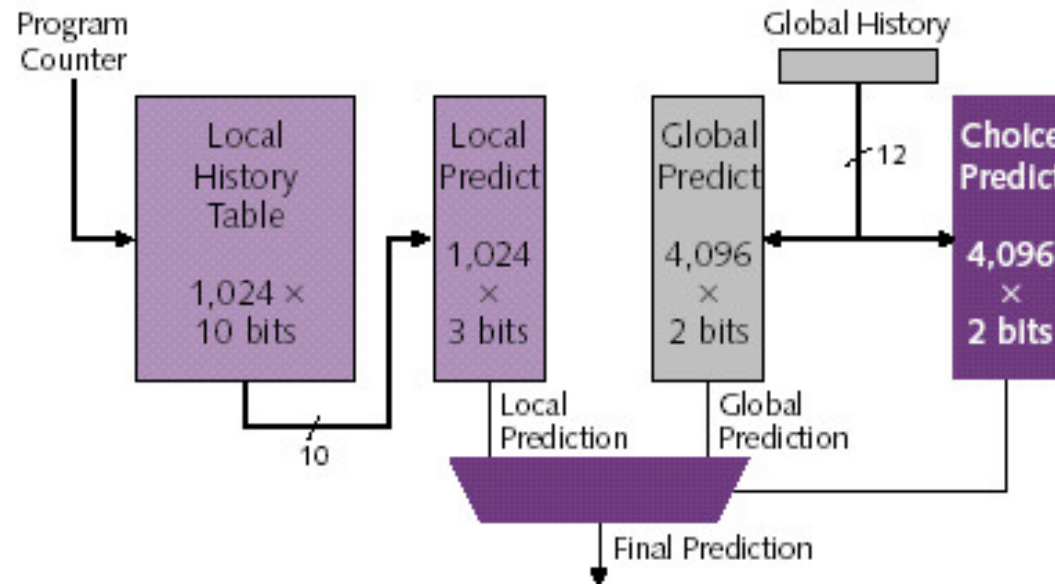
Can We Do Even Better?

- Predictability of branches varies
- Some branches are more predictable using local history
- Some branches are more predictable using global
- For others, a simple two-bit counter is enough
- Yet for others, a single bit is enough
- Observation: There is heterogeneity in predictability behavior of branches
 - No one-size fits all branch prediction algorithm for all branches
- Idea: Exploit that heterogeneity by designing heterogeneous (hybrid) branch predictors

Hybrid Branch Predictors

- Idea: Use more than one type of predictor (i.e., multiple algorithms) and select the “best” prediction
 - E.g., hybrid of 2-bit counters and global predictor
- Advantages:
 - + Better accuracy: different predictors are better for different branches
 - + Reduced **warmup** time (faster-warmup predictor used until the slower-warmup predictor warms up)
- Disadvantages:
 - Need “meta-predictor” or “selector” to decide which predictor to use
 - Longer access latency
 - More hardware & complexity

Alpha 21264 Tournament Predictor



- Minimum branch penalty: 7 cycles
- Typical branch penalty: 11+ cycles
- 48K bits of target addresses stored in I-cache
- Predictor tables are reset on a context switch
- Kessler, "[The Alpha 21264 Microprocessor](#)," IEEE Micro 1999.

Biased Branches and Branch Filtering

- Observation: Many branches are biased in one direction (e.g., 99% taken)
- Problem: These branches *pollute* the branch prediction structures → make the prediction of other branches difficult by causing “interference” in branch prediction tables and history registers
- Solution: Detect such biased branches, and predict them with a simpler predictor (e.g., last time, static, ...)
- Chang et al., “Branch classification: a new mechanism for improving branch predictor performance,” MICRO 1994.

Are We Done w/ Branch Prediction?

- Hybrid branch predictors work well
 - E.g., 90-97% prediction accuracy on average
- Some “difficult” workloads still suffer a lot, though!
 - E.g., gcc
 - Max IPC with tournament prediction: 9
 - Max IPC with perfect prediction: 35

Some Other Branch Predictor Types

- **Loop branch detector and predictor**
 - ❑ Loop iteration count detector/predictor
 - ❑ Works well for loops with small number of iterations, where iteration count is predictable
 - ❑ Used in Intel Pentium M
- **Perceptron branch predictor**
 - ❑ Learns the *direction correlations* between individual branches
 - ❑ Assigns weights to correlations using simple machine learning
 - ❑ Jimenez and Lin, “Dynamic Branch Prediction with Perceptrons,” HPCA 2001.
- **Hybrid history-length based predictor**
 - ❑ Uses different tables with different history lengths
 - ❑ Seznec, “Analysis of the O-Geometric History Length Branch Predictor,” ISCA 2005.

Intel Pentium M Predictors: Loop and Jump

The advanced branch prediction in the Pentium M processor is based on the Intel Pentium[®] 4 processor's [6] branch predictor. On top of that, two additional predictors to capture special program flows, were added: a Loop Detector and an Indirect Branch Predictor.

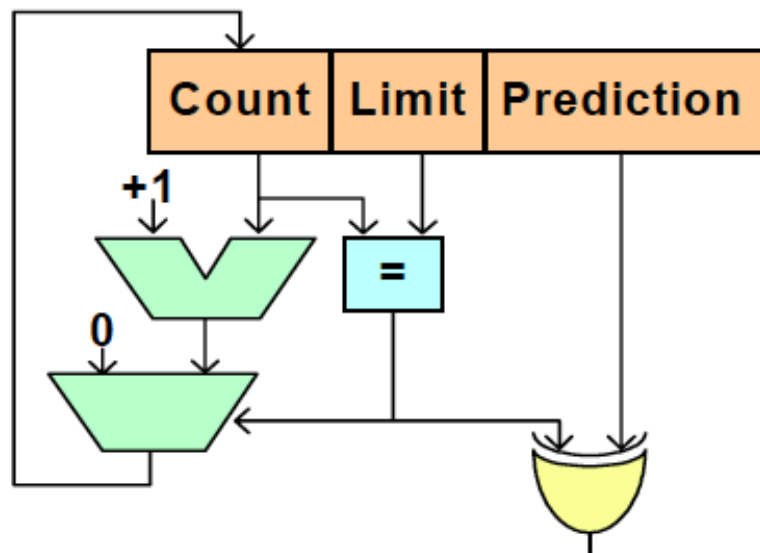


Figure 2: The Loop Detector logic

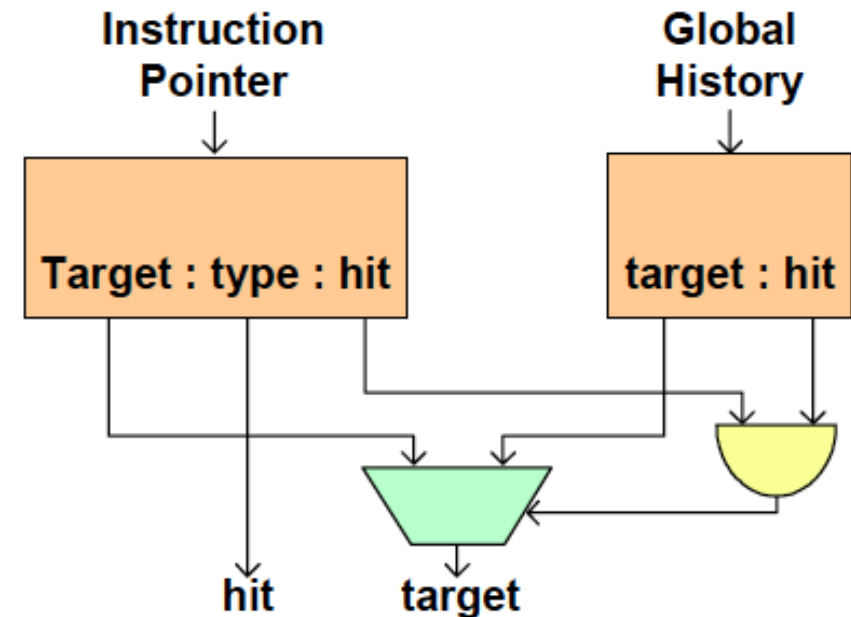


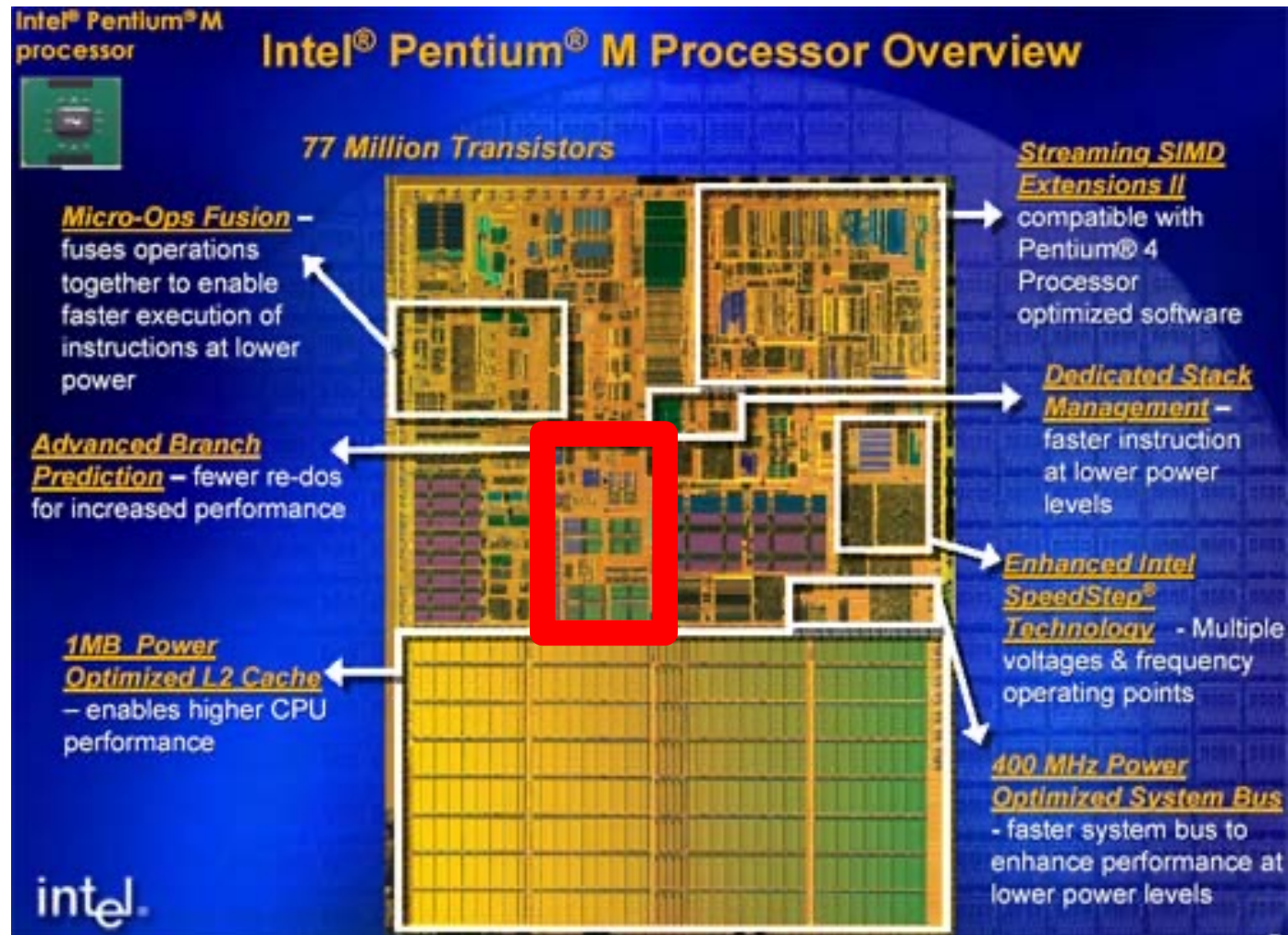
Figure 3: The Indirect Branch Predictor logic

Gochman et al.,

“[The Intel Pentium M Processor: Microarchitecture and Performance](#),”

Intel Technology Journal, May 2003.

Intel Pentium M (2003)



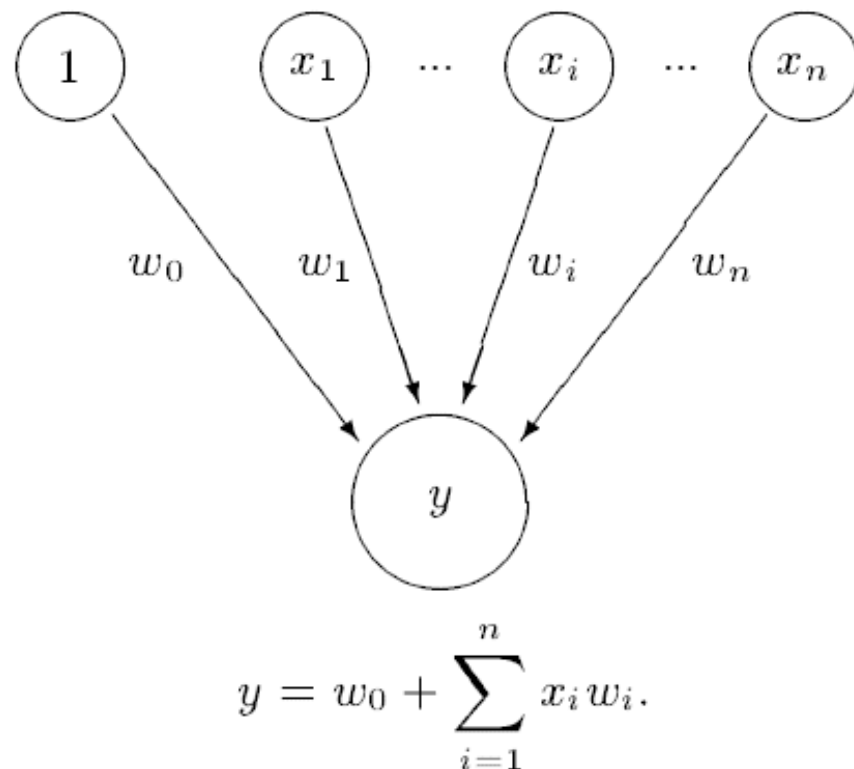
More Advanced Branch Prediction

Perceptrons for Learning Linear Functions

- A perceptron is a simplified model of a biological neuron
- It is also a simple **binary classifier**
- A perceptron maps an input vector X to a 0 or 1
 - ❑ Input = Vector X
 - ❑ Perceptron learns the linear function (if one exists) of how each element of the vector affects the output (stored in an internal Weight vector)
 - ❑ Output = $\text{Weight} \cdot X + \text{Bias} > 0$
- In the branch prediction context
 - ❑ Vector X : Branch history register bits
 - ❑ Output: Prediction for the current branch

Perceptron Branch Predictor (I)

- Idea: Use a perceptron to learn the correlations between branch history register bits and branch outcome
- A perceptron learns a target Boolean function of N inputs



Each branch associated with a perceptron

A perceptron contains a set of weights w_i

→ Each weight corresponds to a bit in the GHR

→ Represents how much the bit is correlated with the direction of the branch

→ Positive correlation: large positive + weight

→ Negative correlation: large negative - weight

Prediction:

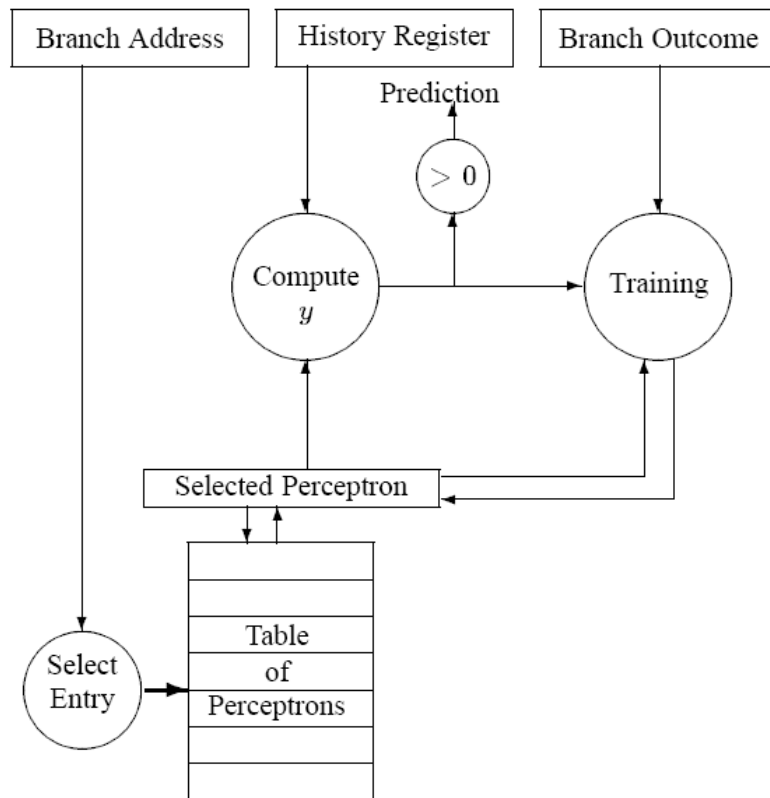
→ Express GHR bits as 1 (T) and -1 (NT)

→ Take dot product of GHR and weights

→ If output > 0 , predict taken

- Jimenez and Lin, “**Dynamic Branch Prediction with Perceptrons,**” HPCA 2001.
- Rosenblatt, “**Principles of Neurodynamics: Perceptrons and the Theory of Brain Mechanisms,**” 1962

Perceptron Branch Predictor (II)



Prediction function:

Dot product of GHR and perceptron weights

$$y = w_0 + \sum_{i=1}^n x_i w_i$$

Output compared to 0

Bias weight (bias of branch, independent of the history)

Training function:

```
if sign( $y_{out}$ )  $\neq$   $t$  or  $|y_{out}| \leq \theta$  then
  for  $i := 0$  to  $n$  do
     $w_i := w_i + tx_i$ 
  end for
end if
```

Perceptron Branch Predictor (III)

■ Advantages

- + More sophisticated learning mechanism → better accuracy
- + Enables long branch history lengths → better accuracy

■ Disadvantages

- Complexity (adder tree to compute perceptron output)
- Can learn only linearly-separable functions
e.g., cannot learn XOR type of correlation between 2 history bits and branch outcome

A successful example of use of machine learning in processor design

See, e.g., Grayson+, “[Evolution of the Samsung Exynos CPU Microarchitecture](#),” ISCA 2020.

Recommended Reading

2020 ACM/IEEE 47th Annual International Symposium on Computer Architecture (ISCA)

Evolution of the Samsung Exynos CPU Microarchitecture

Industrial Product

Brian Grayson*, Jeff Rupley[†], Gerald Zuraski Jr.[‡], Eric Quinnell[§], Daniel A. Jiménez[¶], Tarun Nakra^{‡‡},
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Abstract—The Samsung Exynos family of cores are high-performance “big” processors developed at the Samsung Austin Research & Design Center (SARC) starting in late 2011. This paper discusses selected aspects of the microarchitecture of these cores - specifically perceptron-based branch prediction, Spectre v2 security enhancements, micro-operation cache algorithms, prefetcher advancements, and memory latency optimizations. Each micro-architecture item evolved over time, both as part of continuous yearly improvement, and in reaction to changing mobile workloads.

Index Terms—microprocessor, superscalar, branch prediction, prefetching

- Deep technical details within the microarchitecture.

The remainder of this paper discusses several aspects of front-end microarchitecture (including branch prediction microarchitecture, security mitigations, and instruction supply) as well as details on the memory subsystem, in particular with regards to prefetching and DRAM latency optimization. The overall generational impact of these and other changes is presented in a cross-workload view of IPC.

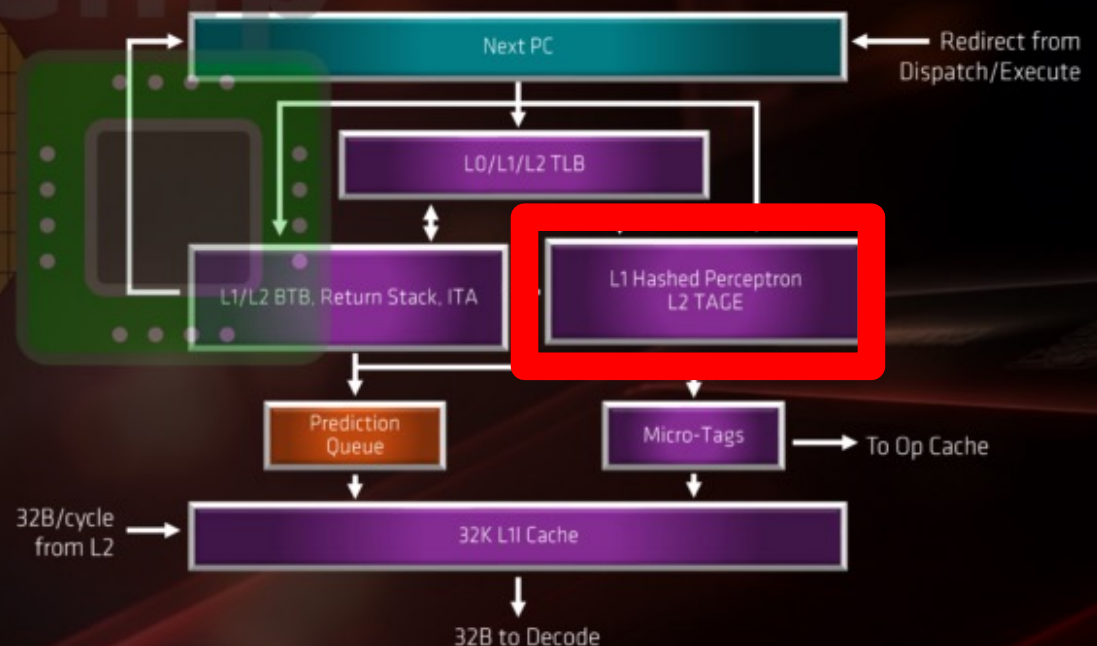
II. METHODOLOGY

AMD Piledriver/Zen/Zen2 (2012-Present)

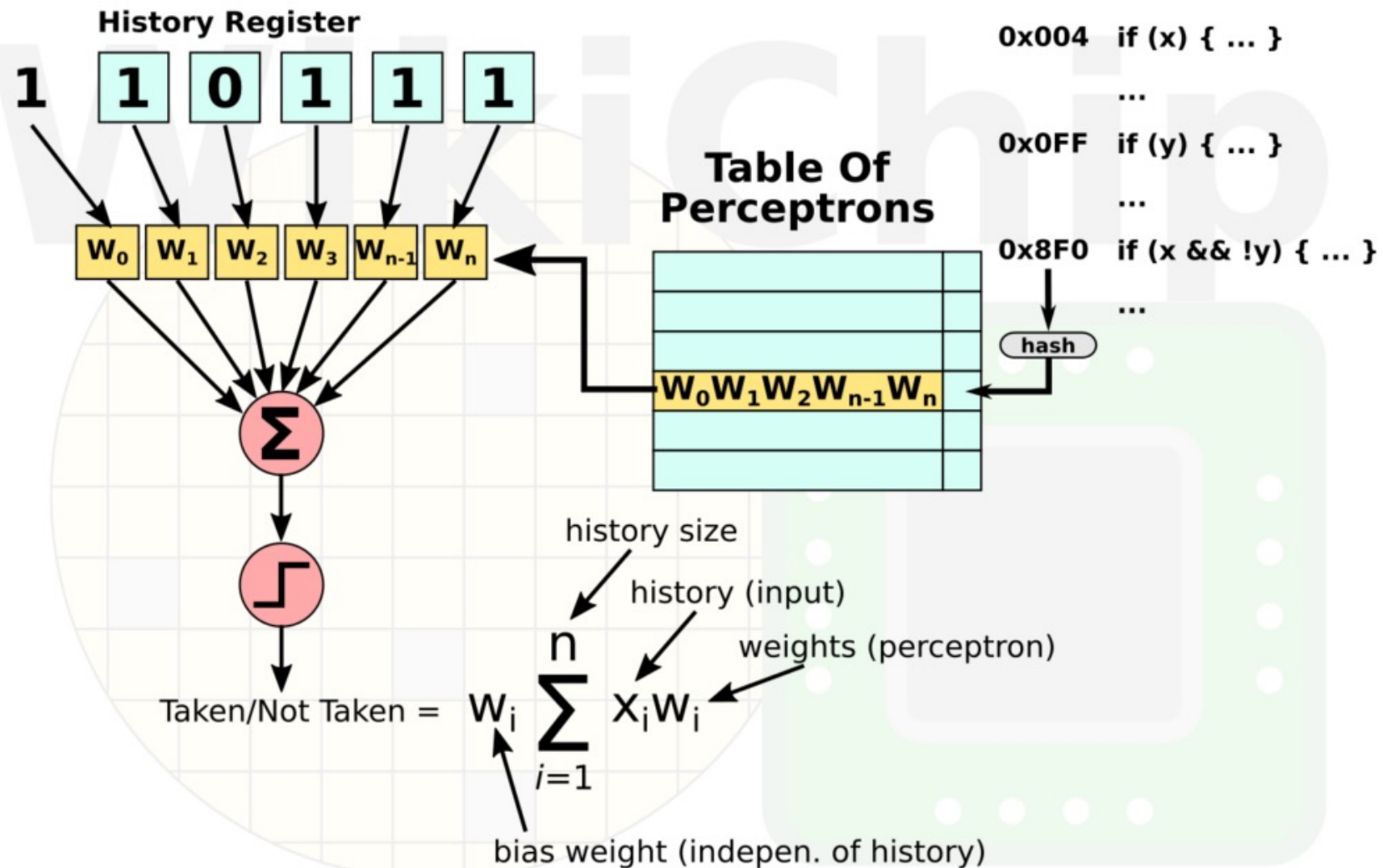
- These processors employ a perceptron branch predictor

FETCH

- Improved branch prediction
 - New TAGE branch predictor
 - Larger BTBs
 - L0 BTB, 16 entries
 - L1 BTB, 512 entries (up from 256)
 - L2 BTB, 7K entries (up from 4K)
 - Larger 1K indirect target array (ITA)
 - ~30% lower mispredict rate target
- Optimized 32KB, 8-way L1I cache
 - Higher associativity
 - Improved prefetching
 - Improved utilization



AMD Zen2 Perceptron Predictor (2019)



Another Idea: TAGE

Prediction Using Multiple History Lengths

- Observation: Different branches require different history lengths for better prediction accuracy
- Idea: Have multiple PHTs indexed with GHRs with different history lengths and intelligently allocate PHT entries to different branches

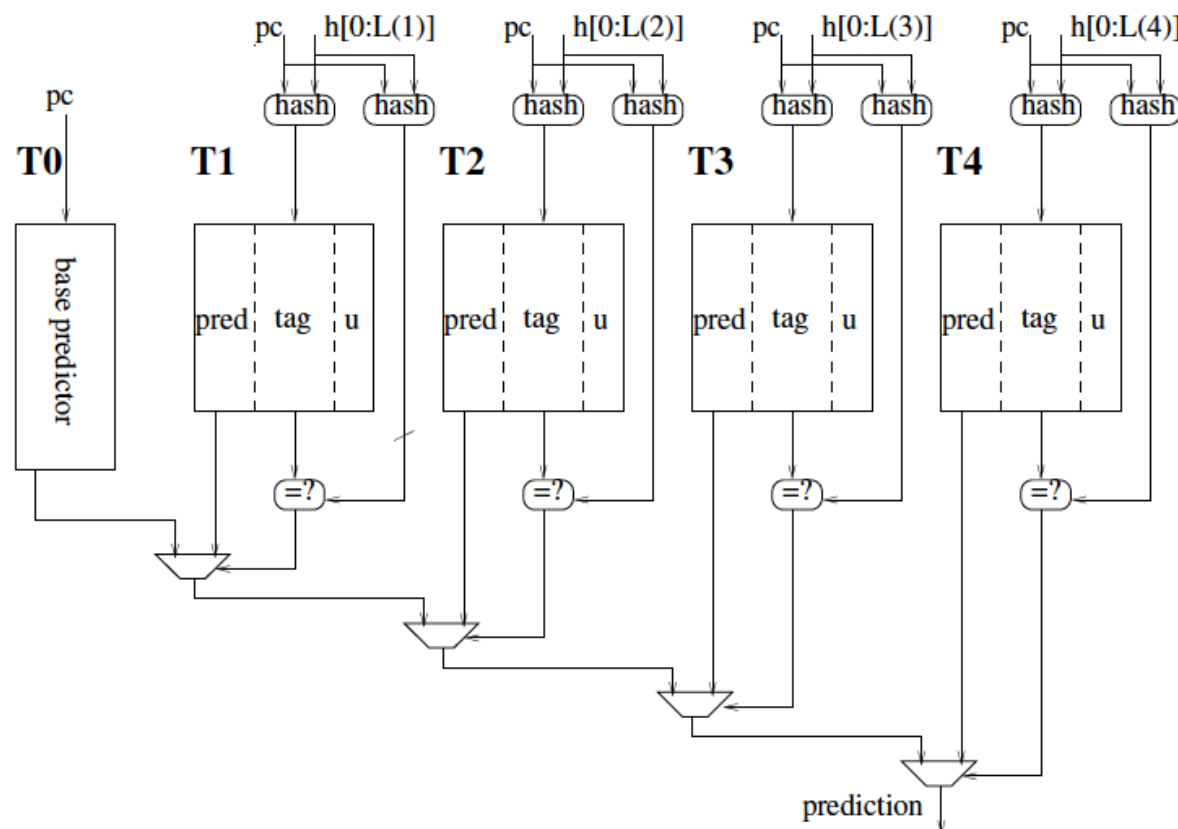
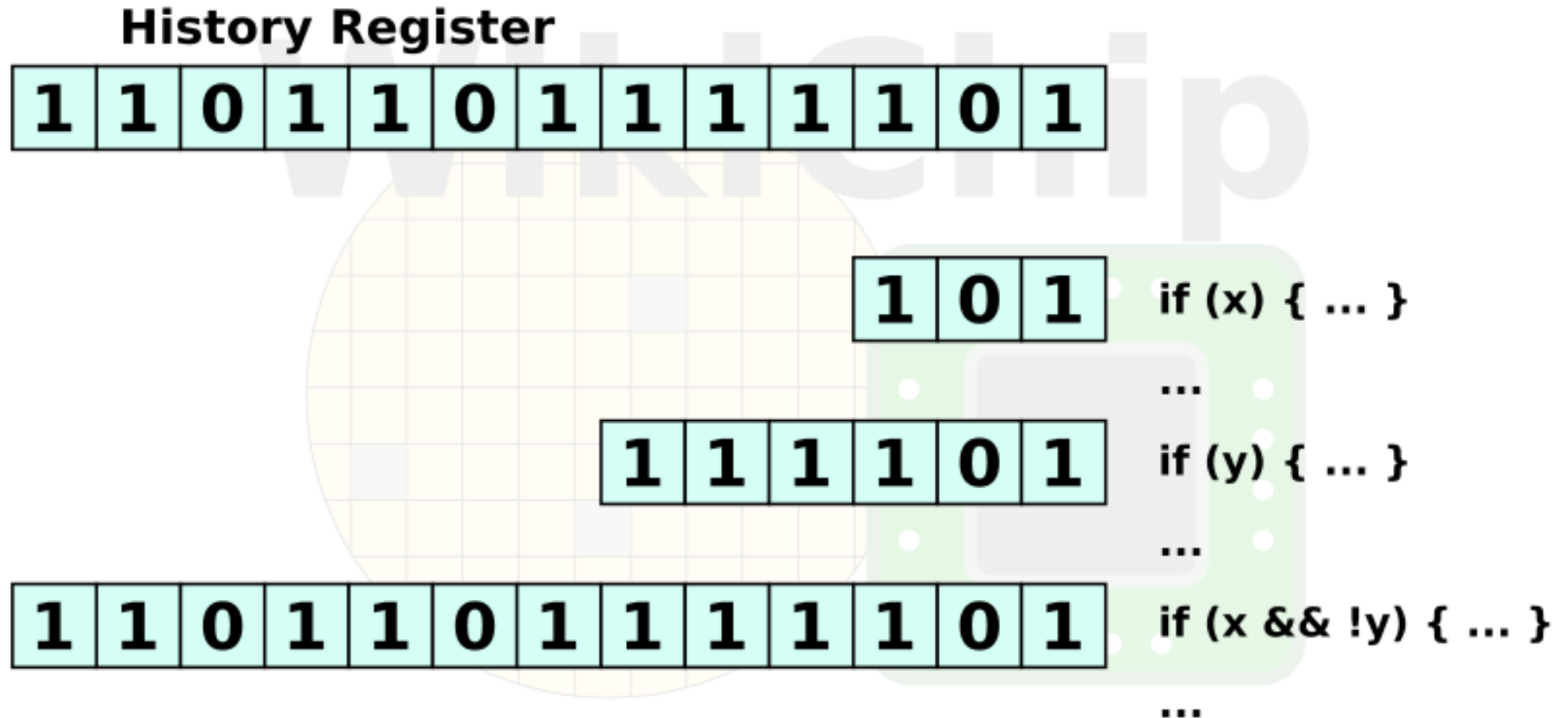


Figure 1: A 5-component TAGE predictor synopsis: a base predictor is backed with several tagged predictor components indexed with increasing history lengths

Seznec and Michaud, “[A case for \(partially\) tagged Geometric History Length Branch Prediction](#),” JILP 2006.

Different Branches: Different History Lengths



TAGE Branch Predictor

- Advantages

- + Chooses the “best” history length to predict each branch → better accuracy
- + Enables long branch history lengths → better accuracy

- Disadvantages

- Hardware (design) complexity is not low
- Need to choose good hash functions and table sizes to maximize accuracy and minimize latency

A successful recent idea that is used in many modern processor designs

AMD Zen2 Perceptron + TAGE (2019)

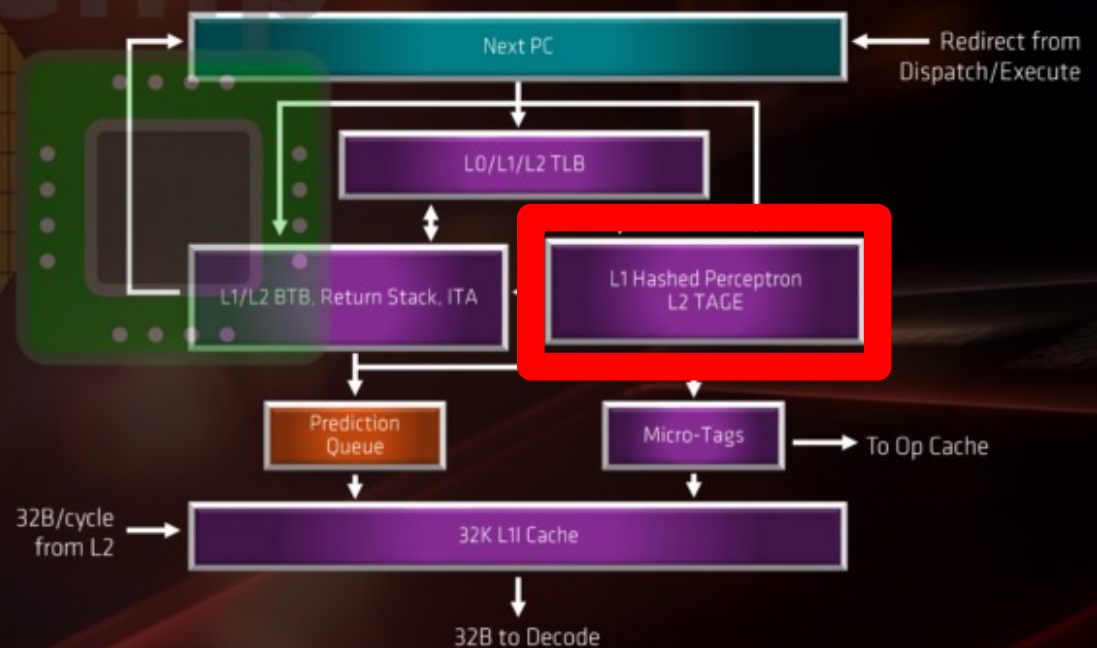
- A multi-level branch predictor
 - Perceptron (L1)
 - TAGE (L2)

The “Zen 2” core is a significant update to the historic “Zen” architecture. Now featuring:

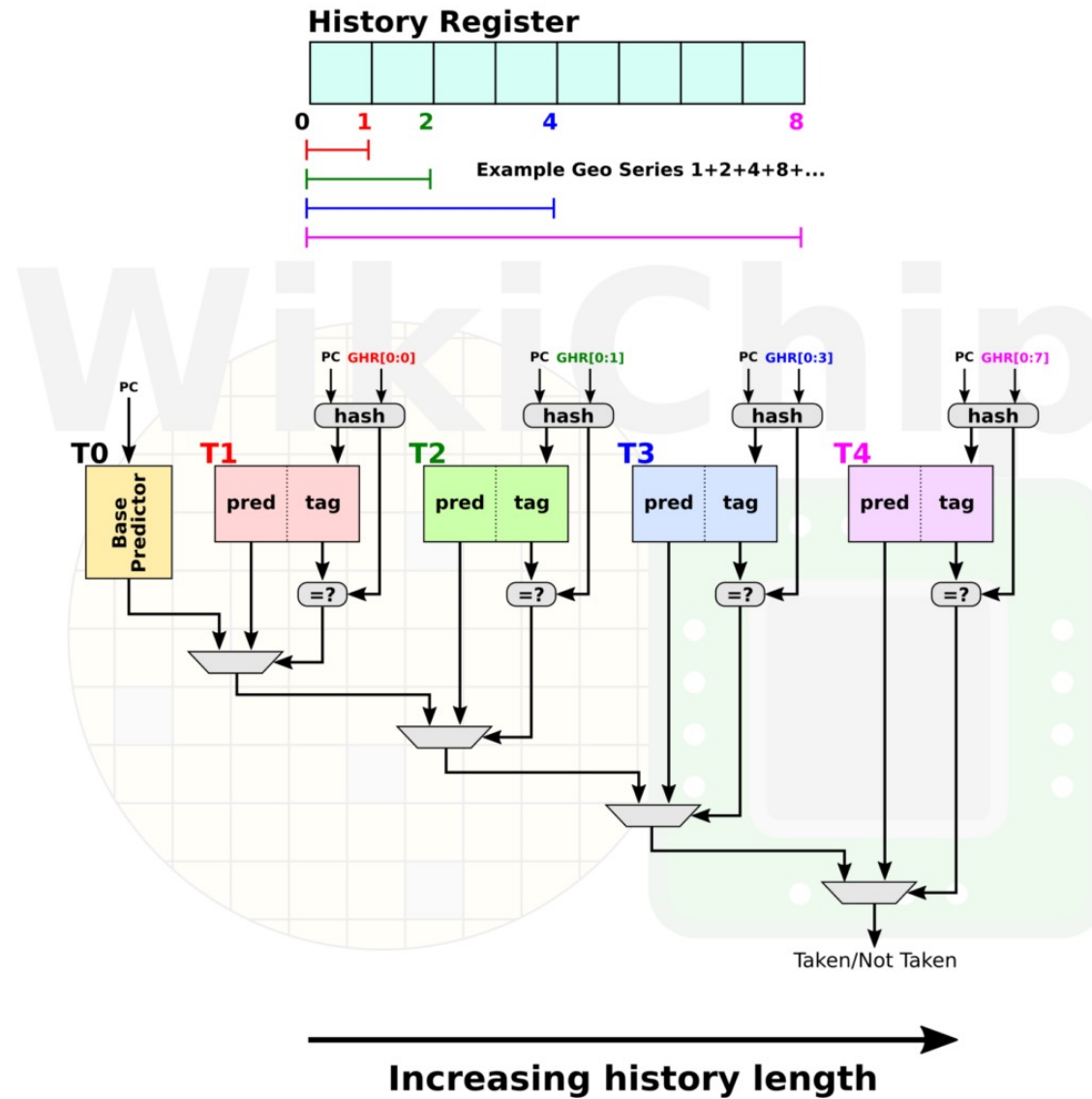
- Up to 15% more instructions-per-cycle²
- 2X L3 cache capacity (up to 32MB)
- 2X floating point throughput (256-bit)
- 2X OpCache capacity (4K)
- 2X Infinity Fabric bandwidth (512-bit)
- New TAGE branch predictor

FETCH

- Improved branch prediction
 - New TAGE branch predictor
 - Larger BTBs
 - L0 BTB, 16 entries
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 - Improved utilization



AMD Zen2 TAGE Predictor (2019)



Aside: Global Branch Correlation Analysis

branch Y: if (cond1)

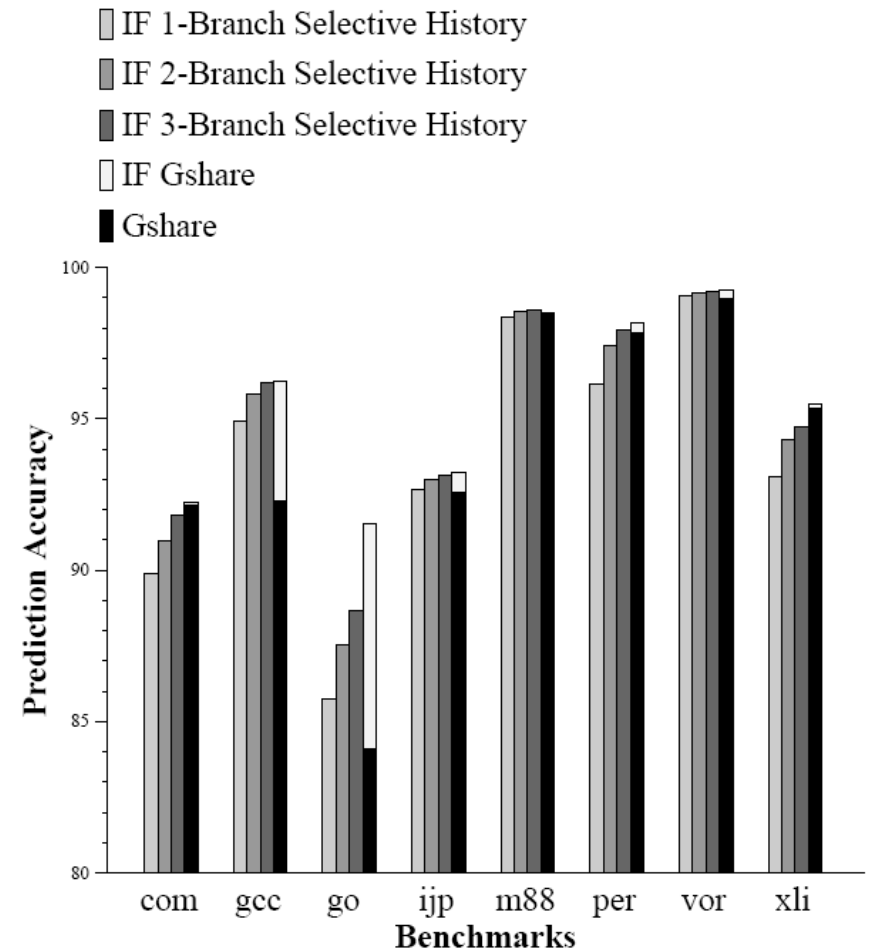
...

branch Z: if (cond2)

...

branch X: if (cond1 AND cond2)

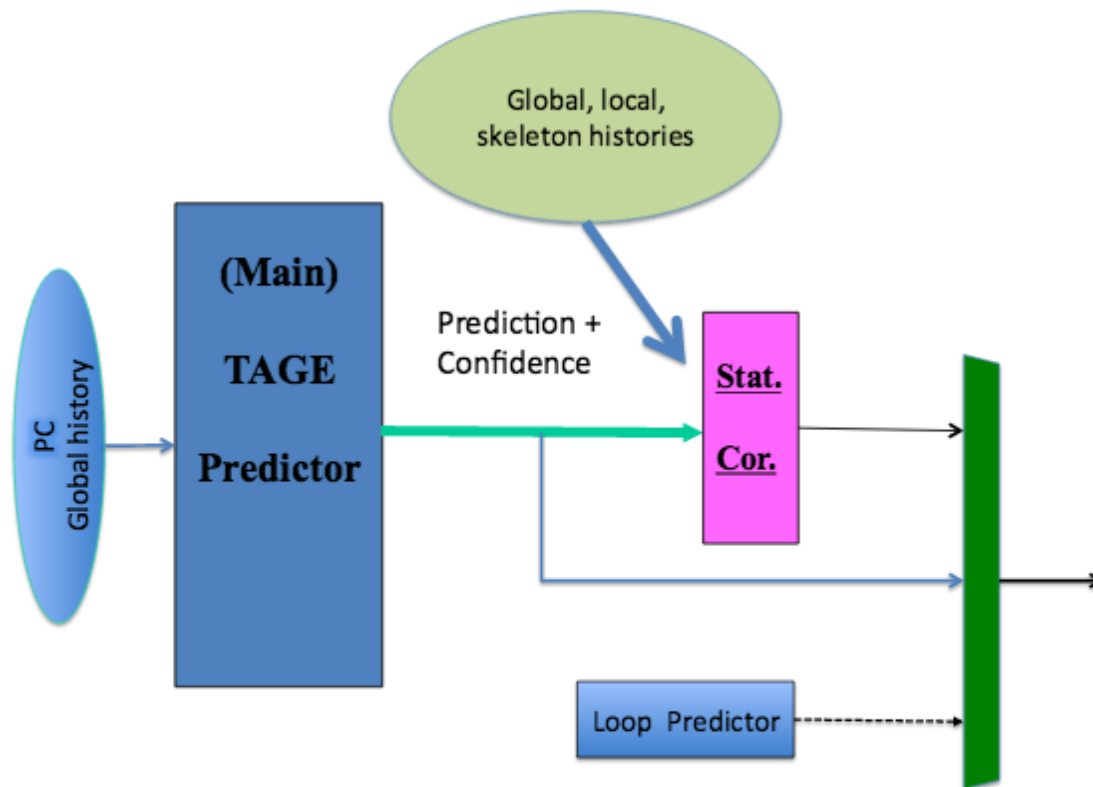
- If Y and Z both taken, then X also taken
- If Y or Z not taken, then X also not taken
- Only 3 past branches' directions *really* matter
- Evers et al., “An Analysis of Correlation and Predictability: What Makes Two-Level Branch Predictors Work,” ISCA 1998.



Can We Do Better?

State of the Art in Branch Prediction

- See the Branch Prediction Championship
 - <https://www.jilp.org/cbp2016/program.html>

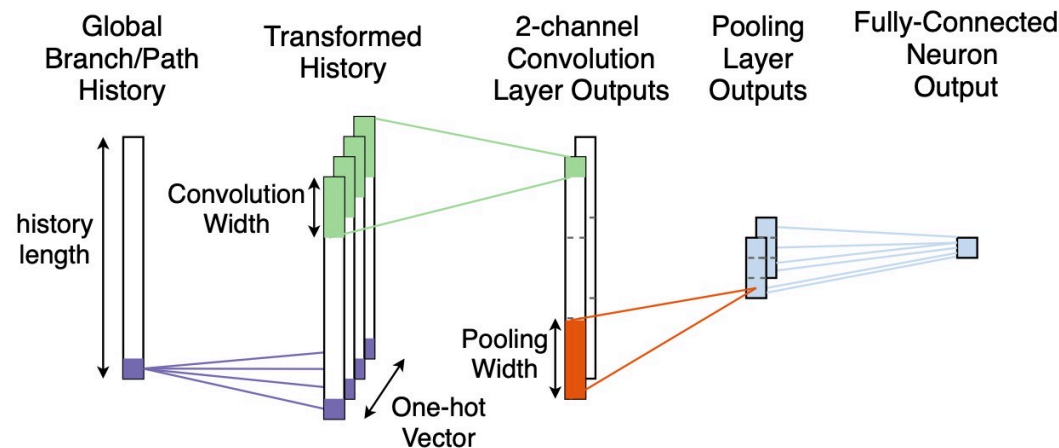


Andre Seznec,
"TAGE-SC-L branch predictors,"
CBP 2014.

Figure 1. The TAGE-SC-L predictor: a TAGE predictor backed with a Statistical Corrector predictor and a loop predictor

Convolutional Neural Nets in Branch Prediction

- Problem: identifying the correlation between a branch's outcome and a large, noisy global history is hard
- Key idea: use convolution over global history to expose hidden correlations, for hard-to-predict branches



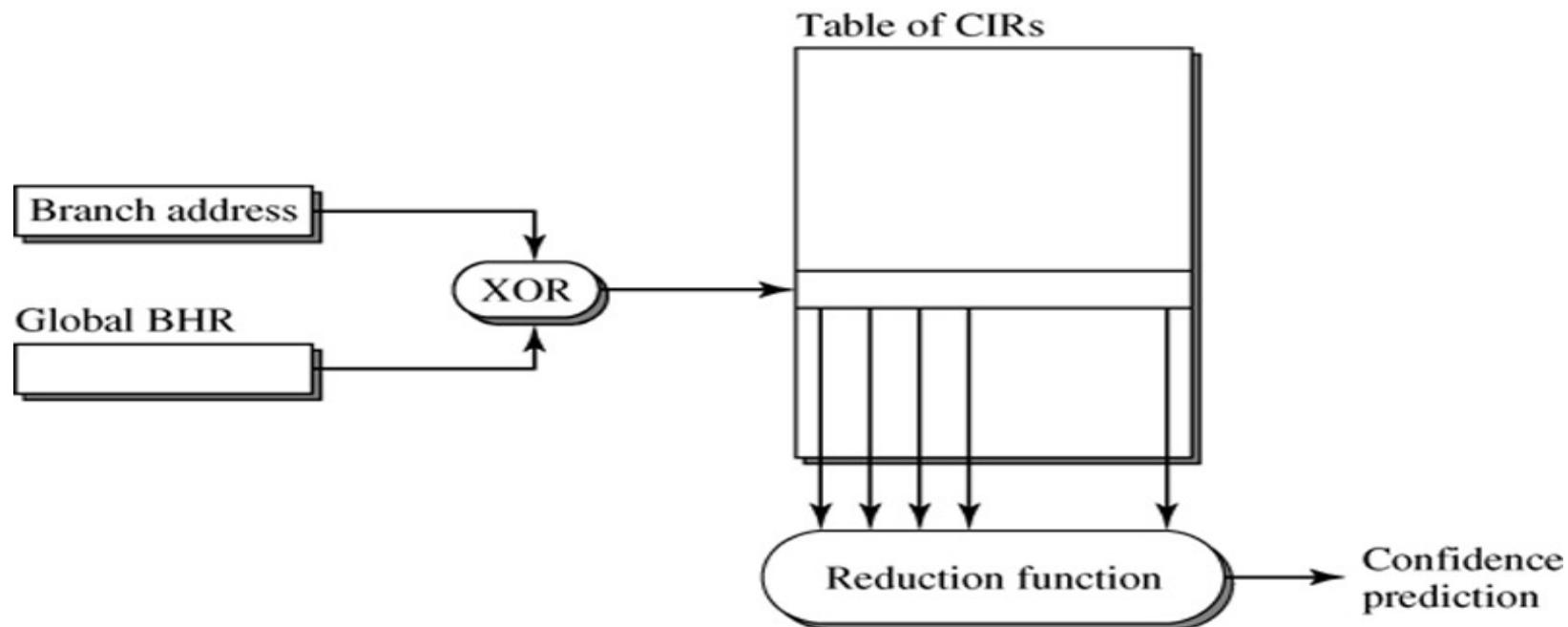
- **BranchNet Idea:** Convolutional Neural Network predictor for hard-to-predict branches + TAGE-SC-L for other branches

Branch Confidence Estimation

- Idea: Estimate if the prediction is likely to be correct
 - i.e., estimate how “confident” you are in the prediction
- Why?
 - Could be very useful in deciding how to speculate:
 - What predictor/PHT/table to choose/use
 - Whether to keep fetching on this path
 - Whether to switch to some other way of handling the branch, e.g. dual-path execution (eager execution) or predicated execution
 - ...
- Jacobsen et al., “Assigning Confidence to Conditional Branch Predictions,” MICRO 1996.

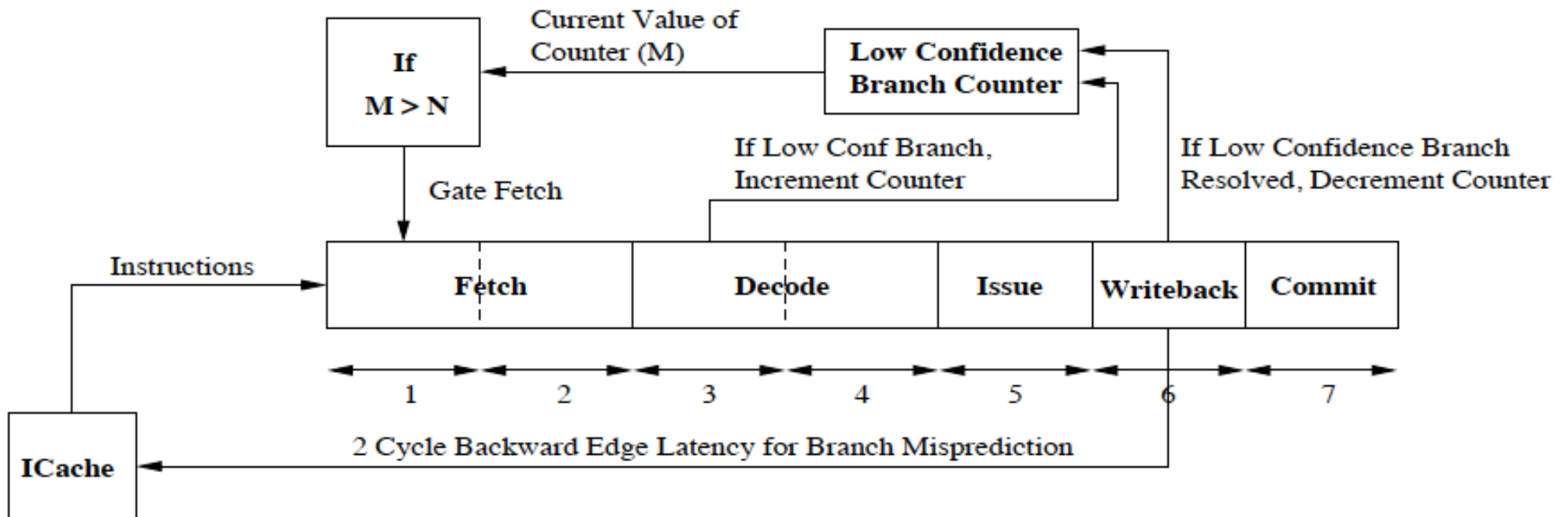
How to Estimate Confidence

- An example estimator:
 - ❑ Keep a record of correct/incorrect outcomes for the past N instances of the “branch”
 - ❑ Based on the correct/incorrect patterns, guess if the current prediction will likely be correct/incorrect



What to Do With Confidence Estimation?

- An example application: Pipeline Gating



Manne et al., “**Pipeline Gating: Speculation Control for Energy Reduction**,” ISCA 1998.

Are We Done with Branch Prediction?

- Community is actively working on it

6th Championship Branch Prediction (CBP2025)

in conjunction with:

ISCA-52 <http://iscaconf.org/isca2025/index.html>

Sponsored by ARM

The organizers of the 6th Championship Branch Prediction (CBP2025) invite you to participate in a competition for the best branch predictor.



In search of designing better **branch direction predictors**
with misprediction lower than the state-of-the-art

Submission deadline: May 2, 2025

<https://ericrotenberg.wordpress.ncsu.edu/cbp2025/>

Are We Done with Branch Prediction?

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Pushing the boundaries of energy-efficient and high-performance branch prediction

TENTATIVE SCHEDULE

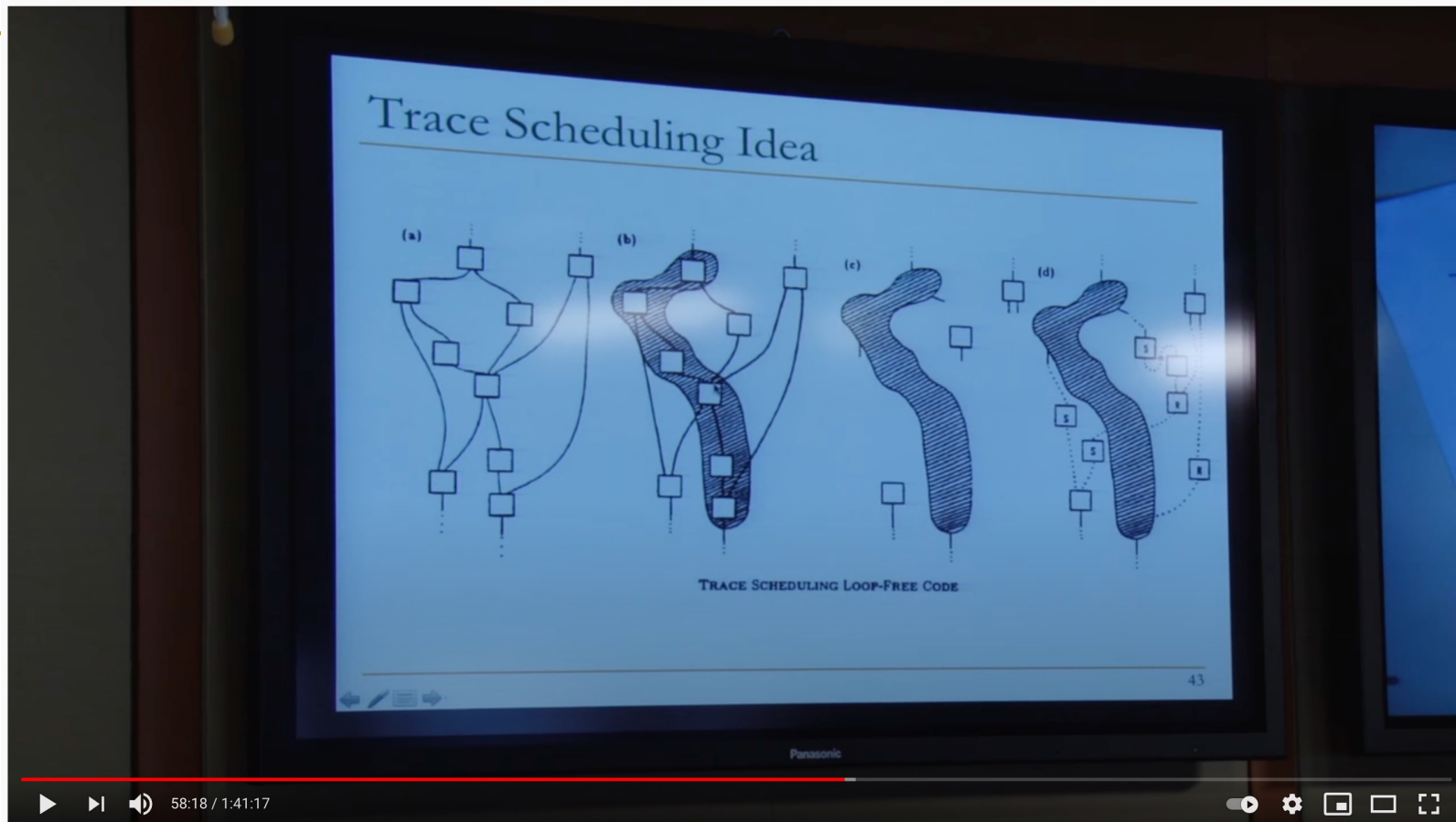
General Availability	January 2026
Submissions Due	May 8, 2026
Acceptance Notifications	May 15, 2026
Camera-ready Papers Due	June 15, 2026
Results/Awards Announced	June 28, 2026 (Raleigh, NC)

Other Ways of Handling Branches

How to Handle Control Dependences

- Critical to keep the pipeline full with correct sequence of dynamic instructions.
- Potential solutions if the instruction is a control-flow instruction:
 - **Stall** the pipeline until we know the next fetch address
 - Guess the next fetch address (**branch prediction**)
 - Employ delayed branching (**branch delay slot**)
 - Do something else (**fine-grained multithreading**)
 - Eliminate control-flow instructions (**predicated execution**)
 - Fetch from both possible paths (if you know the addresses of both possible paths) (**multipath execution**)

Lecture on Static Instruction Scheduling



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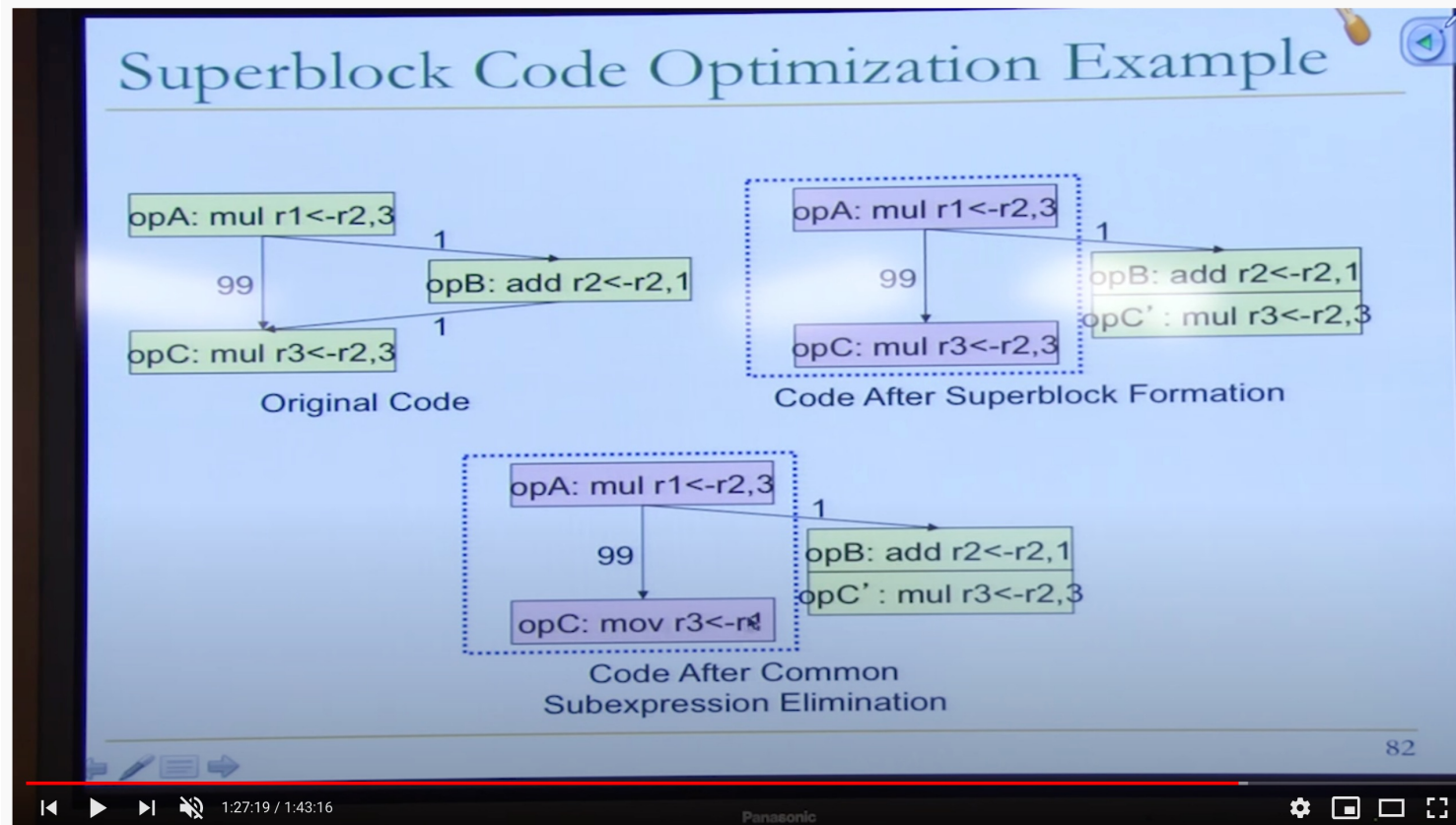


<https://www.youtube.com/onurmutlulectures>

Lectures on Static Instruction Scheduling

- **Computer Architecture, Spring 2015, Lecture 16**
 - Static Instruction Scheduling (CMU, Spring 2015)
 - <https://www.youtube.com/watch?v=isBEVkiJgGA&list=PL5PHm2jkkXmi5CxxI7b3JCL1TWybTDtKq&index=18>
- **Computer Architecture, Spring 2013, Lecture 21**
 - Static Instruction Scheduling (CMU, Spring 2013)
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- See Backup Slides

Digital Design & Computer Arch.

Lecture 17: Branch Prediction

Prof. Onur Mutlu

ETH Zürich
Spring 2026
23 April 2026

**Further Slides for Your Own Study
(May Be Covered in Future Lectures)**

Backup Slides

(These are for Your Further Study)

How to Handle Control Dependences

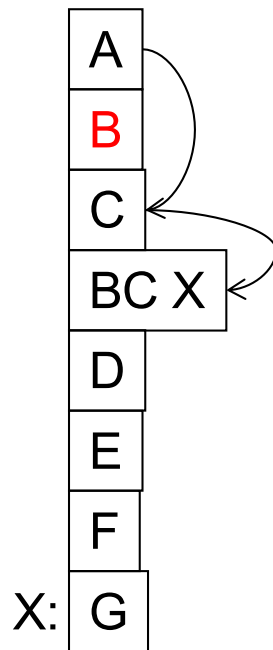
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 - Eliminate control-flow instructions (**predicated execution**)
 - Fetch from both possible paths (if you know the addresses of both possible paths) (**multipath execution**)

Delayed Branching (I)

- Change the semantics of a branch instruction
 - Branch after N instructions
 - Branch after N cycles
- Idea: Delay the execution of a branch. N instructions (delay slots) that come after the branch are **always** executed regardless of branch direction
- Problem: How do you find instructions to fill the delay slots?
 - Branch must be independent of delay slot instructions
- Unconditional branch: Easier to find instructions to fill the delay slot
- Conditional branch: Condition computation should not depend on instructions in delay slots → difficult to fill the delay slot

Delayed Branching (II)

Normal code:

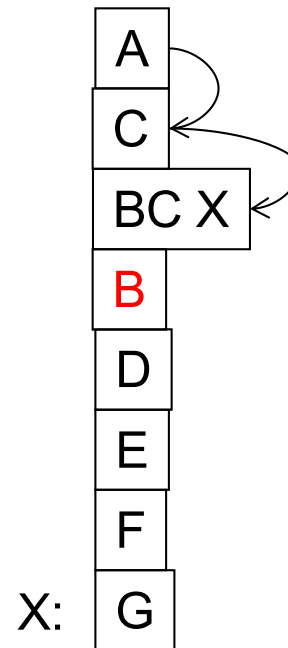


Timeline:

if	ex
A	
B	A
C	B
BC	C
--	BC
G	--

6 cycles

Delayed branch code:



Timeline:

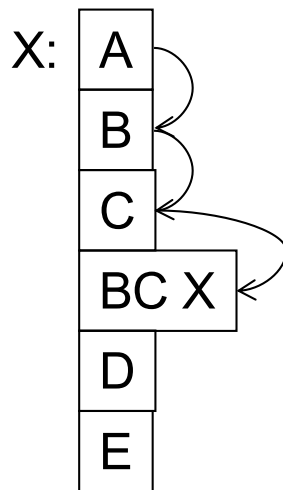
if	ex
A	
C	A
BC	C
B	BC
G	B

5 cycles

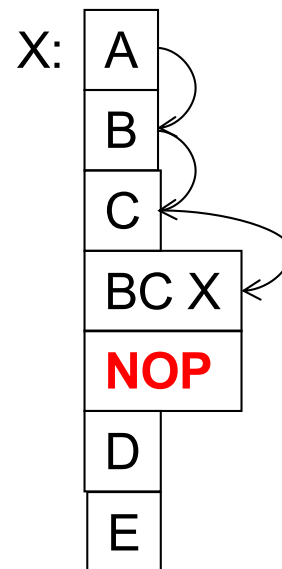
Fancy Delayed Branching (III)

- Delayed branch with squashing
 - ❑ In SPARC ISA
 - ❑ Semantics: If the branch falls through (i.e., it is **not taken**), the delay slot instruction is **not** executed
 - ❑ Why could this help?

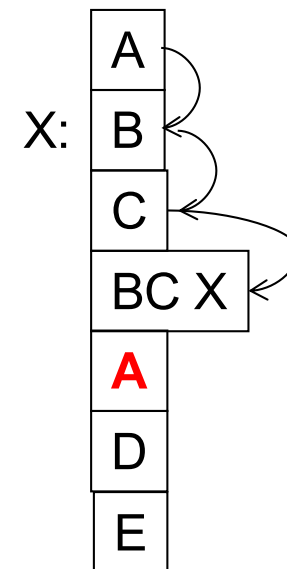
Normal code:



Delayed branch code:



Delayed branch w/ squashing:



Delayed Branching (IV)

- Advantages:

- + Keeps the pipeline full with useful instructions in a simple way assuming

- 1. Number of delay slots == number of instructions to keep the pipeline full before the branch resolves

- 2. All delay slots can be filled with useful instructions

- Disadvantages:

- Not easy to fill the delay slots (even with a 2-stage pipeline)

- 1. Number of delay slots increases with pipeline depth, superscalar execution width

- 2. Number of delay slots should be variable with variable latency operations. Why?

- Ties ISA semantics to hardware implementation

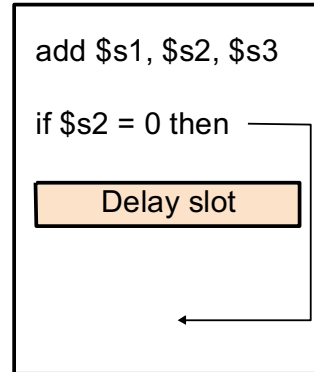
- SPARC, MIPS, HP-PA: 1 delay slot

- What if pipeline implementation changes with the next design?

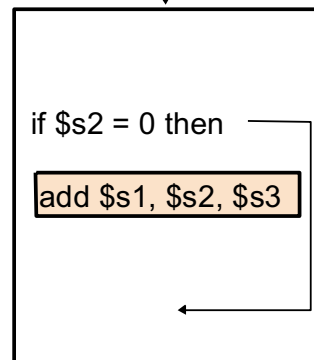
An Aside: Filling the Delay Slot

reordering
independent
instructions
does not change
program semantics

a. From before

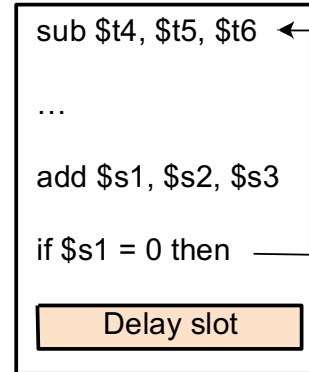


Becomes

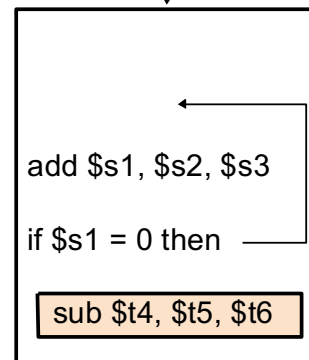


within same
basic block

b. From target

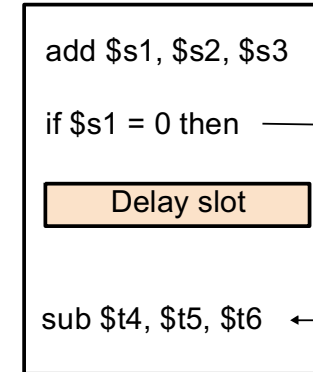


Becomes

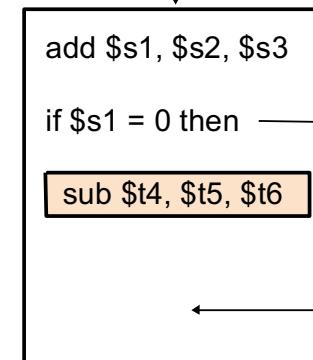


For correctness:
add a new instruction
to the not-taken path?

c. From fall through



Becomes



For correctness:
add a new instruction
to the taken path?

Safe?

Lecture on Static Instruction Scheduling

Trace Scheduling Idea

(a) (b) (c) (d)

TRACE SCHEDULING LOOP-FREE CODE

43

Panasonic

58:18 / 1:41:17

Lecture 16. Static Instruction Scheduling - Carnegie Mellon - Comp. Arch. 2015 - Onur Mutlu

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Lecturer: Prof. Onur Mutlu (<http://users.ece.cmu.edu/~omutlu/>)

Date: Feb 23rd, 2015

Lecture 16 slides (pdf): <http://www.ece.cmu.edu/~ece447/s15/li...>

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Lectures on Static Instruction Scheduling

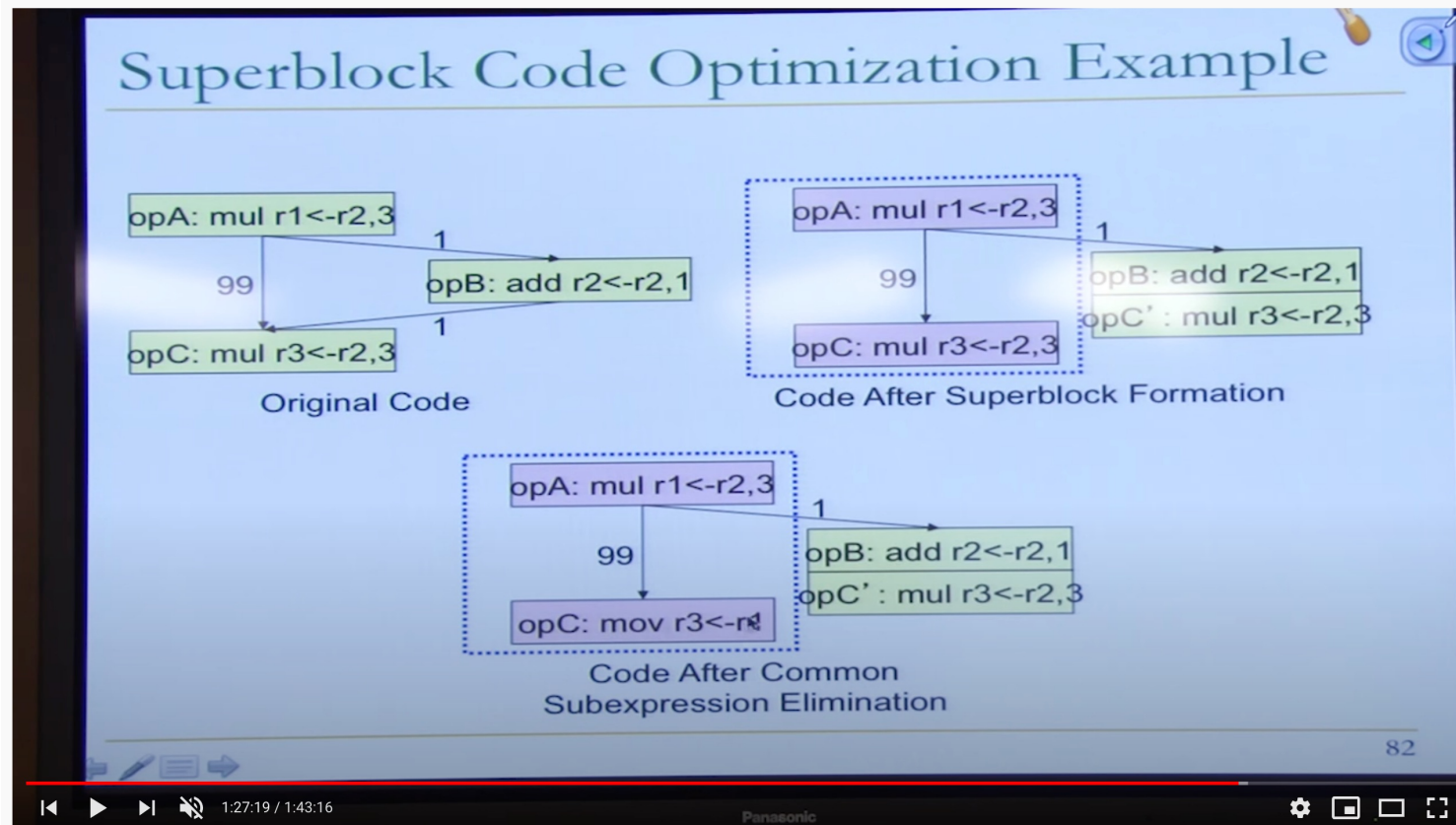
- Computer Architecture, Spring 2015, Lecture 16
 - Static Instruction Scheduling (CMU, Spring 2015)
 - <https://www.youtube.com/watch?v=isBEVkiJgGA&list=PL5PHm2jkkXmi5CxxI7b3JCL1TWybTDtKq&index=18>

- Computer Architecture, Spring 2013, Lecture 21
 - Static Instruction Scheduling (CMU, Spring 2013)
 - <https://www.youtube.com/watch?v=XdDUn2WtkRg&list=PL5PHm2jkkXmidJOd59REog9jDnPDTG6IJ&index=21>

How to Handle Control Dependences

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- See Backup Slides

How to Handle Control Dependences

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Predicate Combining (*not* Predicated Execution)

- Complex predicates are converted into multiple branches
 - ❑ if ((a == b) && (c < d) && (a > 5000)) { ... }
 - 3 conditional branches
- Problem: This increases the number of control dependencies
- Idea: Combine predicate operations to feed a single branch instruction instead of having one branch for each
 - ❑ Predicates stored and operated on using condition registers
 - ❑ A single branch checks the value of the combined predicate
- + Fewer branches in code → fewer mipredictions/stalls
- Possibly unnecessary work
 - If the first predicate is false, no need to compute other predicates
- Condition registers exist in IBM RS6000 and the POWER architecture

Predication (Predicated Execution)

- Idea: Convert control dependence to data dependence
- Simple example: Suppose we had a Conditional Move instruction...
 - ❑ CMOV condition, R1 $\leftarrow$ R2
 - ❑ R1 = (condition == true) ? R2 : R1
 - ❑ Employed in most modern ISAs (x86, Alpha)
- Code example with branches vs. CMOVs
if (a == 5) {b = 4;} else {b = 3;}

CMPEQ condition, a, 5;

CMOV condition, b $\leftarrow$ 4;

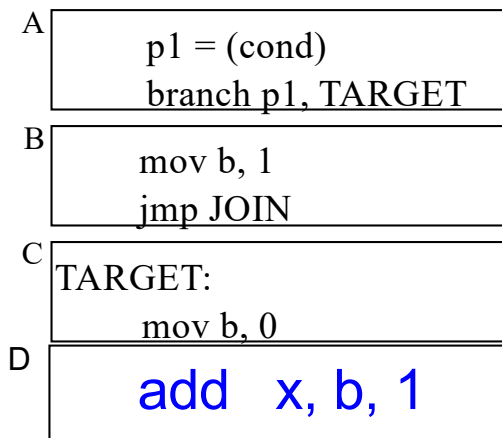
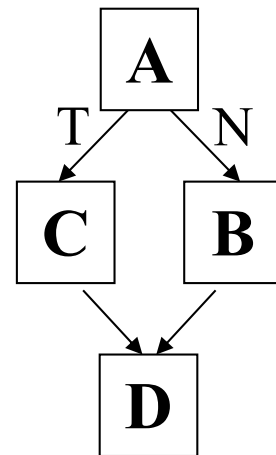
CMOV !condition, b $\leftarrow$ 3;

Predication (Predicated Execution)

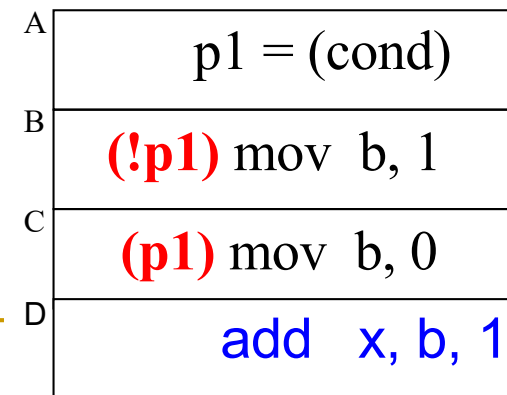
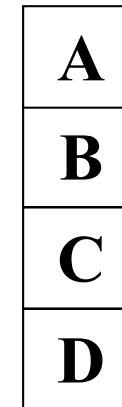
- Idea: Compiler converts control dependence into data dependence
→ **branch is eliminated**
 - Each instruction has a **predicate bit** set based on the predicate computation
 - **Only instructions with TRUE predicates are committed** (others become NOPs)

(normal branch code)

```
if (cond) {  
    b = 0;  
}  
else {  
    b = 1;  
}
```



(predicated code)



Predicated Execution References

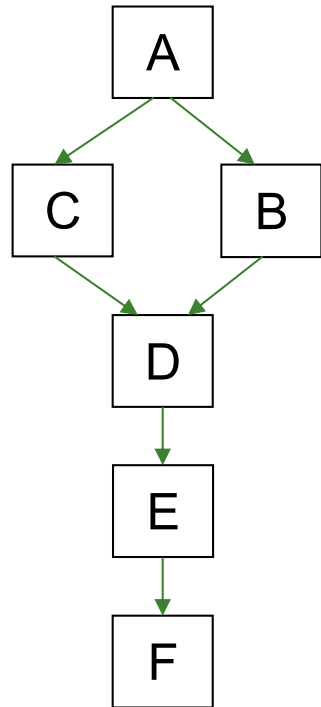
- Allen et al., “Conversion of control dependence to data dependence,” POPL 1983.
- Kim et al., “Wish Branches: Combining Conditional Branching and Predication for Adaptive Predicated Execution,” MICRO 2005.
- Kim et al., “Diverge-Merge Processor (DMP): Dynamic Predicated Execution of Complex Control-Flow Graphs Based on Frequently Executed Paths,” MICRO 2006.

Conditional Move Operations

- Very limited form of predicated execution
- $\text{CMOV } R1 \leftarrow R2$
 - $R1 = (\text{ConditionCode} == \text{true}) ? R2 : R1$
 - Employed in most modern ISAs (x86, Alpha)

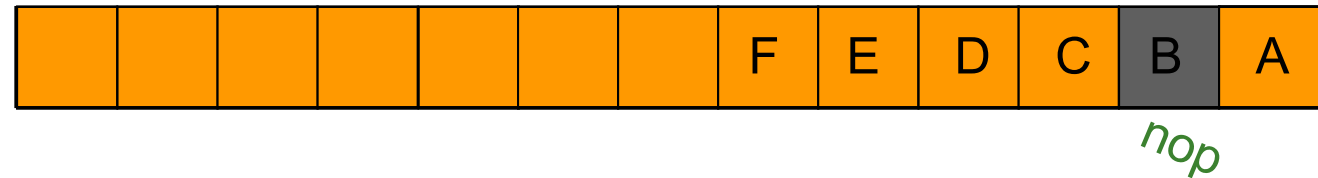
Predicated Execution (II)

- Predicated execution can be high performance and energy-efficient



Predicated Execution

Fetch Decode Rename Schedule RegisterRead Execute



Branch Prediction

Fetch Decode Rename Schedule RegisterRead Execute



Pipeline flush!!

Predicated Execution

- Eliminates branches → enables straight line code (i.e., larger basic blocks in code)
- Advantages
 - Eliminates hard-to-predict branches
 - Always-not-taken prediction works better (no branches)
 - Compiler has more freedom to optimize code (no branches)
 - control flow does not hinder inst. reordering optimizations
 - code optimizations hindered only by data dependencies
- Disadvantages
 - Useless work: some instructions fetched/executed but discarded (especially bad for easy-to-predict branches)
 - Requires additional ISA (and hardware) support
 - Can we eliminate all branches this way?

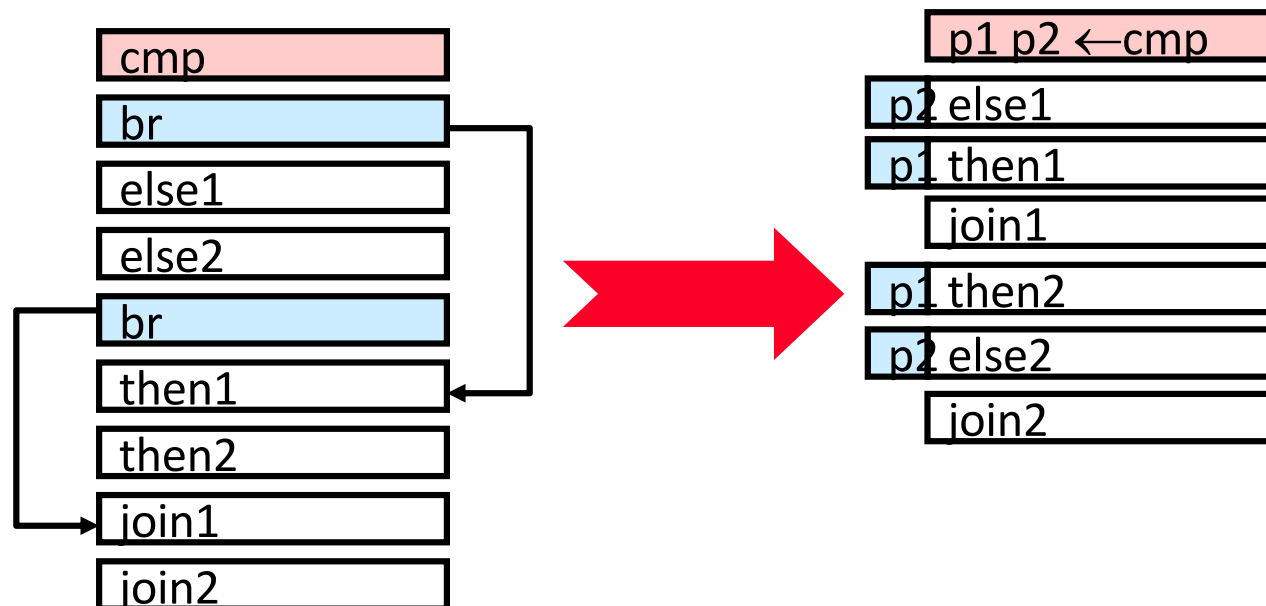
Predicated Execution vs. Branch Prediction

- + Eliminates mispredictions for hard-to-predict branches
 - + No need for branch prediction for some branches
 - + Good if misprediction cost > useless work due to predication

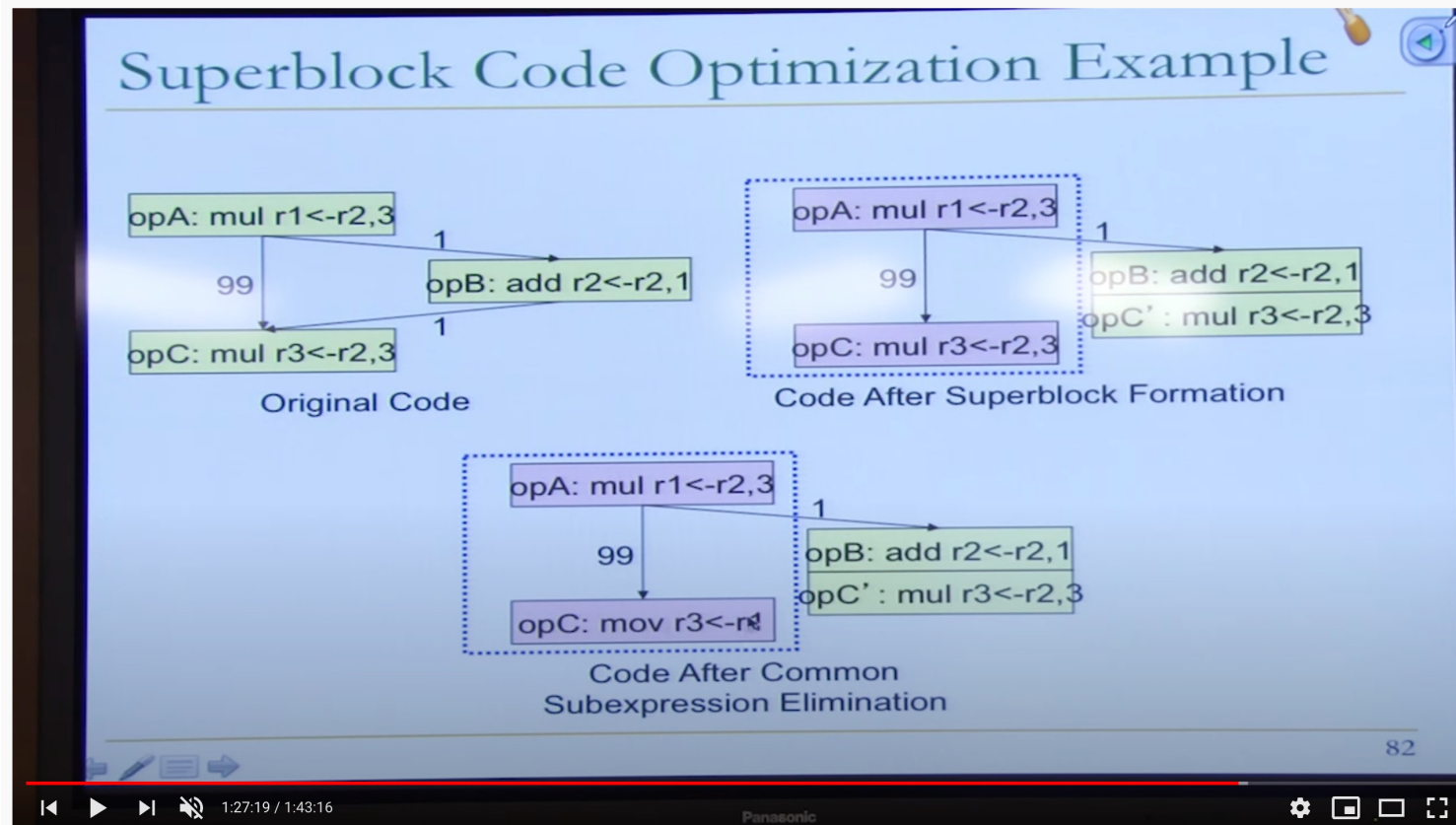
- Causes useless work for branches that are easy to predict
 - Reduces performance if misprediction cost < useless work
 - **Adaptivity**: Static predication is not adaptive to run-time branch behavior. Branch behavior changes based on input set, program phase, control-flow path.

Predicated Execution in Intel Itanium

- Each instruction can be separately predicated
- 64 one-bit predicate registers
 - each instruction carries a 6-bit predicate field
- An instruction is effectively a NOP if its predicate is false



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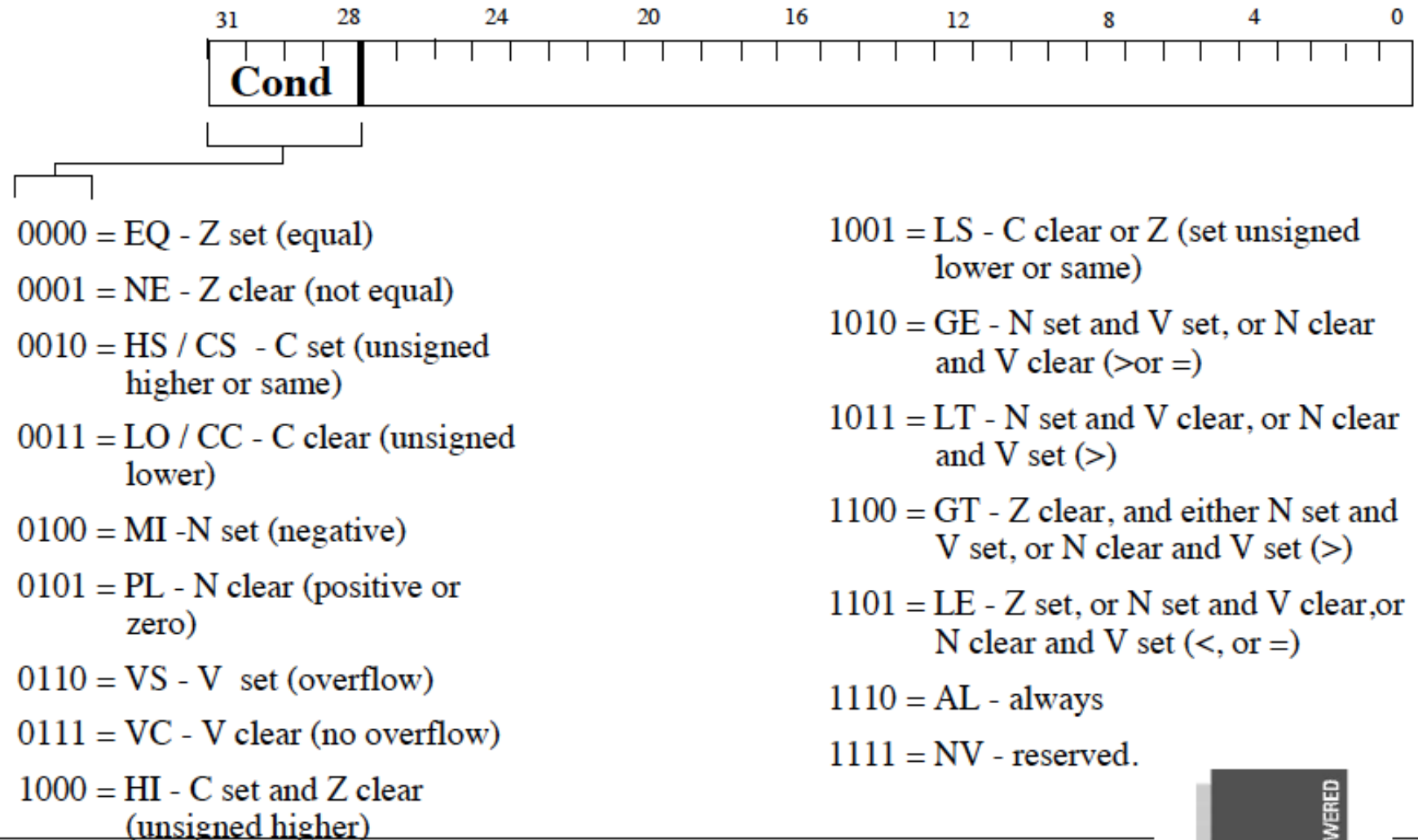
Conditional Execution in the ARM ISA

- Almost all ARM instructions could include an optional condition code
 - Prior to ARM v8
- An instruction with a condition code is executed **only if**
 - the condition code flags in the CPSR (Current Program Status Register) meet the specified condition

Conditional Execution in ARM ISA

31	2827	1615	87	0	Instruction type							
Cond	0 0 I	Opcode	S	Rn	Rd	Operand2				Data processing / PSR Transfer		
Cond	0 0 0 0 0 0	A S	Rd	Rn	Rs	1 0 0 1	Rm			Multiply		
Cond	0 0 0 0 1	U A S	RdHi	RdLo	Rs	1 0 0 1	Rm			Long Multiply (v3M / v4 only)		
Cond	0 0 0 1 0	B 0 0	Rn	Rd	0 0 0 0	1 0 0 1	Rm			Swap		
Cond	0 1 I P U B	W L	Rn	Rd	Offset				Load/Store Byte/Word			
Cond	1 0 0 P U S	W L	Rn	Register List						Load/Store Multiple		
Cond	0 0 0 P U 1	W L	Rn	Rd	Offset1	1 S H 1	Offset2			Halfword transfer : Immediate offset (v4 only)		
Cond	0 0 0 P U 0	W L	Rn	Rd	0 0 0 0	1 S H 1	Rm			Halfword transfer: Register offset (v4 only)		
Cond	1 0 1 L	Offset							Branch			
Cond	0 0 0 1	0 0 1 0	1 1 1 1	1 1 1 1	1 1 1 1	0 0 0 1	Rn			Branch Exchange (v4T only)		
Cond	1 1 0 P U	N W L	Rn	CRd	CPNum	Offset				Coprocessor data transfer		
Cond	1 1 1 0	Op1	CRn	CRd	CPNum	Op2	0	CRm			Coprocessor data operation	
Cond	1 1 1 0	Op1	L	CRn	Rd	CPNum	Op2	1	CRm			Coprocessor register transfer
Cond	1 1 1 1	SWI Number									Software interrupt	

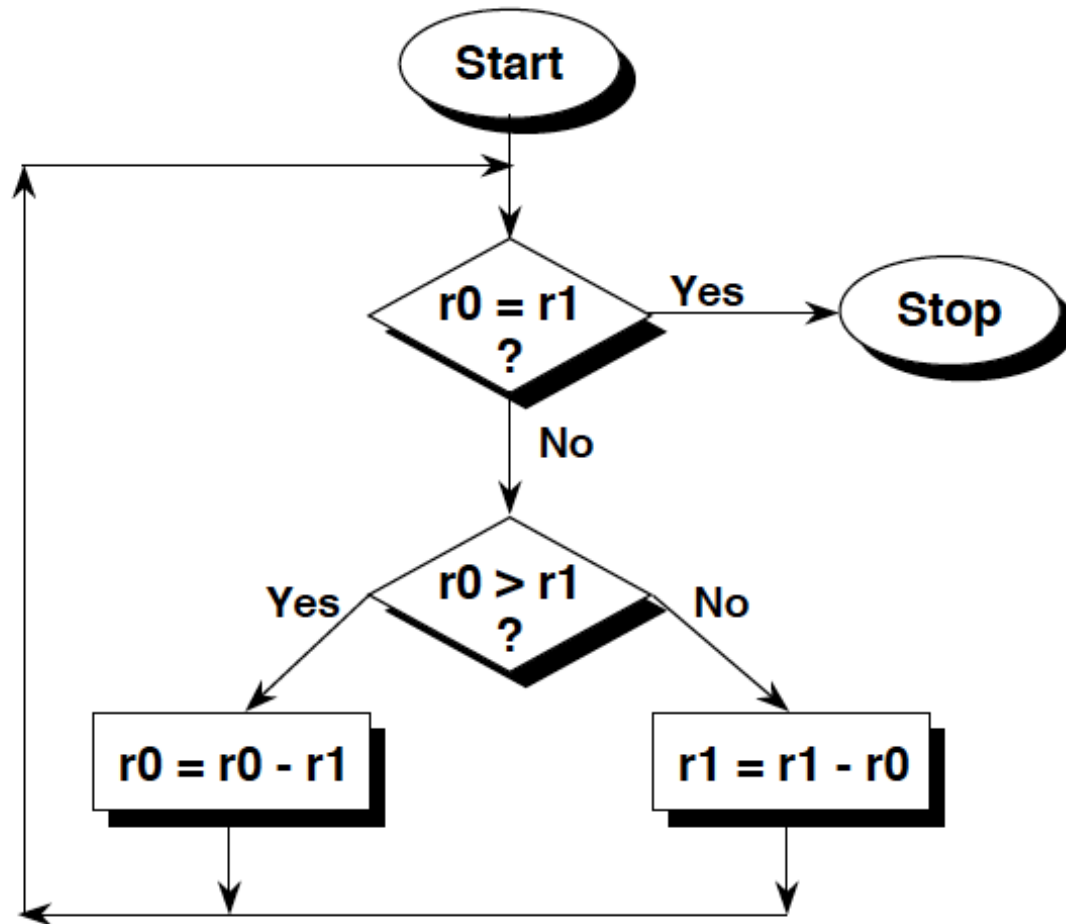
Conditional Execution in ARM ISA



Conditional Execution in ARM ISA

- * **To execute an instruction conditionally, simply postfix it with the appropriate condition:**
 - For example an add instruction takes the form:
 - `ADD r0,r1,r2` ; `r0 = r1 + r2` (ADDAL)
 - To execute this only if the zero flag is set:
 - `ADDEQ r0,r1,r2` ; If zero flag set then...
 - ; ... `r0 = r1 + r2`
- * **By default, data processing operations do not affect the condition flags (apart from the comparisons where this is the only effect). To cause the condition flags to be updated, the S bit of the instruction needs to be set by postfixing the instruction (and any condition code) with an “S”.**
 - For example to add two numbers and set the condition flags:
 - `ADDS r0,r1,r2` ; `r0 = r1 + r2`
 - ; ... and set flags

Conditional Execution in ARM ISA



* **Convert the GCD algorithm given in this flowchart into**

- 1) “Normal” assembler, where only branches can be conditional.
- 2) ARM assembler, where all instructions are conditional, thus improving code density.

* **The only instructions you need are CMP, B and SUB.**

Conditional Execution in ARM ISA

“Normal” Assembler

```
gcd    cmp r0, r1      ;reached the end?
        beq stop
        blt less       ;if r0 > r1
        sub r0, r0, r1  ;subtract r1 from r0
        bal gcd
less   sub r1, r1, r0   ;subtract r0 from r1
        bal gcd
stop
```

ARM Conditional Assembler

```
gcd    cmp    r0, r1      ;if r0 > r1
        subgt r0, r0, r1  ;subtract r1 from r0
        sublt r1, r1, r0  ;else subtract r0 from r1
        bne   gcd        ;reached the end?
```

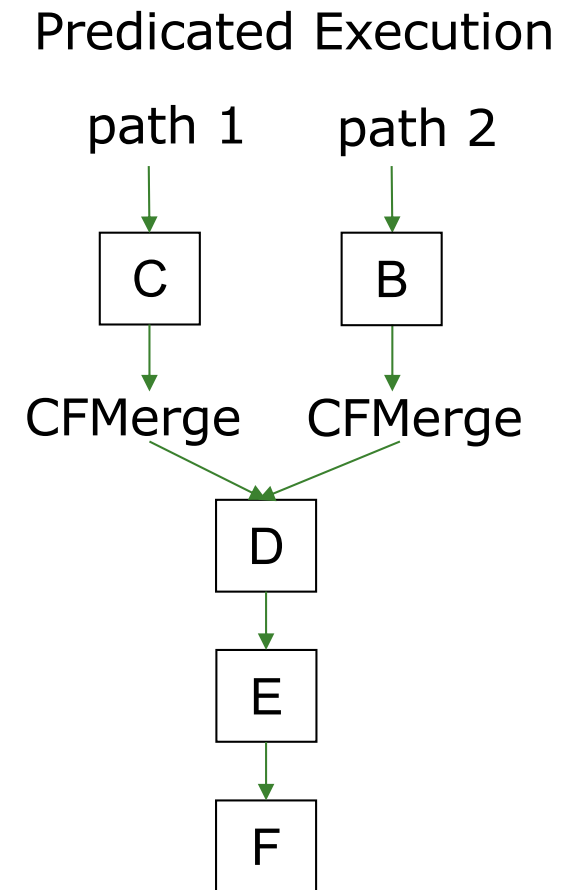
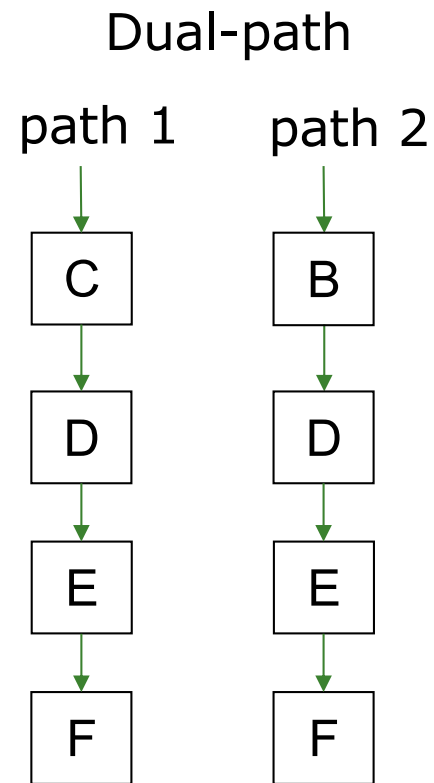
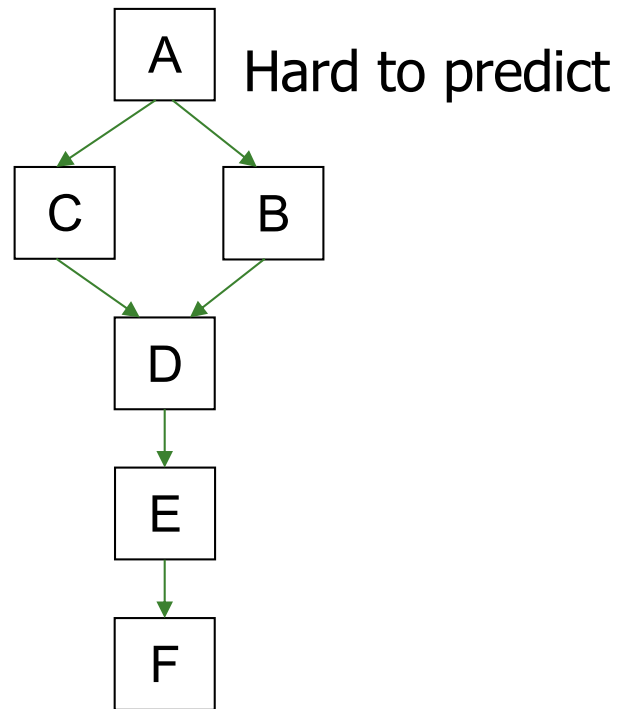
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Multi-Path Execution

- Idea: Execute both paths after a conditional branch
 - For all branches: Riseman and Foster, “The inhibition of potential parallelism by conditional jumps,” IEEE Transactions on Computers, 1972.
 - For a hard-to-predict branch: Use dynamic confidence estimation
- Advantages:
 - + Improves performance if misprediction cost > useless work
 - + No ISA change needed
- Disadvantages:
 - What happens when the machine encounters another hard-to-predict branch? Execute both paths again?
 - Paths followed quickly become exponential
 - Each followed path requires its own context (registers, PC, GHR)
 - Wasted work (and reduced performance) if paths merge

Dual-Path Execution versus Predication



Handling Other Types of Branches

Remember: Branch Types

Type	Direction at fetch time	Number of possible next fetch addresses?	When is next fetch address resolved?
Conditional	Unknown	2	Execution (register dependent)
Unconditional	Always taken	1	Decode (PC + offset)
Call	Always taken	1	Decode (PC + offset)
Return	Always taken	Many	Execution (register dependent)
Indirect	Always taken	Many	Execution (register dependent)

How can we predict an indirect branch with many target addresses?

Call and Return Prediction

- Direct calls are easy to predict

- Always taken, single target
- Call marked in BTB, target predicted by BTB

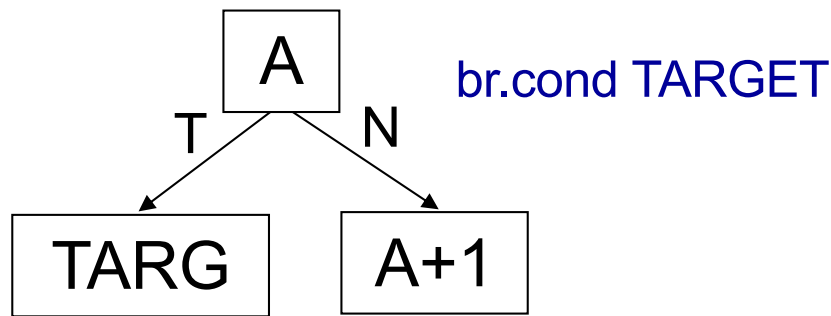
- Returns are indirect branches

- A function can be called from many points in code
- A return instruction can have many target addresses
 - Next instruction after each call point for the same function
- Observation: Usually a return matches a call
- Idea: Use a stack to predict return addresses (Return Address Stack)
 - A fetched call: pushes the return (next instruction) address on the stack
 - A fetched return: pops the stack and uses the address as its predicted target
 - Accurate most of the time: 8-entry stack → > 95% accuracy

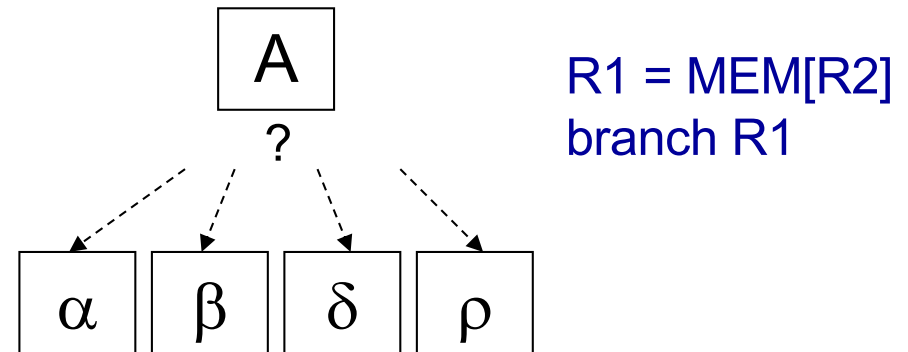
Call X
...
Call X
...
Call X
...
Return
Return
Return

Indirect Branch Prediction (I)

- Register-indirect branches have multiple targets



Conditional (Direct) Branch



Indirect Jump

- Used to implement
 - ❑ Switch-case statements
 - ❑ Virtual function calls
 - ❑ Jump tables (of function pointers)
 - ❑ Interface calls

Indirect Branch Prediction (II)

- No direction prediction needed
- Idea 1: Predict the last resolved target as the next fetch address
 - + Simple: Use the BTB to store the target address
 - Inaccurate: 50% accuracy (empirical). Many indirect branches switch between different targets
- Idea 2: Use history based target prediction
 - E.g., Index the BTB with GHR XORed with Indirect Branch PC
 - Chang et al., “Target Prediction for Indirect Jumps,” ISCA 1997.
 - + More accurate
 - An indirect branch maps to (too) many entries in BTB
 - Conflict misses with other branches (direct or indirect)
 - Inefficient use of space if branch has few target addresses

Intel Pentium M Indirect Branch Predictor

The advanced branch prediction in the Pentium M processor is based on the Intel Pentium[®] 4 processor's [6] branch predictor. On top of that, two additional predictors to capture special program flows, were added: a Loop Detector and an Indirect Branch Predictor.

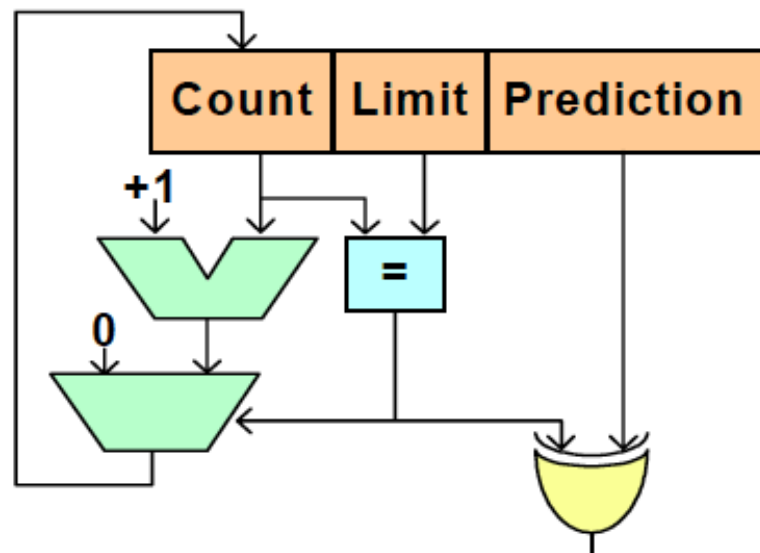


Figure 2: The Loop Detector logic

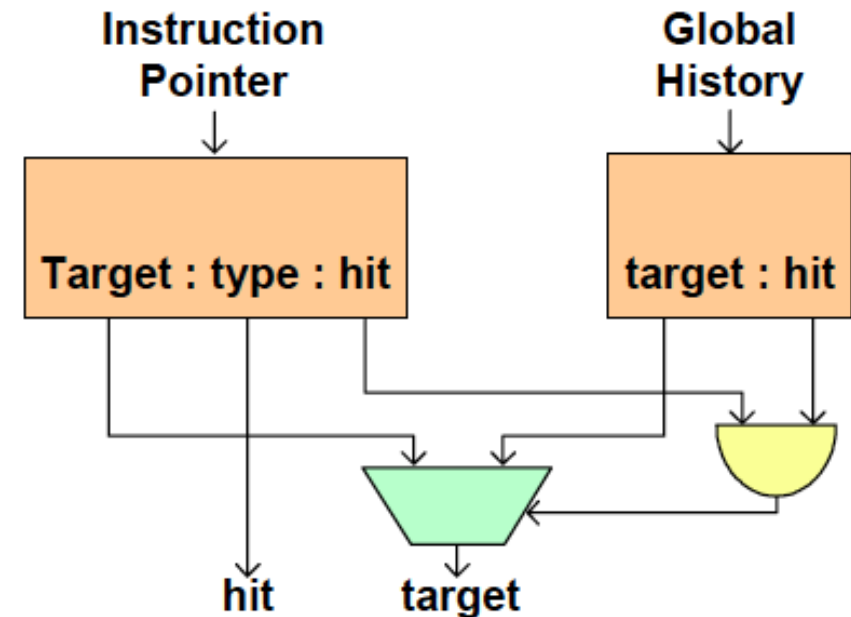


Figure 3: The Indirect Branch Predictor logic

Gochman et al.,

“**The Intel Pentium M Processor: Microarchitecture and Performance,**”

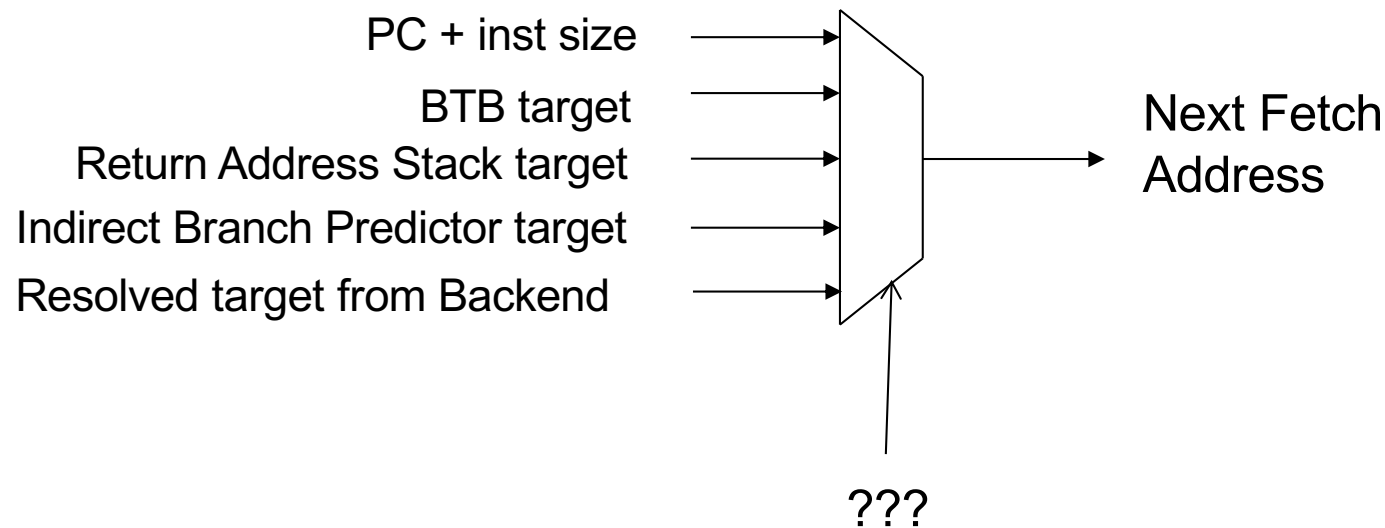
Intel Technology Journal, May 2003.

Issues in Branch Prediction (I)

- Need to identify a branch before it is fetched
- How do we do this?
 - ❑ BTB hit → indicates that the fetched instruction is a branch
 - ❑ BTB entry contains the “type” of the branch
 - ❑ Pre-decoded “branch type” information stored in the instruction cache identifies type of branch
- What if no BTB?
 - ❑ Bubble in the pipeline until target address is computed
 - ❑ E.g., IBM POWER4

Latency of Branch Prediction

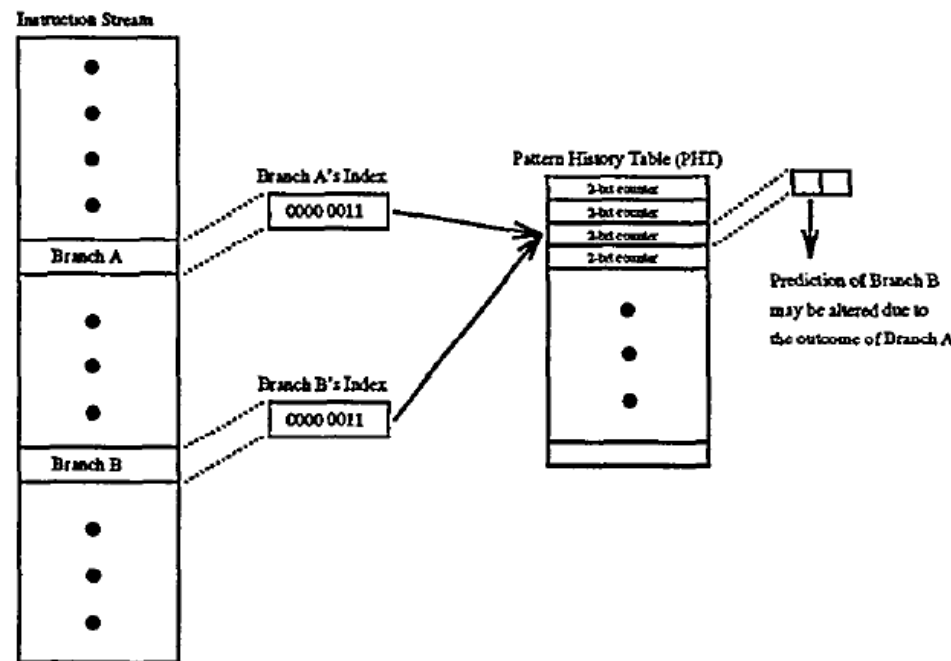
- **Latency:** Prediction is latency critical
 - ❑ Need to generate next fetch address for the next cycle
 - ❑ Bigger, more complex predictors are more accurate but slower



Interference in Branch Predictors

An Issue: Interference in the PHTs

- Sharing the PHTs between histories/branches leads to interference
 - Different branches map to the same PHT entry and modify it
 - Interference can be positive, **negative**, or neutral



- Interference can be eliminated by dedicating a PHT per branch
 - Too much hardware cost
- How else can you eliminate or reduce interference?

Reducing Interference in PHTs (I)

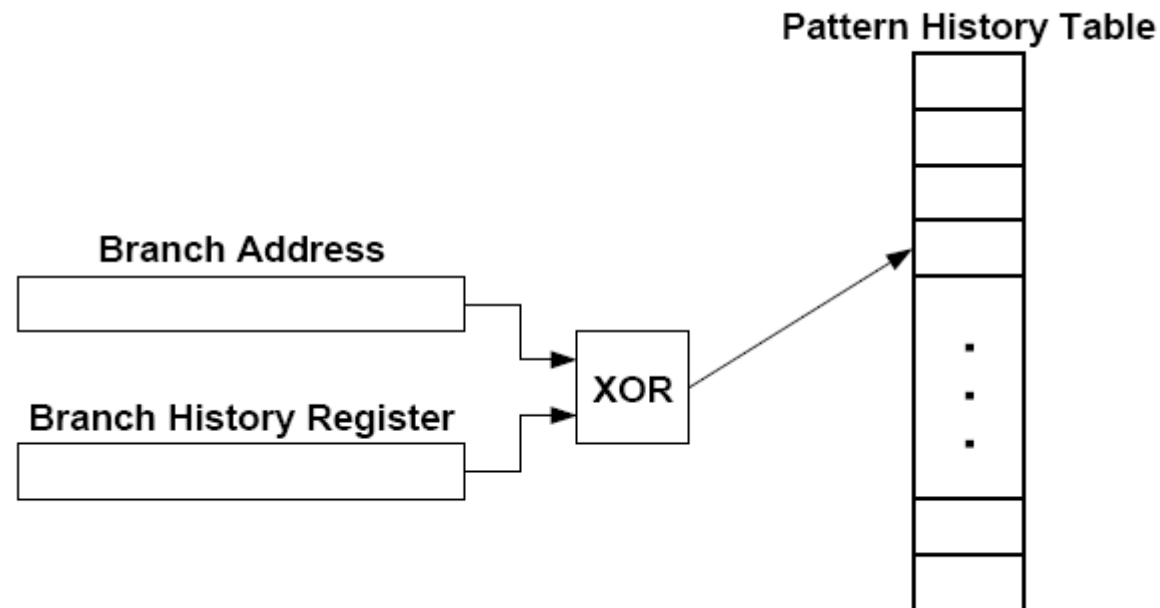
- Increase size of PHT
- Branch filtering
 - ❑ Predict highly-biased branches separately so that they do not consume PHT entries
 - ❑ E.g., static prediction or BTB based prediction
- Hashing/index-randomization
 - ❑ Gshare
 - ❑ Gskew
- Agree prediction

Biased Branches and Branch Filtering

- Observation: Many branches are biased in one direction (e.g., 99% taken)
- Problem: These branches *pollute* the branch prediction structures → make the prediction of other branches difficult by causing “interference” in branch prediction tables and history registers
- Solution: Detect such biased branches, and predict them with a simpler predictor (e.g., last time, static, ...)
- Chang et al., “Branch classification: a new mechanism for improving branch predictor performance,” MICRO 1994.

Reducing Interference: Gshare

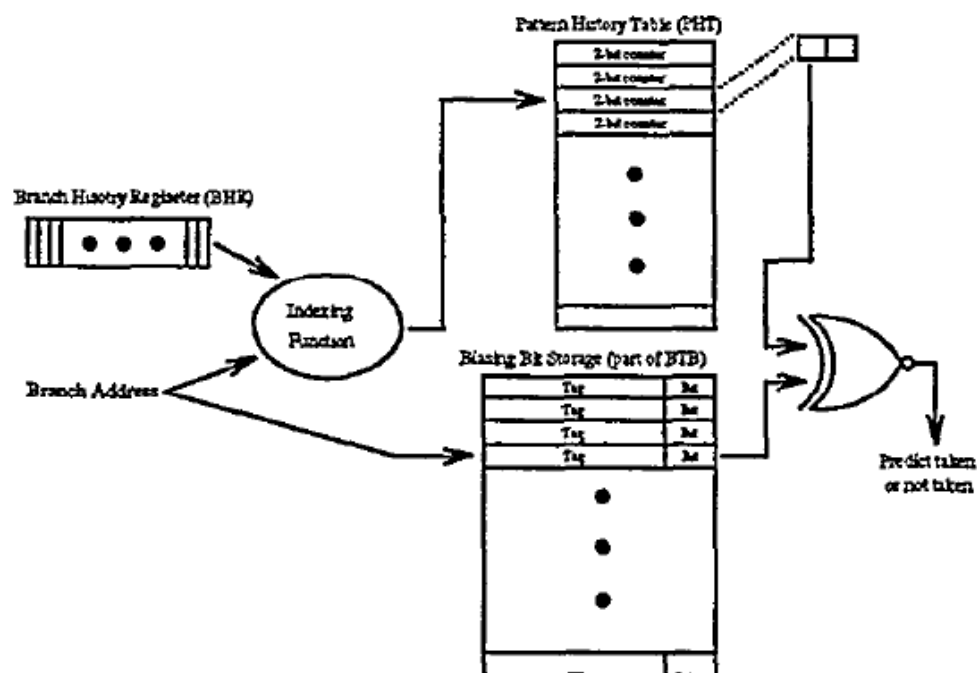
- Idea 1: Randomize the indexing function into the PHT such that probability of two branches mapping to the same entry reduces
 - Gshare predictor: GHR hashed with the Branch PC
 - + Better utilization of PHT + More context information
 - Increases access latency



- McFarling, “Combining Branch Predictors,” DEC WRL Tech Report, 1993.

Reducing Interference: Agree Predictor

- Idea 2: Agree prediction
 - Each branch has a “bias” bit associated with it in BTB
 - Ideally, most likely outcome for the branch
 - High bit of the PHT counter indicates whether or not the prediction agrees with the bias bit (not whether or not prediction is taken)
- + Reduces negative interference (Why???)
- Requires determining bias bits (compiler vs. hardware)



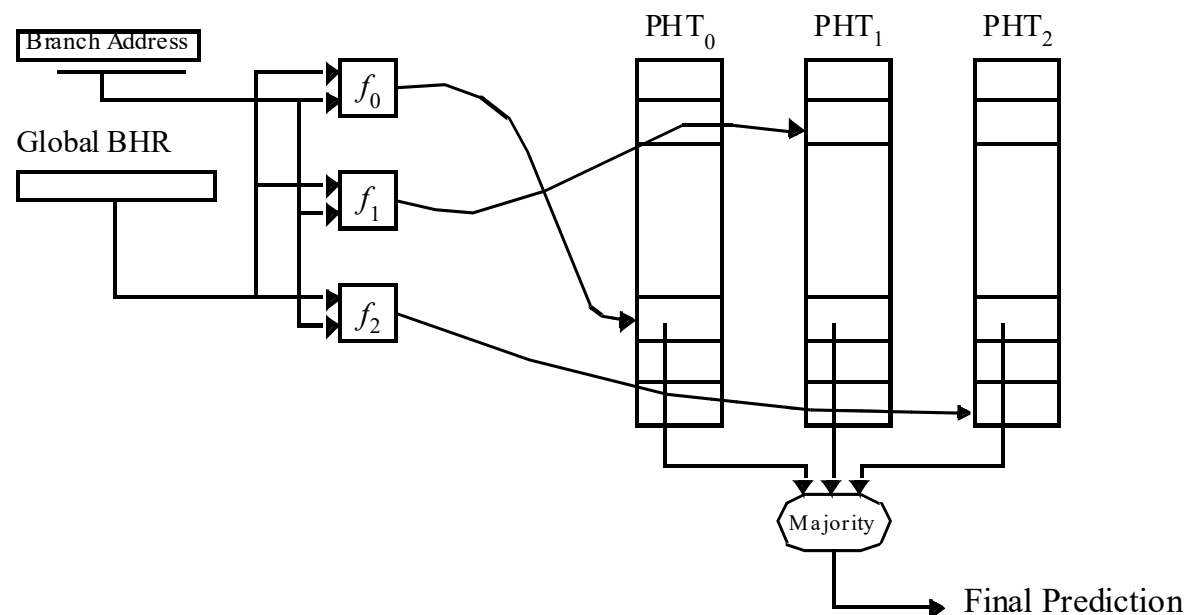
Sprangle et al., “The Agree Predictor: A Mechanism for Reducing Negative Branch History Interference,” ISCA 1997.

Why Does Agree Prediction Make Sense?

- Assume two branches have taken rates of 85% and 15%.
- Assume they conflict in the PHT
- Let's compute the **probability they have opposite outcomes**
 - Baseline predictor:
 - $P(b1\ T, b2\ NT) + P(b1\ NT, b2\ T)$
 $= (85\% * 85\%) + (15\% * 15\%) = 74.5\%$
 - Agree predictor:
 - Assume bias bits are set to T (b1) and NT (b2)
 - $P(b1\ agree, b2\ disagree) + P(b1\ disagree, b2\ agree)$
 $= (85\% * 15\%) + (15\% * 85\%) = 25.5\%$
- Works because most branches are biased (not 50% taken)

Reducing Interference: Gskew

- Idea 3: Gskew predictor
 - ❑ Multiple PHTs
 - ❑ Each indexed with a different type of hash function
 - ❑ Final prediction is a majority vote
- + Distributes interference patterns in a more randomized way
(interfering patterns less likely in different PHTs at the same time)
- More complexity (due to multiple PHTs, hash functions)



Seznec, “An optimized 2bcgskew branch predictor,” IRISA Tech Report 1993.

Michaud, “Trading conflict and capacity aliasing in conditional branch predictors,” ISCA 1997

More Techniques to Reduce PHT Interference

- The bi-mode predictor
 - ❑ Separate PHTs for mostly-taken and mostly-not-taken branches
 - ❑ Reduces negative aliasing between them
 - ❑ Lee et al., “The bi-mode branch predictor,” MICRO 1997.
- The YAGS predictor
 - ❑ Use a small tagged “cache” to predict branches that have experienced interference
 - ❑ Aims to not to mispredict them again
 - ❑ Eden and Mudge, “The YAGS branch prediction scheme,” MICRO 1998.
- Alpha EV8 (21464) branch predictor
 - ❑ Seznec et al., “Design tradeoffs for the Alpha EV8 conditional branch predictor,” ISCA 2002.

Another Direction: Helper Threading

- Idea: Pre-compute the outcome of the branch with a separate, customized thread (i.e., a helper thread)

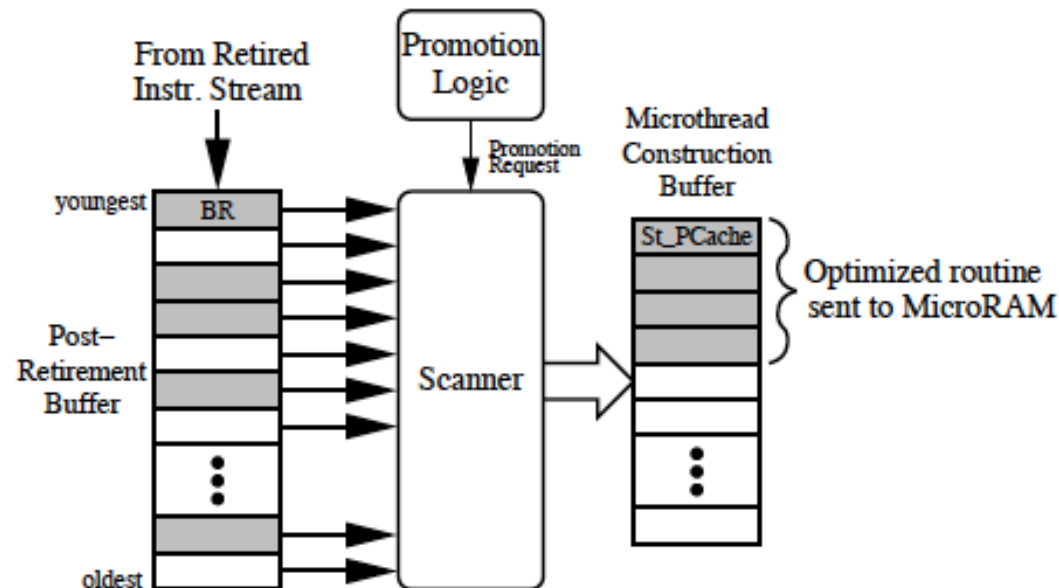


Figure 3. The Microthread Builder

- Chappell et al., “Difficult-Path Branch Prediction Using Subordinate Microthreads,” ISCA 2002.
- Chappell et al., “Simultaneous Subordinate Microthreading,” ISCA 1999.

Issues in Wide & Fast Fetch

I-Cache Line and Way Prediction

- Problem: Complex branch prediction can take too long (many cycles)
- Goal
 - Quickly generate (a reasonably accurate) next fetch address
 - Enable the fetch engine to run at high frequencies
 - Override the quick prediction with more sophisticated prediction
- Idea: Get the predicted next cache line and way at the time you fetch the current cache line
- Example Mechanism (e.g., Alpha 21264)
 - Each cache line tells which line/way to fetch next (prediction)
 - On a fill, line/way predictor points to next sequential line
 - On branch resolution, line/way predictor is updated
 - If line/way prediction is incorrect, one cycle is wasted

Alpha 21264 Line & Way Prediction

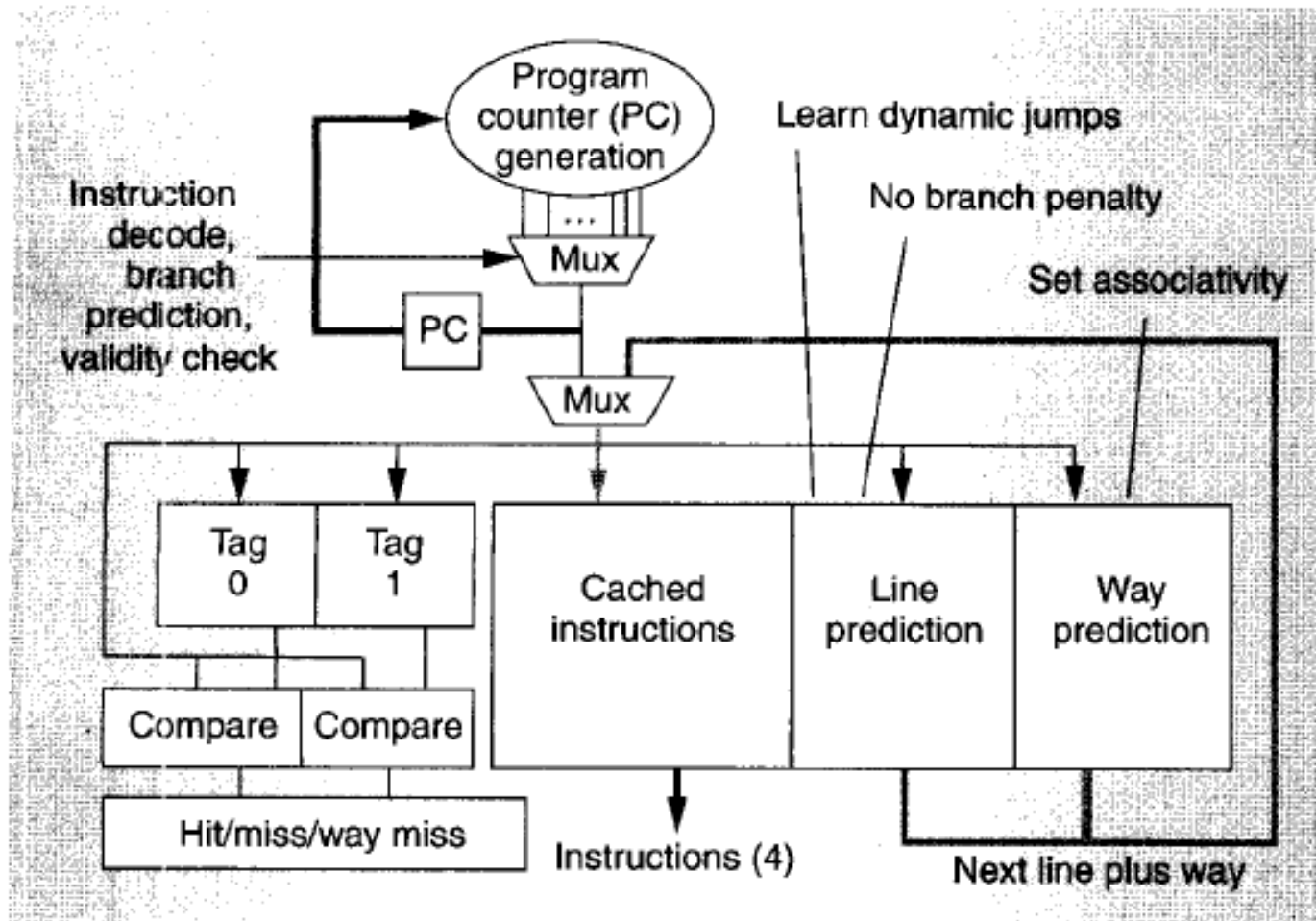
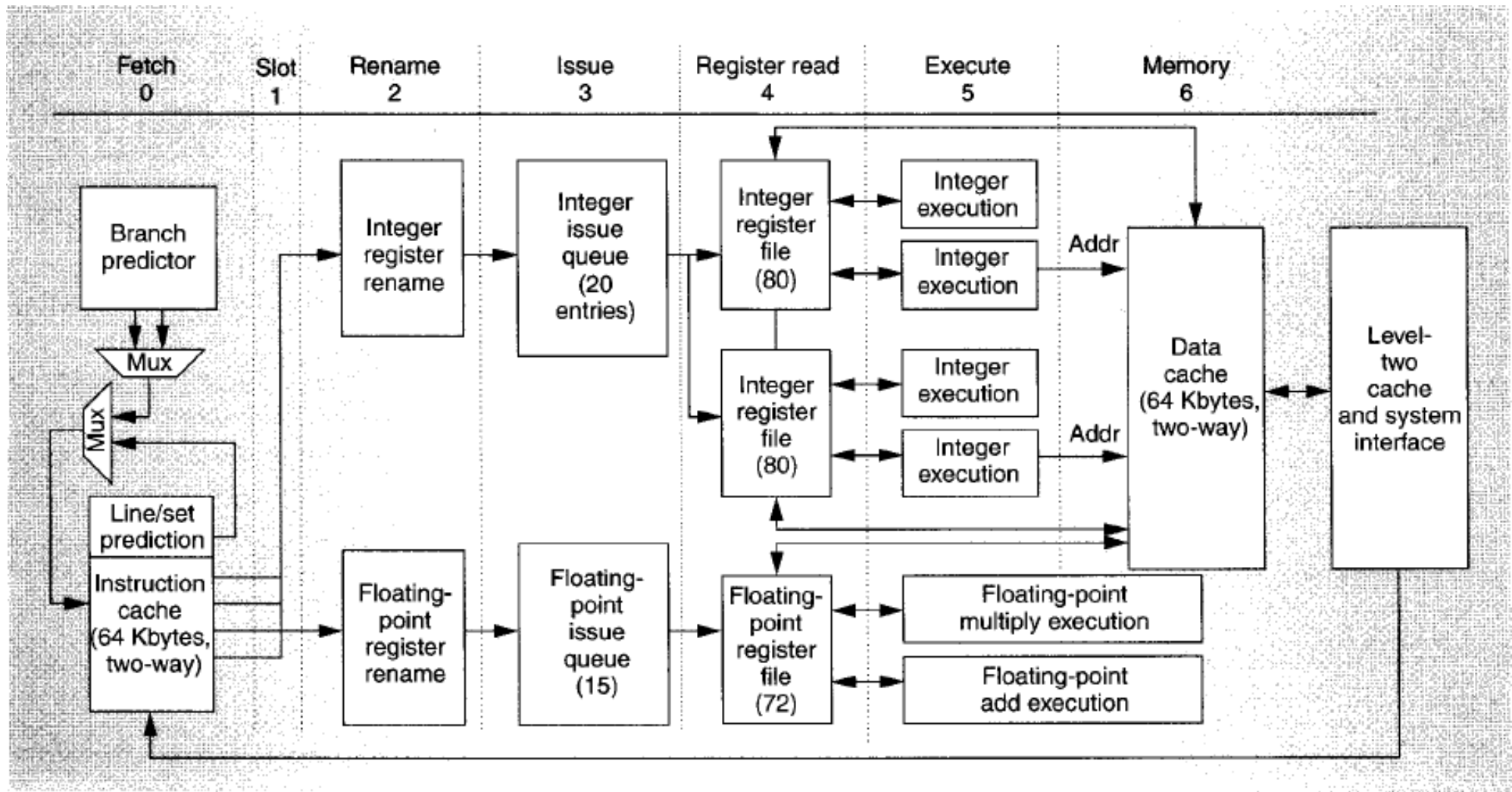


Figure 3. Alpha 21264 instruction fetch. The line and way prediction (wrap-around path on the right side) provides a fast instruction fetch path that avoids common fetch stalls when the predictions are correct.

Alpha 21264 Line & Way Prediction



Issues in Wide Fetch Engines

- Wide Fetch: Fetch multiple instructions per cycle
- Superscalar
- VLIW
- SIMT (GPUs' single-instruction multiple thread model)
- Wide fetch engines suffer from the branch problem:
 - How do you feed the wide pipeline with useful instructions in a single cycle?
 - What if there is a taken branch in the "fetch packet"?
 - What if there are "multiple (taken) branches" in the "fetch packet"?

Fetching Multiple Instructions Per Cycle

- Two problems

1. **Alignment** of instructions in I-cache

- ❑ What if there are not enough (N) instructions in the cache line to supply the fetch width?

2. **Fetch break**: Branches present in the fetch block

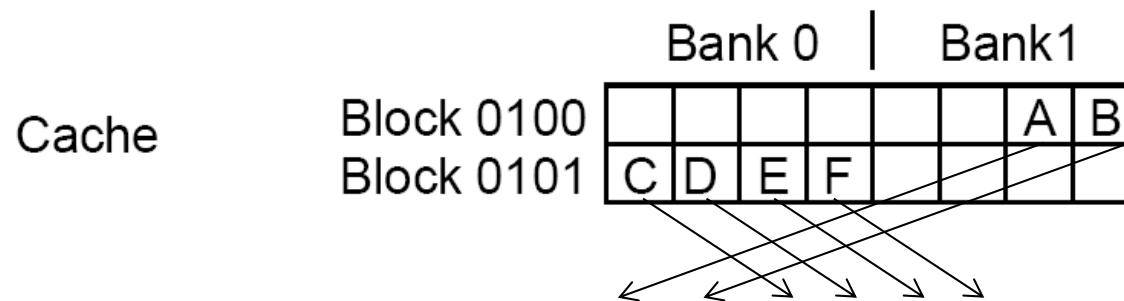
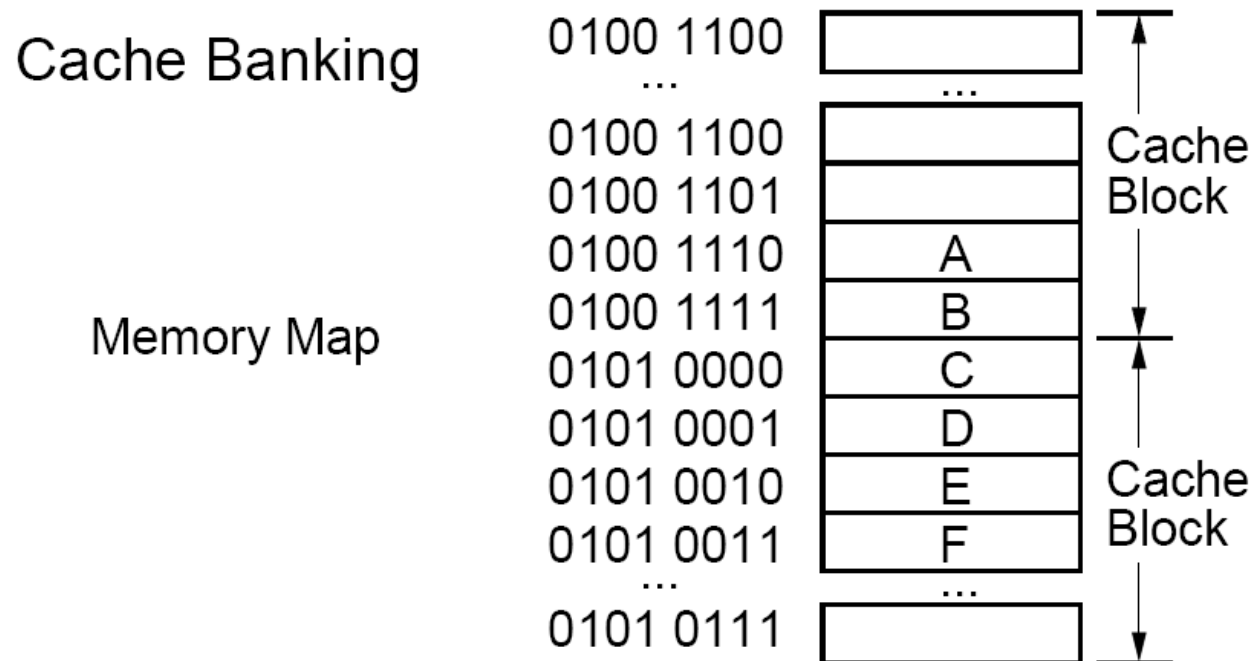
- ❑ Fetching sequential instructions in a single cycle is easy
- ❑ What if there is a control flow instruction in the N instructions?
- ❑ Problem: **The direction of the branch is not known but we need to fetch more instructions**

- These can cause effective fetch width < peak fetch width

Wide Fetch Solutions: Alignment

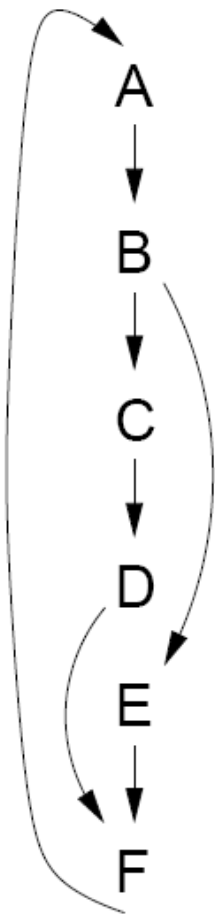
- **Large cache blocks:** Hope N instructions contained in the block
- **Split-line fetch:** If address falls into second half of the cache block, fetch the first half of next cache block as well
 - ❑ Enabled by banking of the cache
 - ❑ Allows sequential fetch across cache blocks in one cycle
 - ❑ Intel Pentium and AMD K5

Split Line Fetch



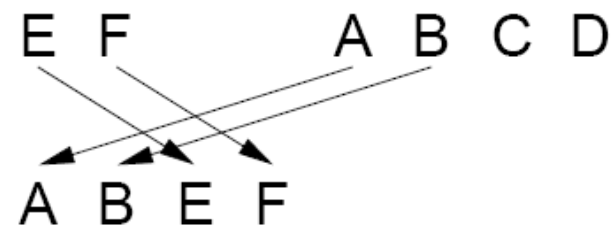
Need alignment logic:

Short Distance Predicted-Taken Branches

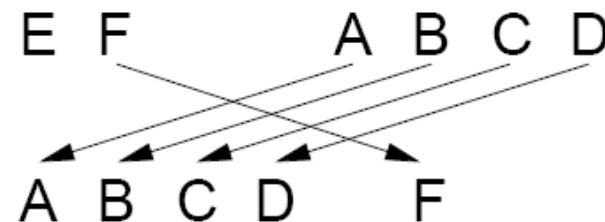


	Bank 0				Bank1			
Block 0100					A	B	C	D
Block 0101	E	F						

First Iteration (Branch B taken to E)



Second Iteration (Branch B fall through to C)

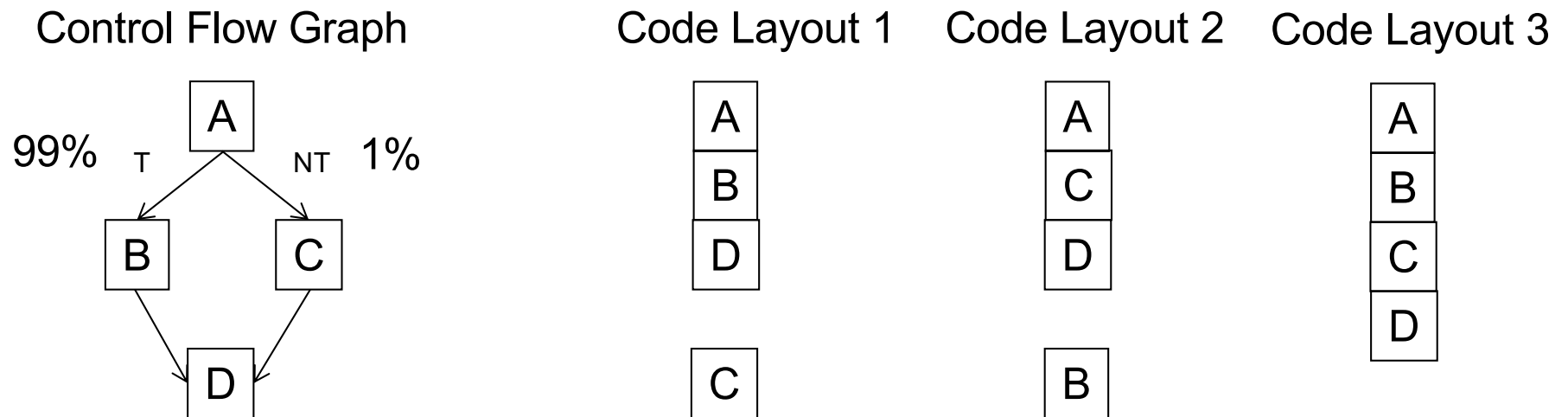


Techniques to Reduce Fetch Breaks

- Compiler
 - ❑ Code reordering (basic block reordering)
 - ❑ Superblock
- Hardware
 - ❑ Trace cache
- Hardware/software cooperative
 - ❑ Block structured ISA

Basic Block Reordering

- Not-taken control flow instructions not a problem: no fetch break: **make the likely path the not-taken path**
- Idea: **Convert taken branches to not-taken ones**
 - ❑ i.e., **reorder basic blocks** (after profiling)
 - ❑ Basic block: code with a single entry and single exit point



- Code Layout 1 leads to the fewest fetch breaks

Basic Block Reordering

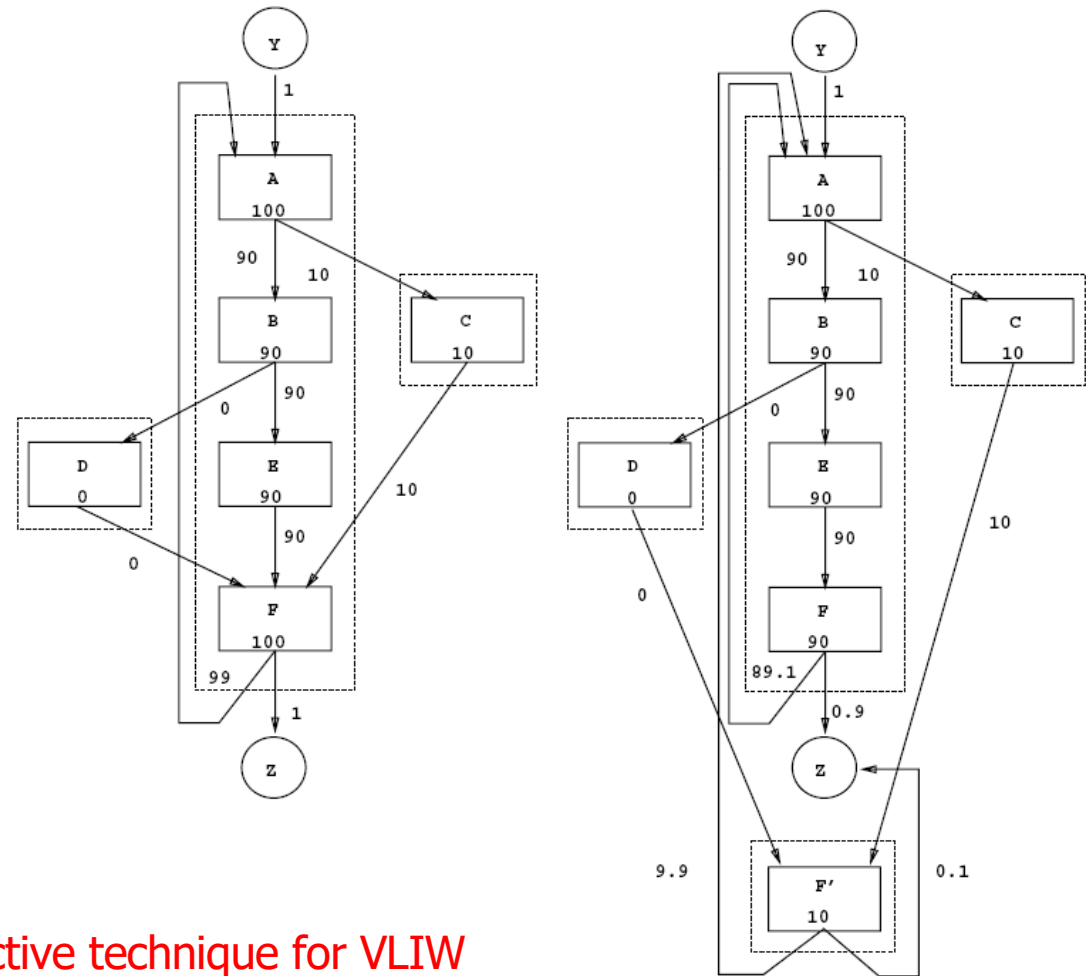
- Pettis and Hansen, “**Profile Guided Code Positioning**,” PLDI 1990.
- Advantages:
 - + Reduced fetch breaks (assuming profile behavior matches runtime behavior of branches)
 - + Increased I-cache hit rate
 - + Reduced page faults
- Disadvantages:
 - Dependent on compile-time profiling
 - Does not help if branches are not biased
 - Requires recompilation

Superblock

- Idea: Combine frequently executed basic blocks such that they form a **single-entry multiple exit larger block**, which is likely executed as straight-line code

- + Helps wide fetch
- + Enables aggressive compiler optimizations and code reordering within the superblock

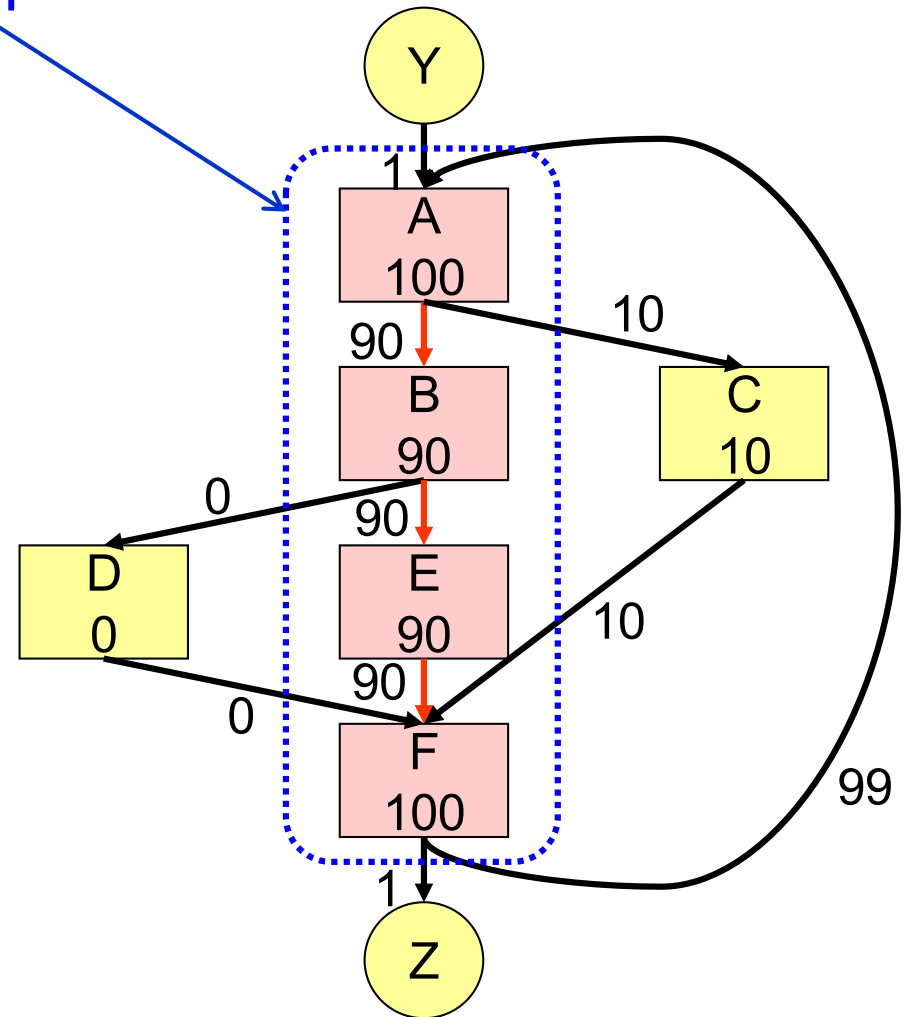
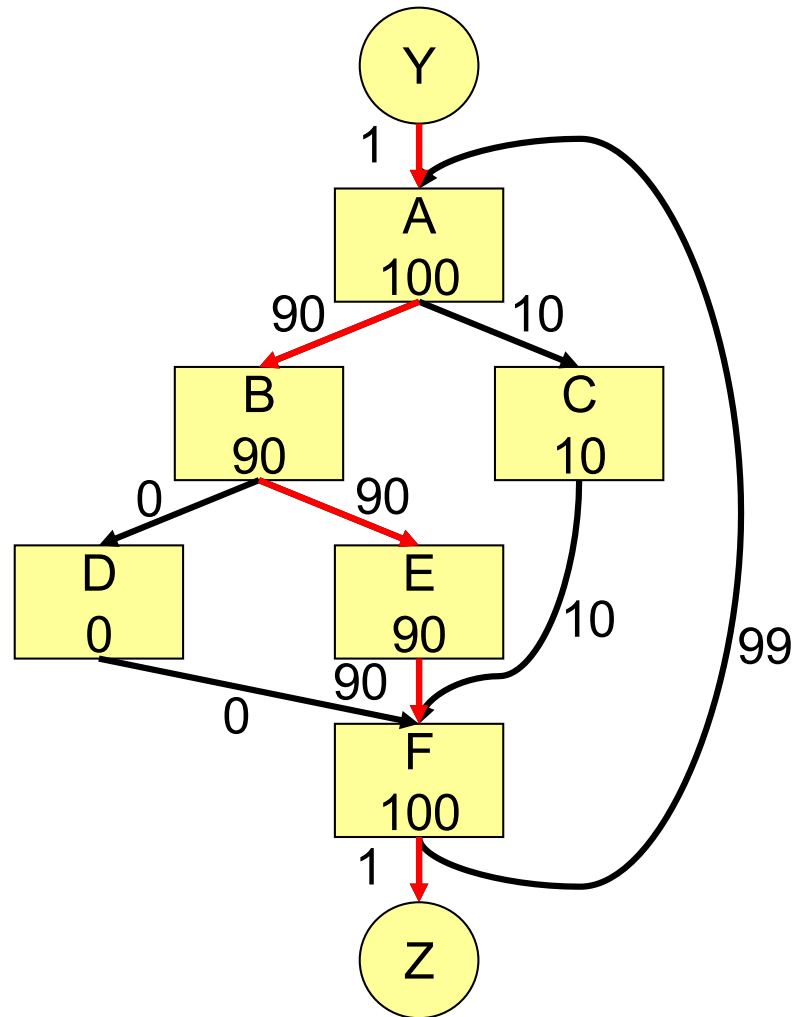
- Increased code size
- Profile dependent
- Requires recompilation



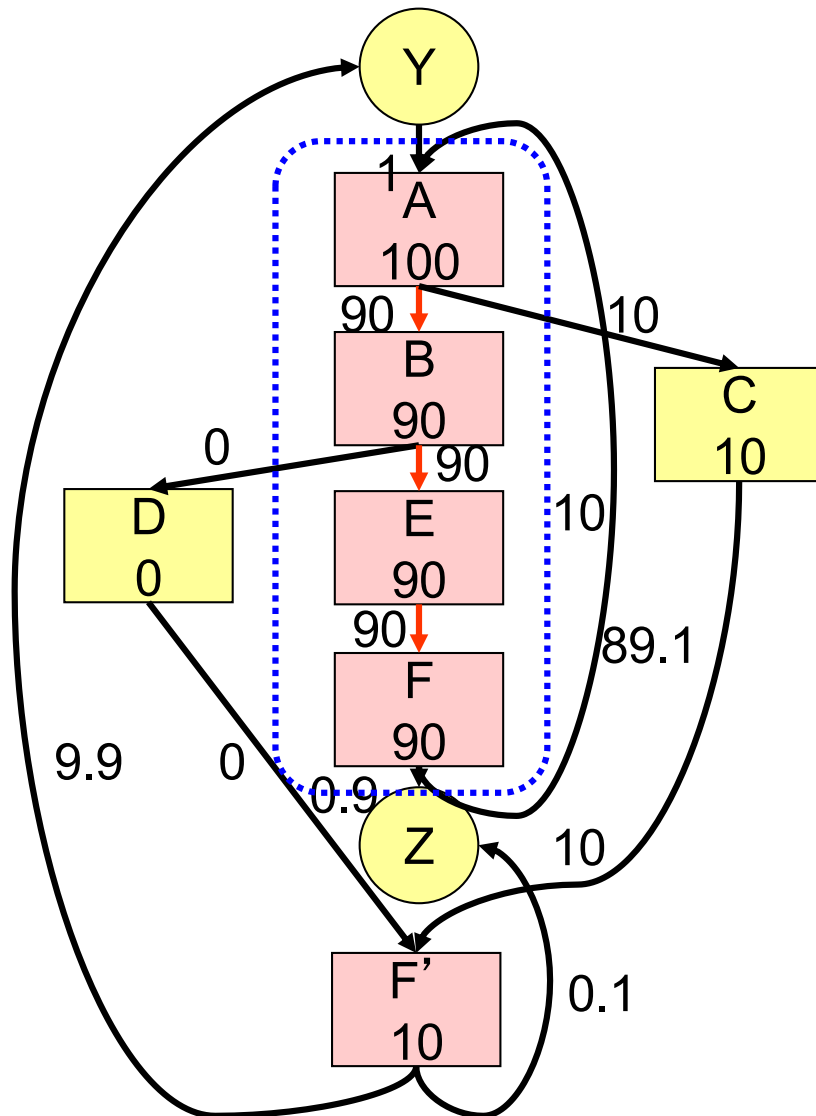
- Hwu et al. “**The Superblock: An effective technique for VLIW and superscalar compilation,**” Journal of Supercomputing, 1993.

Superblock Formation (I)

Is this a superblock?



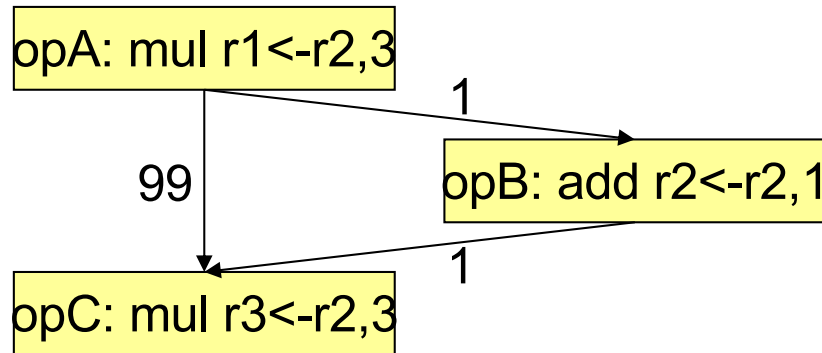
Superblock Formation (II)



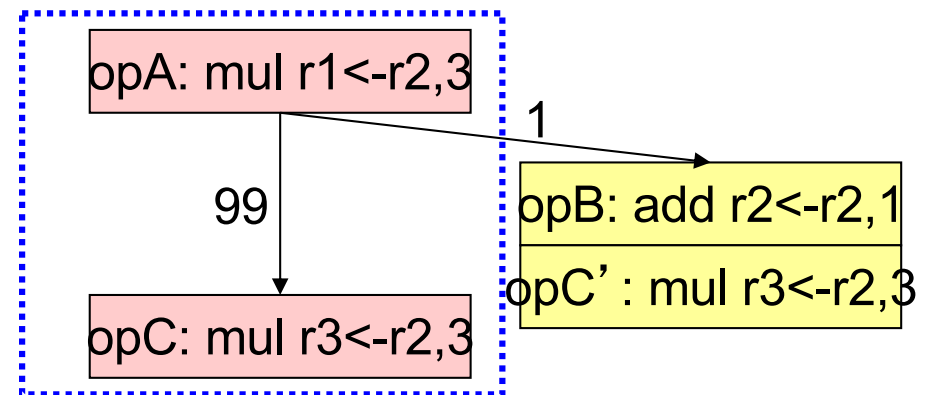
Tail duplication:

duplication of basic blocks
after a side entrance to
eliminate side entrances
→ transforms
a **trace** into a **superblock**.

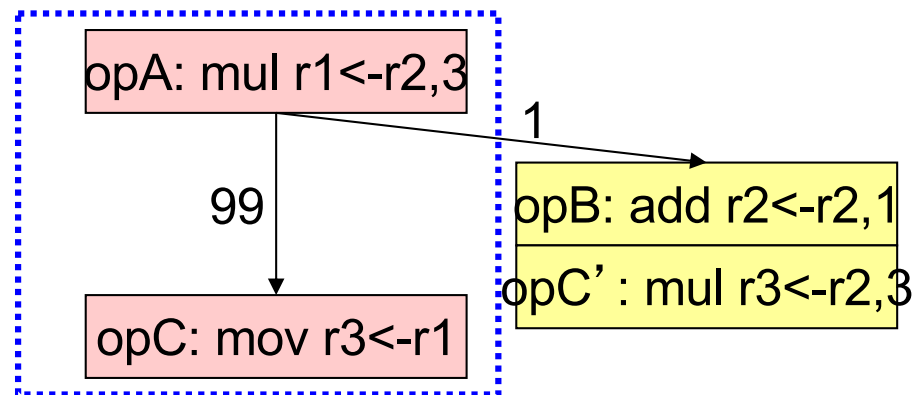
Superblock Code Optimization Example



Original Code



Code After Superblock Formation



Code After Common
Subexpression Elimination

Techniques to Reduce Fetch Breaks

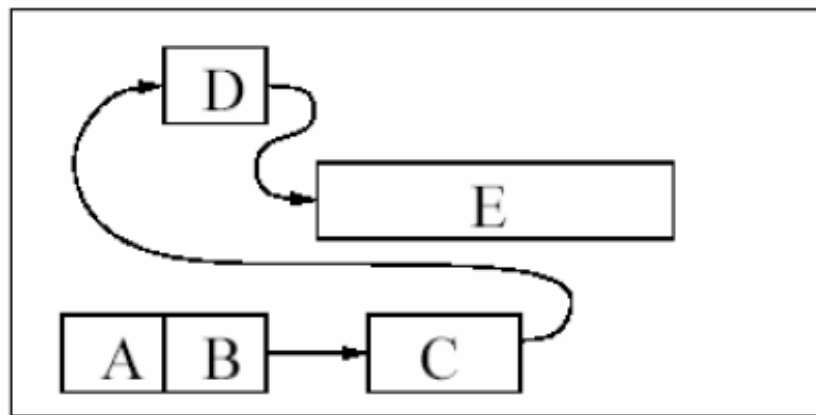
- Compiler
 - ❑ Code reordering (basic block reordering)
 - ❑ Superblock

- Hardware
 - ❑ Trace cache

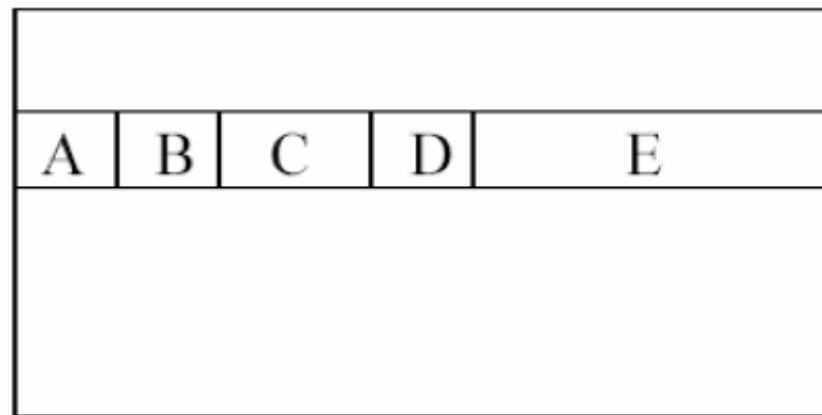
- Hardware/software cooperative
 - ❑ Block structured ISA

Trace Cache: Basic Idea

- A trace is a sequence of executed instructions.
- It is specified by a start address and the branch outcomes of control transfer instructions.
- Traces repeat: programs have frequently executed paths
- Trace cache idea: Store the dynamic instruction sequence in the same physical location.



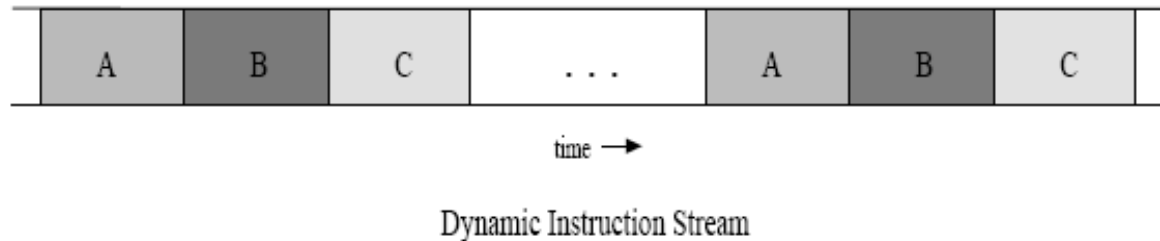
(a) Instruction cache.



(b) Trace cache.

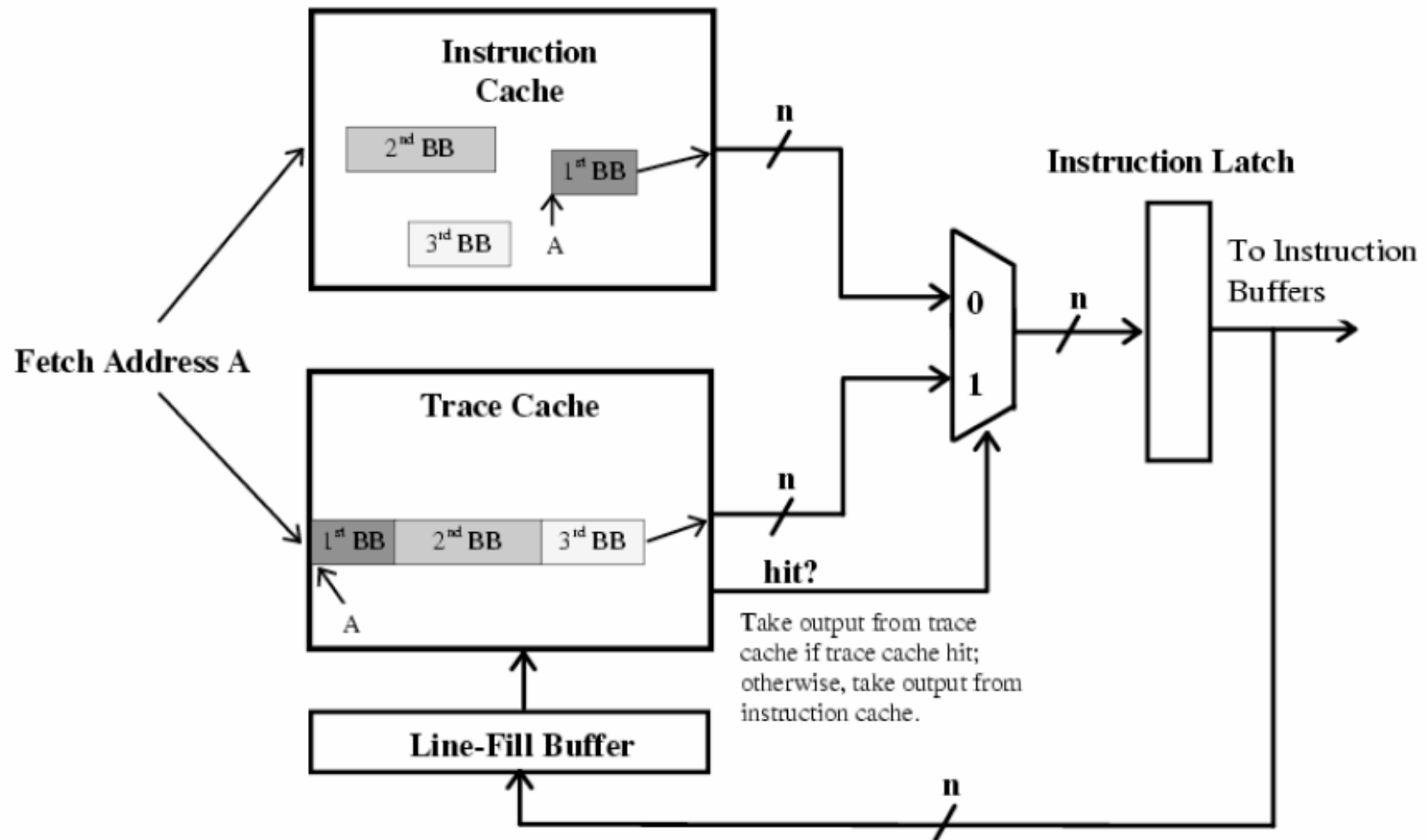
Reducing Fetch Breaks: Trace Cache

- Dynamically determine the basic blocks that are executed consecutively
- Trace: Consecutively executed basic blocks
- Idea: Store consecutively-executed basic blocks in physically-contiguous internal storage (called trace cache)

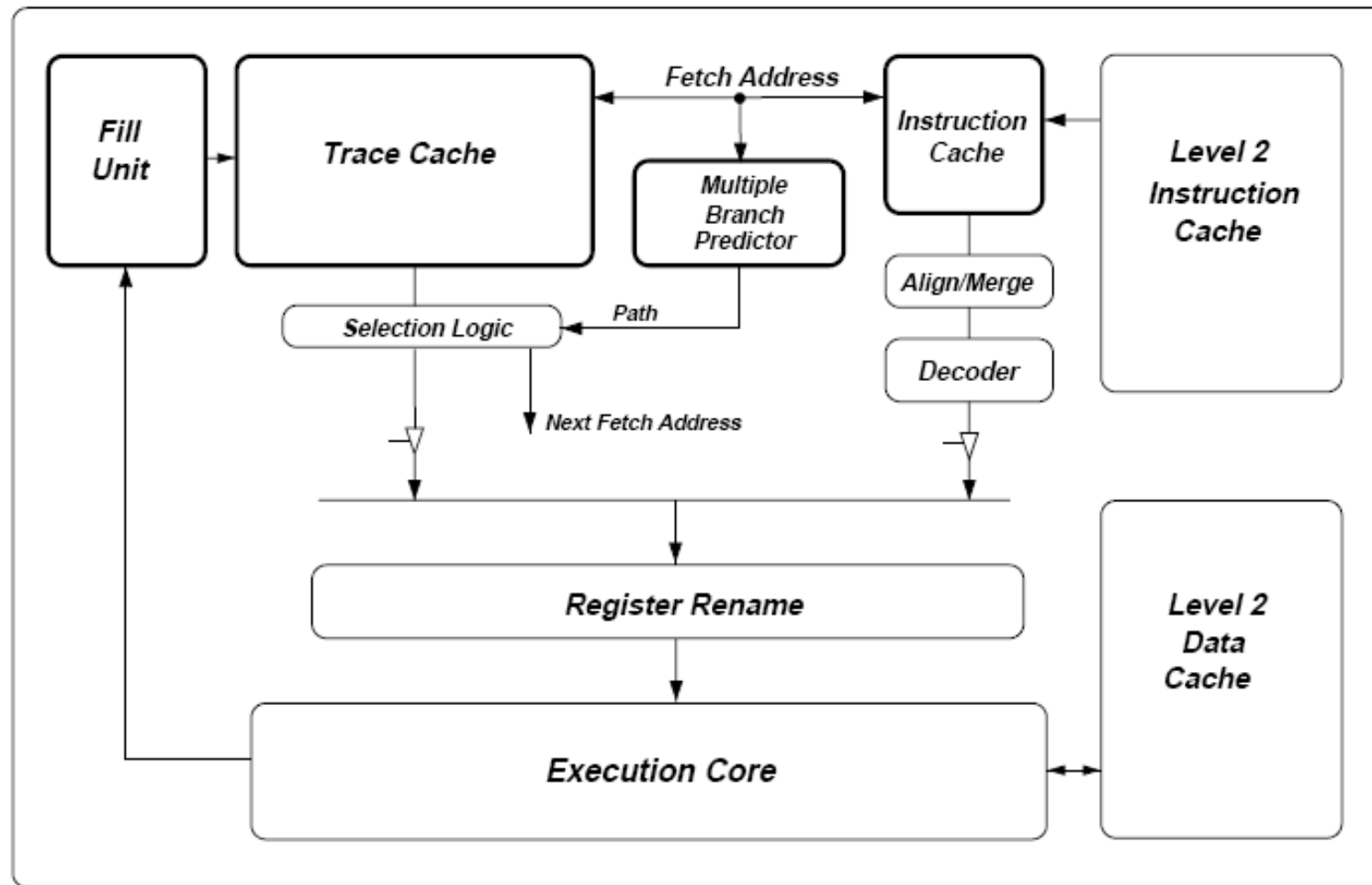


- Basic trace cache operation:
 - ❑ Fetch from consecutively-stored basic blocks (predict next trace or branches)
 - ❑ Verify the executed branch directions with the stored ones
 - ❑ If mismatch, flush the remaining portion of the trace
- Rotenberg et al., “Trace Cache: a Low Latency Approach to High Bandwidth Instruction Fetching,” MICRO 1996.
- Patel et al., “Critical Issues Regarding the Trace Cache Fetch Mechanism,” Umich TR, 1997.

Trace Cache: Example



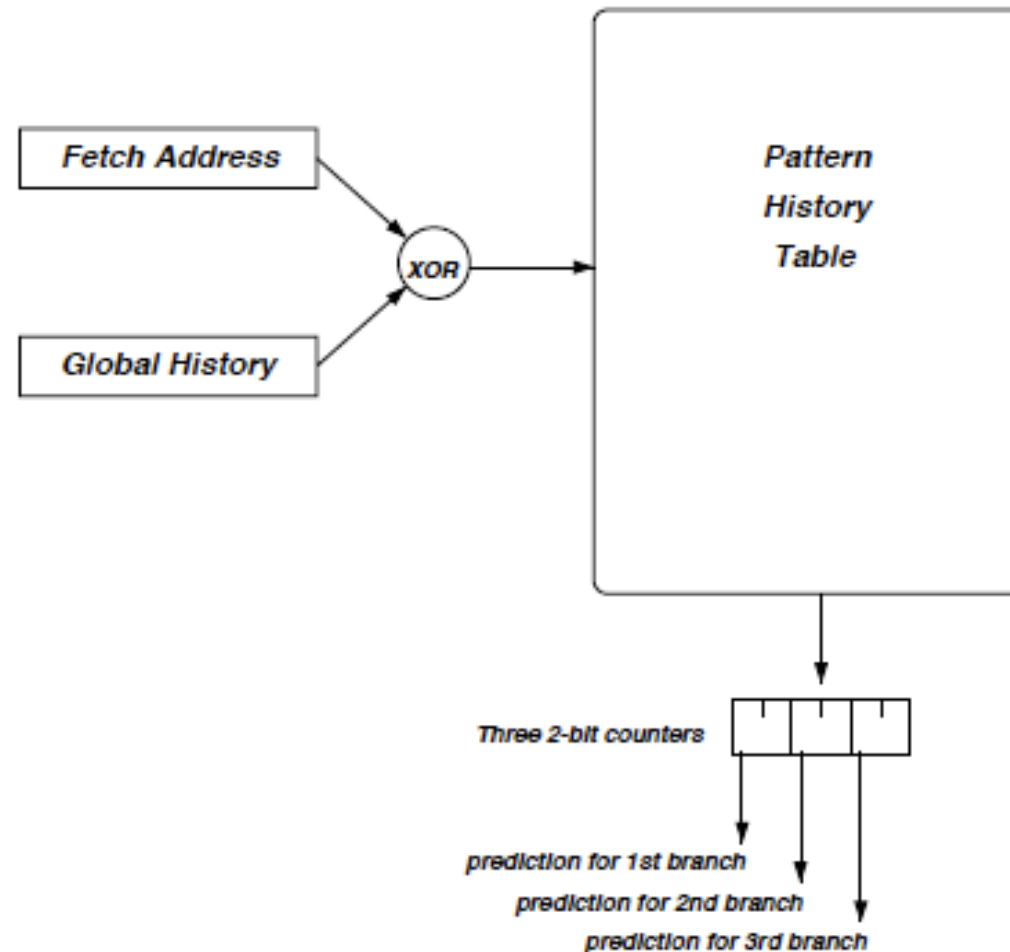
An Example Trace Cache Based Processor



- From Patel's PhD Thesis: "Trace Cache Design for Wide Issue Superscalar Processors," University of Michigan, 1999.

Multiple Branch Predictor

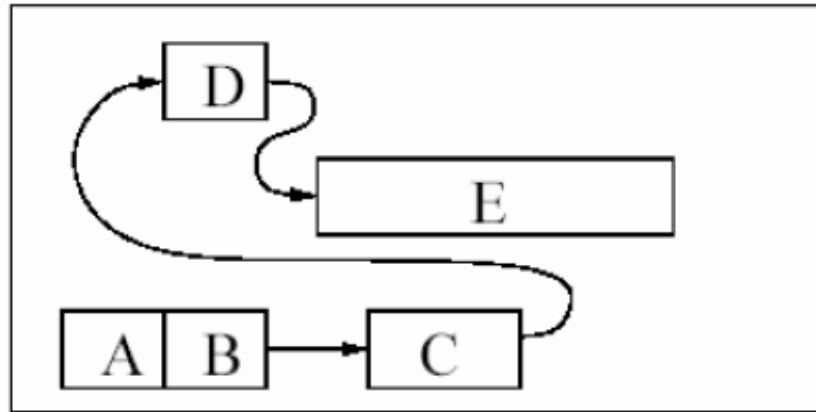
- S. Patel, “Trace Cache Design for Wide Issue Superscalar Processors,” PhD Thesis, University of Michigan, 1999.



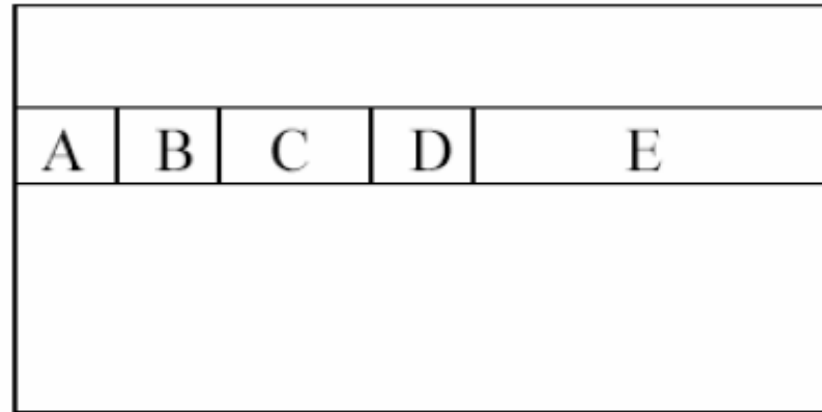
What Does A Trace Cache Line Store?

- 16 slots for instructions. Instructions are stored in decoded form and occupy approximately five bytes for a typical ISA. Up to three branches can be stored per line. Each instruction is marked with a two-bit tag indicating to which block it belongs.
 - Four target addresses. With three basic blocks per segment and the ability to fetch partial segments, there are four possible targets to a segment. The four addresses are explicitly stored allowing immediate generation of the next fetch address, even for cases where only a partial segment matches.
 - Path information. This field encodes the number and directions of branches in the segment and includes bits to identify whether a segment ends in a branch and whether that branch is a return from subroutine instruction. In the case of a return instruction, the return address stack provides the next fetch address.
-
- Patel et al., “Critical Issues Regarding the Trace Cache Fetch Mechanism,” Umich TR, 1997.

Trace Cache: Advantages/Disadvantages



(a) Instruction cache.



(b) Trace cache.

- + Reduces fetch breaks (assuming branches are biased)
- + No need for decoding (instructions can be stored in decoded form)
- + Can enable dynamic optimizations within a trace
- Requires hardware to form traces (more complexity) → called fill unit
- Results in duplication of the same basic blocks in the cache
- Can require the prediction of multiple branches per cycle
 - If multiple cached traces have the same start address
 - What if XYZ and XYT are both likely traces?

Intel Pentium 4 Trace Cache

- A 12K-uop trace cache replaces the L1 I-cache
- Trace cache stores decoded and cracked instructions
 - Micro-operations (uops): returns 6 uops every other cycle
- x86 decoder can be simpler and slower
- A. Peleg, U. Weiser; "[Dynamic Flow Instruction Cache Memory Organized Around Trace Segments Independent of Virtual Address Line](#)", United States Patent No. 5,381,533, Jan 10, 1995

